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Sheaf cohomology

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Introduction

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PART I

Category theory

Introduction

Why categories? Historical introduction?

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a class $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection of sets $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then

$$(f \circ g) \circ h = f \circ (g \circ h)$$

- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then

$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Definition 1.1.2 (Small category). A category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Remark 1.1.1 (Size issues). We will be sketchy about foundational issues (set-theoretic issues, universes, ...).
Therefore all categories will be $\mathcal{U}$-categories for a large enough *universe* $\mathcal{U}$ (i.e: for any pair of objects $X, Y \in \mathsf{C}$ the Hom-set $\mathrm{Hom}(X, Y)$ is $\mathcal{U}$-small) and small categories will be $\mathcal{U}$-small (the set of objects $\mathrm{Ob}(\mathsf{C})$ is $\mathcal{U}$-small).
We will refer as "big" categories those that are $\mathcal{V}$-categories for a larger universe $\mathcal{V}$. By Grothendieck's axiom (for any set X there exists a universe $\mathcal{U}$ such that $X \in \mathcal{U}$) every category C is a $\mathcal{V}$-category for a big enough universe $\mathcal{V}$, and this universe can be chosen so that C is $\mathcal{V}$-small.
For a basic introduction of what an *universe* is and a more carefull treatment of this type of size issues see [KS06, Section 1.1]. More details can also be found at [SGA4]

Notation. To indicate that X is an object of a category C we write

$$X \in \mathsf{C} \text{ instead of } X \in \mathrm{Ob}(\mathsf{C})$$

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphism from X to Y we write

$$f : X \to Y \text{ or } X \xrightarrow{f} Y$$

We will sometimes abbreviate the composition $f \circ g$ as fg.

Example 1.1.1.

- The category Set has the collection of all sets as objects and mappings as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as morphisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Example 1.1.2 (Category of open sets)**.** Let X be a topological space. There is a category of open sets $\mathsf{Open}(X)$ consisting of

- As objects, the open subsets of X.
- As morphisms, inclusion of open sets: $U \to V$ if and only if $U \subseteq V$

Examples of non-concrete categories?

Example 1.1.3 (Discrete category)**.** Given a category C we can define the **discrete category** C_d taking the same collection of objects as C but only the identity morphisms between them.

Definition 1.1.3 (Subcategory)**.** Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.4.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

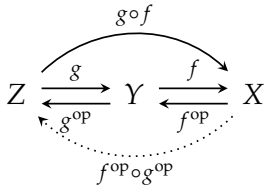
Definition 1.1.4 (Product category)**.** Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}(\mathsf{C} \times \mathsf{D}) \coloneqq \mathrm{Ob}(\mathsf{C}) \times \mathrm{Ob}(\mathsf{D})$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) \coloneqq \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ then
$$(f', g') \circ (f, g) \coloneqq (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

1.1.1 Duality

Definition 1.1.5 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} \colon X \to Y$, $g^{\mathrm{op}} \colon Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$$
where $f : Y \to X$ and $g : Z \to Y$ are the associated morphisms in C. Visually



Example 1.1.5. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed ordering $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.2. Let $f^{\mathrm{op}} \colon Y \to X$ and $g^{\mathrm{op}} \colon Y \to X$ be two morphisms in the opposite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Note that, by definition, for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$.

Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.

In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

Definition 1.2.1 (Monomorphism and epimorphism)**.** Let $f : X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms $g, h : Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$, i.e: f is left-cancellative.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h : Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$, i.e: f is right-cancellative.

Example 1.2.1.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

Example 1.2.2. In Ring, there are epimorphisms that are not surjective ring homomorphisms.
For example, consider the inclussion $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$. For any ring homomorphism $f : \mathbb{Q} \to R$, where R is any ring, since for all $\frac{a}{b} \in \mathbb{Q}$ we have $a, b \in \mathbb{Z}$

$$f\left(\frac{a}{b}\right) = f(a)f(b^{-1}) = f(a)f(b)^{-1}$$

Therefore f is completely determined by its values in $\mathbb{Z}$ and, since $f \circ \iota = f|_{\mathbb{Z}}$ we get that ι is right-cancellative.

Lemma 1.2.1 (Monomorphisms and epimorphisms are dual)**.** Let $f : X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}} : Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h : Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}} : X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Definition 1.2.2 (Isomorphism). A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Proposition 1.2.1. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$
4. Let $h \coloneqq (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand
$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$
On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have
$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$
□

Example 1.2.3.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.
- In Ring, an isomorphism is a bijective ring homomorphism.

Proposition 1.2.2. Let $f : X \to Y$ be an isomorphism. Then f is both a monomorphism and epimorphism.

Proof. Suppose $f: X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1}: Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Consider a pair of morphisms $g, h: Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$

Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h': Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.4. In general, a morphism $f: X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set with more than one point X endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $\mathbb{1}_X: X_d \to X_t$, which is continuous. Since $\mathbb{1}_X$ is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism. However since there does not exists a continuous bijection $X_t \to X_d$, $\mathbb{1}_X$ cannot be an isomorphism.

Proposition 1.2.3 (Llevar?)**.** Let D be a subcategory of a category C. If $f: X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f: X \to Y$ is an isomorphism in D there exists an inverse $f^{-1}: Y \to X$ in D. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

Proposition 1.2.4 (Llevar?)**.** Let D be a full subcategory of a category C. If $f: X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f: X \to Y$ is an isomorphism in C there exists an inverse $f^{-1}: Y \to X$ in D. Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^{-1} \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Definition 1.3.1 (Initial and terminal objects)**.** An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

Example 1.3.1.

- In Set, the empty set $\varnothing$ is initial and the singleton $\{*\}$ is terminal.
- In Grp, trivial groups are both initial and terminal.
- In Ring both Rng, the zero ring is initial. However, the terminal is different: in Ring it is $\mathbb{Z}$ and in Rng is the zero ring.
- In Top, the empty topological space is initial and the singleton $\{*\}$ is terminal.
- There are no initial nor terminal objects, since there are no homomorphisms between fields of different characteristic.

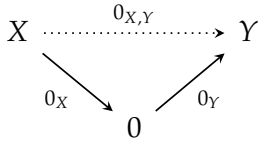
Proposition 1.3.1 (Uniqueness of initial and terminal objects)**.** Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

Proof. Since X is initial in C there exists a unique morphism $f : X \to Y$. Analogously there exist a unique morphism $g : Y \to X$.

Note that $g \circ f : X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g : Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

Definition 1.3.2 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist unique maps $0_X : X \to 0$ and $0_Y : 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0_{X,Y}} Y$:



By abuse of notation, we will denote all of these maps by 0.

Example 1.3.2.

- Set and Top have no zero objects since $\varnothing \neq \{*\}$.
- In Grp and Ab, the zero group is the zero object.
- Ring has no zero objects. However in Rng the zero ring is the zero object.

Corollary 1.3.1 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.1. □

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $Ff : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = Fg \circ Ff$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = Ff \circ Fg$. This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

And a contravariant functor by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

Remark 2.1.1. We will disfavour the covariant/contravariant distinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.1.

- There is a (covariant) endofunctor $P\colon \mathsf{Set} \to \mathsf{Set}$ that sends a set A to its power set $PA = \{A' \subseteq A\}$ and a map $f\colon A \to B$ to the direct image $f_*\colon PA \to PB$, $f_*(A') := f(A')$.
  We also get a contravariant version $P\colon \mathsf{Set}^{\mathrm{op}} \to \mathsf{Set}$ that sends a map

$f: A \to B$ to the inverse image $f^{-1}: PB \to PA$, $f^{-1}(B') := f^{-1}(B)$.
- There is a functor $F: \mathsf{Set} \to \mathsf{Grp}$ that sends a set X to the free group on X.
- There is a functor $-^*: \mathsf{Vect}_{\mathbb{K}}^{\mathrm{op}} \to \mathsf{Vect}_{\mathbb{K}}$ that sends a vector space V to its dual V^*.
- There is a functor $\mathrm{Spec}: \mathsf{CRing}^{\mathrm{op}} \to \mathsf{Top}$ that sends a commutative ring R to its set $\mathrm{Spec}(R)$ of prime ideals endowed with the Zariski topology.

Example 2.1.2 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{rccc} \mathbb{1}: & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & X \\ & f\downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.3 (Constant functor). Let C be a category. For every object $C \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_C: & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & C \\ & f\downarrow & \longmapsto & \downarrow \mathbb{1}_C \\ & Y & & C \end{array}$$

Example 2.1.4 (Bifunctor). Let C, C' and D be three categories. Recall that we can construct the product category $\mathsf{C} \times \mathsf{C}'$.
A **bifunctor** is a functor of the form $F: \mathsf{C} \times \mathsf{C}' \to \mathsf{D}$. This means that for any $X \in \mathsf{C}$ and $X' \in \mathsf{C}$, $F(X, -): \mathsf{C}' \to \mathsf{D}$ and $F(-, X'): \mathsf{C} \to \mathsf{D}$ are functors.
Furthermore for any morphisms $f: X \to Y$ in C and $g: X' \to Y'$ in C' we have a commutative diagram

$$\begin{array}{ccc} F(X,X') & \xrightarrow{F(X,g)} & F(X,Y') \\ {\scriptstyle F(f,X')}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(f,Y')} \\ F(Y,X') & \xrightarrow[F(Y,g)]{} & F(Y,Y') \end{array}$$

Example 2.1.5. Let C be a For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-): & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(-,C): & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \uparrow \circ f \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a bifunctor

$$\begin{array}{ccc} \mathrm{Hom}(-,-)\colon \mathsf{C}^{\mathrm{op}}\times\mathsf{C} & \longrightarrow & \mathsf{Set} \\ (X,Y) & & \mathrm{Hom}(X,Y) \\ {\scriptstyle (f,g)}\downarrow & \longmapsto & \downarrow{\scriptstyle g\circ -\circ f} \\ (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.6. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{ccc} \iota\colon \mathsf{D} & \hookrightarrow & \mathsf{C} \\ X & & X \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f} \\ Y & & Y \end{array}$$

Example 2.1.7 (Preimage functor)**.** Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Open}(X), \mathsf{Open}(Y)$.
For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{ccc} f^{-1}\colon \mathsf{Open}(Y) & \longrightarrow & \mathsf{Open}(X) \\ U & & f^{-1}(U) \\ \downarrow & \longmapsto & \downarrow \\ V & & f^{-1}(V) \end{array}$$

Proposition 2.1.1 (Functors preseve isomorphisms)**.** Let C, D be two categories and $F\colon \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f\colon X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1}\colon Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$\begin{aligned} F(f^{-1} \circ f) = F\mathbb{1}_X &\implies Ff^{-1} \circ Ff = \mathbb{1}_{F(X)} \\ F(f \circ f^{-1}) = F\mathbb{1}_Y &\implies Ff \circ Ff^{-1} = \mathbb{1}_{F(Y)} \end{aligned}$$

i.e: Ff has an inverse Ff^{-1} in D. □

Definition 2.1.2 (Functor types)**.** A functor $F\colon \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

Example 2.1.8.

- The functor $U\colon \mathsf{Grp} \to \mathsf{Set}$ that sends a group to its underlying set is faithful.
- A subcategory D of a category C is full if and only if the inclussion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.

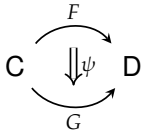
– Given a field $\mathbb{K}$, let $\mathsf{C}_{\mathbb{K}}$ be the category whose objects are $\mathbb{K}^n$ for some $n \in \mathbb{N}$. The inclussion $\mathsf{C}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ is essentially surjective, since all finite dimensional $\mathbb{K}$-vector spaces are isomorphic to $\mathbb{K}^n$ for some n.

2.2 Natural transformations

Definition 2.2.1. Let $F, G \colon \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi \colon F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D, one for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by



If each component ψ_X is an isomorphism in D, we say that ψ is a **natural isomorphism**, and we write $F \cong G$.

Example 2.2.1. In $\mathsf{Vect}_{\mathbb{K}}$, the map $\mathrm{ev} \colon V \to V^{**}$ that sends a vector $v \in V$ to the linear evaluation function $\mathrm{ev}_v \colon V^* \to \mathbb{K}$ defines the components of a natural transformation from the identity functor $\mathbb{1} \colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$ to the double dual functor $-^{**} \colon \mathsf{Vect}_{\mathbb{K}} \to \mathsf{Vect}_{\mathbb{K}}$.

Definition 2.2.2 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F \colon \mathsf{C} \to \mathsf{D}$ and $G \colon \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**. We denote this by $\mathsf{C} \simeq \mathsf{D}$.

Theorem 2.2.1 ([Rie, Theorem 1.5.9])**.** A functor $F \colon \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

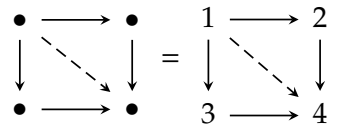
Limits and colimits

3.1 Diagrams, cones and cocones

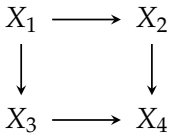
Definition 3.1.1 (Diagram). Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X\colon \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j \coloneqq X(j)$.

Example 3.1.1 (Constant diagram). A diagram $X\colon \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.2. Consider the category

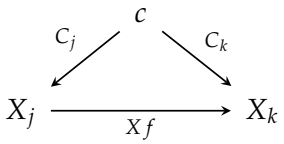
$\mathsf{J} =$  $=$

A diagram with shape J is a commutative square



3.1.1 Cones and cocones

Definition 3.1.2 (Cone). A **cone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C\colon \mathrm{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j\colon c \to X_j\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j\colon c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes



If $C\colon \mathrm{const}_c \Rightarrow X$ and $D\colon \mathrm{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram

commute

$$\begin{array}{ccc} \mathrm{const}_c & \Longrightarrow & \mathrm{const}_d \\ & {\scriptstyle C}\searrow \quad \swarrow{\scriptstyle D} & \\ & X & \end{array}$$

Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.3 (Cocone)**.** A **cocone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C\colon X \Rightarrow \mathrm{const}_c$ from the constant diagram to X. The components $\{C_j\colon X_j \to c\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j\colon X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

$$\begin{array}{ccc} X_j & \xrightarrow{Xf} & X_k \\ & {\scriptstyle C_j}\searrow \quad \swarrow{\scriptstyle C_k} & \\ & c & \end{array}$$

If $C\colon X \Rightarrow \mathrm{const}_c$ and $D\colon X \Rightarrow \mathrm{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram commute

$$\begin{array}{ccc} & X & \\ & {\scriptstyle C}\swarrow \quad \searrow{\scriptstyle D} & \\ \mathrm{const}_c & \Longrightarrow & \mathrm{const}_d \end{array}$$

Therefore for any diagram $X\colon \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

Notation. For a diagram $X\colon \mathsf{J} \to \mathsf{C}$ we will denote a cone $C\colon \mathrm{const}_c \Rightarrow X$ by

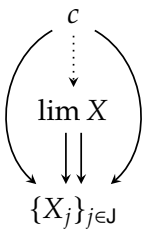
$$\begin{array}{c} c \\ \downarrow\downarrow \\ \{X_j\}_{j\in\mathsf{J}} \end{array}$$

and a cocone $D\colon X \Rightarrow \mathrm{const}_d$ by

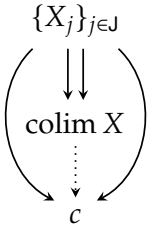
$$\begin{array}{c} \{X_j\}_{j\in\mathsf{J}} \\ \downarrow\downarrow \\ d \end{array}$$

3.2 Limits and colimits

Definition 3.2.1 (Limit)**.** A **limit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X\colon \mathrm{const}_{\lim X} \Rightarrow X$ such that for any other cone $C\colon \mathrm{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute



Definition 3.2.2 (Colimit). A **colimit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$, denoted $\operatorname{colim} X$, is a terminal object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\operatorname{colim} X\colon X \Rightarrow \operatorname{const}_{\operatorname{colim} X}$ such that for any other cocone $C\colon X \Rightarrow \operatorname{const}_c$ there exists a unique morphism $\operatorname{colim} X \to c$ making the following diagram commute



Proposition 3.2.1 (Uniqueness of limits and colimits). If a diagram $X\colon \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Proof. Direct from Proposition 1.3.1. □

Definition 3.2.3 (Functors and limits). Let $F\colon \mathsf{C} \to \mathsf{D}$ be a functor and $X\colon \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

We now define some special cases of limits and colimits that are generally useful. However, it is important to take into account that this constructions do not exist in general for all categories.

Products and coproducts

Let J be a small and discrete category.

Since the only morphisms in J are the identity morphisms, a diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists in a collection of objects $\{X_j\}_{j\in\mathsf{J}}$.

Therefore a cone with apex c over X is an object $c \in \mathsf{C}$ together with morphisms $c \to X_j$ for each $j \in \mathsf{J}$ and a cocone with nadir $c \in \mathsf{C}$ over X is an object c together with morphisms $X_j \to c$ for each $j \in \mathsf{J}$.

Note that any collection of objects $\{X_j\}_{j\in\mathsf{J}}$ in C may be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Definition 3.2.4 (Product). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **product** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\prod_{j\in \mathsf{J}} X_j$, as the limit of the diagram $X\colon \mathsf{J}\to \mathsf{C}$.
Less abstractly, a **product** is an object $\prod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\pi_k\colon \prod_{j\in \mathsf{J}} X_j \to X_k$ for every $j\in \mathsf{J}$ such that, for any map $Y\to X_j$ there exists a unique morphism $Y\to \prod_{j\in \mathsf{J}} X_j$ making the following diagram commute

$$\begin{array}{ccc} Y & \dashrightarrow & \prod_{j\in \mathsf{J}} X_j \longrightarrow X_j \\ & \searrow & \uparrow \\ & & \end{array}$$

Definition 3.2.5 (Coproduct). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **coproduct** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\coprod_{j\in \mathsf{J}} X_j$, as the colimit of the diagram $X\colon \mathsf{J}\to \mathsf{C}$.
Less abstractly, a **coproduct** is an object $\coprod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\iota_k\colon X_k \to \coprod_{j\in \mathsf{J}} X_j$ for every $j\in \mathsf{J}$ such that, for any map $X_j\to Y$ there exists a unique morphism $\coprod_{j\in \mathsf{J}} X_j \to Y$ making the following diagram commute

$$\begin{array}{ccc} Y & \dashleftarrow & \coprod_{j\in \mathsf{J}} X_j \longleftarrow X_j \\ & \nwarrow & \downarrow \\ & & \end{array}$$

Remark 3.2.1. Consider the case where $\mathsf{J}=\varnothing$ is the empty category, i.e: $\mathrm{Ob}(\mathsf{J})=\varnothing$.
Suppose that products exist in C.
Let's see how are $\varnothing$-shaped products (empty products).
Note that the family of objecs $\{X_j\}_{j\in\varnothing}\subseteq \mathsf{C}$ is empty, and so there are no projection maps $\pi_k : \prod_{j\in\varnothing} X_j \to X_k$ nor morphisms $\{f_j : Y\to X_j\}_{j\in\varnothing}$ for all $Y\in\mathsf{C}$. Therefore the product condition is vacuosly true, i.e: there exists a unique map $Y\to \prod_{j\in\varnothing} X_j$ for all $Y\in\mathsf{C}$. This means that $\prod_{j\in\varnothing} X_j$ is a terminal object in C.
Dually, if C has coproducts then the empty coproduct is initial in C.

Equalisers and coequalisers

Consider the *parallel pair* category

$$\mathsf{J} = \quad \bullet \rightrightarrows \bullet$$

A diagram $X\colon \mathsf{J}\to\mathsf{C}$ corresponds to a pair of morphisms $f,g\colon A\rightrightarrows B$ in C.

Therefore a cone $C\colon \mathrm{const}_c \Rightarrow X$ is a morphism $c\colon C\to A$ such that $fc=gc$:

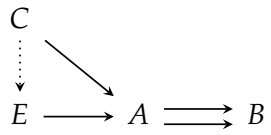
$$C \xrightarrow{c} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

Analogously a cocone $D\colon X\Rightarrow \mathrm{const}_d$ is a morphism $d\colon B\to D$ such that $df=dg$:

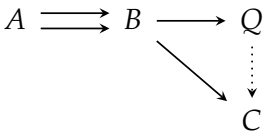
$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{d} D$$

Note that any pair of morphisms $f,g\colon A\rightrightarrows B$ can be considered as a diagram $X\colon \mathsf{J}\to\mathsf{C}$.

Definition 3.2.6 (Equaliser). Let $f,g\colon A\rightrightarrows B$ be two morphisms in C, considered as a diagram $X\colon \mathsf{J}\to\mathsf{C}$. The **equaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $E\in\mathsf{C}$ together with a morphism $\mathrm{eq}\colon E\to A$ such that for any morphism $c\colon C\to A$ with $fc=gc$ there exists a unique map $C\to E$ the following diagram commute



Definition 3.2.7 (Coequaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **coequaliser** of f and g is the limit of this diagram. Less abstractly, it is an object $Q \in \mathsf{C}$ together with a morphism $\mathrm{coeq} : B \to Q$ such that for any morphism $d : B \to C$ with $df = dg$ there exists a unique map $Q \to C$ making the following diagram commute



Proposition 3.2.2. Equaliser arrows are monomorphisms. Dually, coequaliser arrows are epimorphisms.

Proof. Let $\mathrm{eq} : E \to X$ be a equaliser. Consider two morphisms $f, g : C \rightrightarrows E$ in C such that $\mathrm{eq} \circ f = \mathrm{eq} \circ g$. By the uniqueness of the map $\mathrm{eq} : E \to X$ we have $f = g$. □

Example 3.2.1 (Equalisers and coequalisers in Set). In Set the equaliser of two maps $f, g : A \rightrightarrows B$ is $E = \{a \in A \mid f(a) = g(a)\}$ together with the inclussion $E \hookrightarrow A$. The coequaliser of f and g is $Q = {}^{B}\!/_{\sim}$ where $\sim$ is the smallest equivalence relation such that for all $a \in A \quad f(a) \sim g(a)$ together with the natural projection

$$\begin{aligned} \mathrm{coeq} : B &\twoheadrightarrow Q \\ b &\mapsto [b] \end{aligned}$$

A special case of equalisers and coequalisers are kernels and cokernels.

Definition 3.2.8 (Kernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **kernel** $\ker f$ of a morphism $f : A \to B$ is the equaliser

$$\ker f \rightarrowtail A \underset{0}{\overset{f}{\rightrightarrows}} B$$

Less abstractly, it is an object $\ker f$ together with a morphism $i : \ker f \to X$ such that for every $C \in \mathsf{C}$ and every morphism $g : C \to X$ such that $f \circ g = 0$ there is a unique morphism $\phi : C \to \ker f$ such that $g = p \circ \phi$

Definition 3.2.9 (Cokernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **cokernel** $\mathrm{coker}\, f$ of a morphism $f : A \to B$ is the coequaliser

$$A \underset{0}{\overset{f}{\rightrightarrows}} B \twoheadrightarrow \mathrm{coker}\, f$$

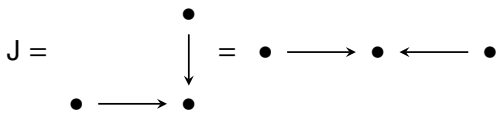
Less abstractly, it is an object $\mathrm{coker}\, f$ together with a morphism $p : Y \to \mathrm{coker}\, f$ such that for every $C \in \mathsf{C}$ and every morphism $g : Y \to C$ such that $g \circ f = 0$ there is a unique morphism $\phi : \mathrm{coker}\, f \to C$ such that $g = \phi \circ p$.
We denote

$${}^{A}\!/_{B} := \mathrm{coker}(A \to B)$$

Remark 3.2.2. By Proposition 3.2.2, kernels are monomorphisms and cokernels are epimorphisms.

Pullbacks and pushouts

Consider the category



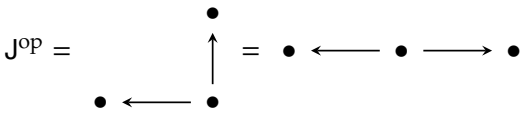
A diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists of a pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$.

Therefore a cone $\mathrm{const}_C \Rightarrow X$ consists of an object $C \in \mathsf{C}$ together with morphisms $C \to X$ and $C \to Y$ making the following diagram commute.

$$\begin{array}{ccc} C & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow[g]{} & Z \end{array}$$

Note that any pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Consider now the opposite category



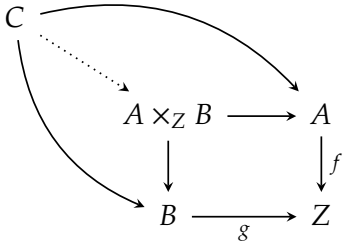
A diagram $Y\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$ consists of a pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$.

Therefore a cocone $X \Rightarrow \mathrm{const}_D$ consists of an object $D \in \mathsf{D}$ together with morphisms $A \to D$ and $B \to D$ making the following diagram commute

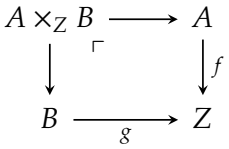
$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

Note that any pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.

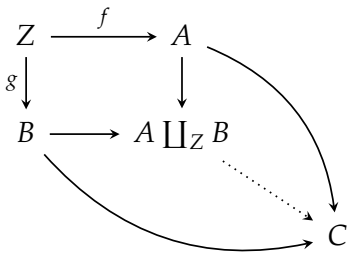
Definition 3.2.10 (Pullback). Let $A \xrightarrow{f} Z \xleftarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.
The **pullback** (or **fibre product**) of f and g is the limit of this diagram, denoted by $A \times_Z B$.
Less abstractly, it is an object $A \times_Z B \in \mathsf{C}$ together with a pair of morphisms $\pi_A\colon A \times_Z B \to A, \pi_B : A \times_Z B \to B$ such that for any pair of morphisms $C \to A, C \to B$ making the square commute, there exists a unique morphism $C \to A \times_Z B$ such that



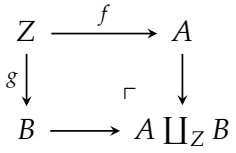
Pullback diagrams are often denoted by



Definition 3.2.11 (Pushout). Let $A \xleftarrow{f} Z \xrightarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X \colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.
The **pushout** of f and g is the colimit of this diagram, denoted by $X \coprod_Z Y$.
Less abstractly, it is an object $A \coprod_Z B \in \mathsf{C}$ together with a pair of morphisms $\iota_A \colon A \to A \coprod_Z B, \iota_B : B \to A \coprod_Z B$ such that for any pair of morphisms $A \to C, B \to C$ making the square commute, there exists a unique morphism $A \coprod_Z B \to C$ such that



Pushout diagrams are often denoted by



Example 3.2.2 (Pullbacks and pushouts in Set). In Set the pullback of two morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ is the set

$$A \times_Z B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$$

together with the projection maps

$$\begin{aligned} \pi_A : A \times_Z B &\to A \\ (a, b) &\mapsto a \end{aligned} \qquad \begin{aligned} \pi_B : A \times_Z B &\to B \\ (a, b) &\mapsto b \end{aligned}$$

The pushout of two morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ is the set

$$A \coprod_Z B = {}^{A \coprod B}\!/_{\sim}$$

where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$ together

the inclusion maps

$$\iota_A : A \to A \coprod_Z B \qquad \iota_B : B \to A \coprod_Z B$$
$$a \mapsto [a, 0] \qquad b \mapsto [b, 1]$$

Remark 3.2.3. Notice how pullbacks in Set can be seen as a equaliser in Set from the product maps $f \circ \pi_A, g \circ \pi_B : A \times B \to Z$. Considering pushouts in Set we get a dual observation.
Indeed, it is not difficult to check that if C has products then the pullback

$$\begin{array}{ccc} A \times_Z B & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow f \\ B & \xrightarrow[g]{} & Z \end{array}$$

is equivalent to the equaliser

$$A \times_Z B \longrightarrow A \times B \underset{g\circ\pi_B}{\overset{f\circ\pi_A}{\rightrightarrows}} Z$$

Dually if C has coproducts the pushout

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & \ulcorner & \downarrow \\ B & \longrightarrow & A \coprod_Z B \end{array}$$

is equivalent to the coequaliser

$$Z \underset{\iota_B\circ g}{\overset{\iota_A\circ f}{\rightrightarrows}} A \coprod B \longrightarrow A \coprod_Z B$$

For a proof see [Rie, Lemmas 3.4.16, 3.4.17].

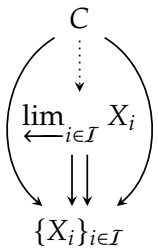
Inverse and direct limits

Definition 3.2.12 (Directed set)**.** A **directed set** is a preordered set $(\mathcal{I}, \leq)$ such that for every $i, j \in \mathcal{I}$, there exists $k \in \mathcal{I}$ with $i \leq k$ and $j \leq k$. We can view this as a category following Example 1.1.5.

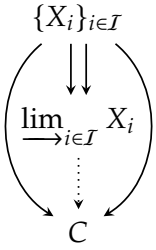
Example 3.2.3. Let $(X, \mathcal{T})$ be a topological space. Then $(\mathcal{T}, \subseteq)$ forms a directed set. This means that the category of open sets $\mathsf{Open}(X)$ can be seen as a directed set.

Definition 3.2.13 (Direct and inverse systems)**.** A **direct system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $f_{ij} : X_i \to X_j$ for all $i \leq j$, such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leq j \leq k$.
Dually, an **inverse system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I}^{\mathrm{op}} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $p_{ij} : X_j \to X_i$ for all $i \leq j$, such that $p_{ii} = \mathrm{id}_{X_i}$ and $p_{ij} \circ p_{jk} = p_{ik}$ for all $i \leq j \leq k$.

Definition 3.2.14 (Inverse limit)**.** The **inverse limit** (or **projective limit**) of an inverse system $X : \mathcal{I} \to \mathsf{C}$ is the limit of this diagram, denoted as $\varprojlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varprojlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\pi_i : \varprojlim_{i\in\mathcal{I}} X_i \to X_i$ such that $p_{ij} \circ \pi_j = \pi_i$ for all $i \leq j$. For any other object C and morphisms $c_i : C \to X_i$ compatible with the system ($p_{ij} \circ c_j = c_i$), there exists a unique morphism $C \to \varprojlim_{i\in\mathcal{I}} X_i$ making the following diagram commute:



Definition 3.2.15 (Direct limit)**.** The **direct limit** (or **inductive limit**) of a direct system $X\colon \mathcal{I} \to \mathsf{C}$ is the colimit of this diagram, denoted as $\varinjlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varinjlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\iota_i\colon X_i \to \varinjlim_{i\in\mathcal{I}} X_i$ such that $\iota_j \circ f_{ij} = \iota_i$ for all $i \leq j$. For any other object C and morphisms $c_i\colon X_i \to C$ compatible with the system ($c_j \circ f_{ij} = c_i$), there exists a unique morphism $\varinjlim_{i\in\mathcal{I}} X_i \to C$ making the following diagram commute:



Example 3.2.4 (Inverse limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{p_{ij}\}_{i\leq j})$ is an inverse system indexed by a directed set $(\mathcal{I}, \leq)$, its inverse limit is

$$\varprojlim_{i\in\mathcal{I}} X_i = \left\{ (x_i)_{i\in\mathcal{I}} \in \prod_{i\in\mathcal{I}} X_i \mid \forall i \leq j \quad x_i = p_{ij}(x_j) \right\}$$

together with the natural projection maps $\pi_k((x_i)_{i\in\mathcal{I}}) = x_k$.

Example 3.2.5 (Direct limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{f_{ij}\}_{i\leq j})$ is a direct system indexed by a directed set $(\mathcal{I}, \leq)$, its direct limit is

$$\varinjlim_{i\in\mathcal{I}} X_i = \coprod_{i\in\mathcal{I}} X_i \Big/ \sim$$

where $\sim$ is the equivalence relation defined by $x_i \sim x_j$ if and only if there exists $k \in \mathcal{I}$ with $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j)$$

together with the natural insertion maps $\iota_k(x_k) = [x_k]$.

We finish this chapter by giving a condition for the existence of finite limits and colimits

Theorem 3.2.1. A category has all finite limits if and only if it has a terminal object, binary products and equalisers.
Dually it has all finite colimits if and only if it has a initial object, binary coproducts and coequalisers.

Proof.

$\Longrightarrow$ Direct since all of these are particular cases of limits.

$\Longleftarrow$ Consider a diagram $X\colon \mathsf{J} \Rightarrow \mathsf{C}$.
Since binary products exist in C we can construct inductively

$$P \coloneqq \prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$$

For every morphism $u\colon j \to k$ in J we can consider two maps from P to X_k: the projection $\pi_k\colon P \to X_k$ and the composition $Xu \circ \pi_j\colon P \to X_j \to X_k$. Therefore we can consider a pair of parallel morphisms

$$\prod_{j\in\mathsf{J}} X_j \underset{t}{\overset{s}{\rightrightarrows}} \prod_{u:j\to k\in\mathsf{J}} X_k$$

where s is defined in each component as π_k and t is $Xu \circ \pi_j$ in each component. The equaliser of s and t (which exists by hypothesis) is then the limit of X. Indeed, let $\mathrm{eq}\colon L \to \prod_{j\in\mathsf{J}} X_j$ be this equaliser. This comes with a set of morphisms $\{\phi_j\colon L \to X_j\}_{j\in\mathsf{J}}$ such that

$$\phi_j = \pi_j \circ \mathrm{eq}$$

By definition $s \circ \mathrm{eq} = t \circ \mathrm{eq}$, so for every $u\colon j \to k$ in J we have

$$\pi_k \circ \mathrm{eq} = (Xu \circ \pi_j) \circ \mathrm{eq}$$

Therefore

$$\phi_k = Xu \circ \phi_j$$

and so L is a cone over X.
To see that it is a limit cone consider another cone $(W, \{\psi_j\colon W \to X_j\})$ such that for any $u\colon j \to k$ we have $\psi_k = Xu \circ \psi_j$. By definition of the product there exists a unique morphism $h\colon W \to \prod_{j\in\mathsf{J}} W_j$ such that $\pi_j \circ h = \psi_j$.
Note that $s \circ h = t \circ h$. Indeed the components of $s \circ h$ are $\pi_k \circ h = \psi_k$ and the components of $t \circ h$ are $Xu \circ \pi_j \circ h = Xu \circ \psi_j = \psi_k$.
Therefore by the definition of equaliser there exists a unique morphism $\widetilde{h}\colon W \to L$ such that $\mathrm{eq} \circ \widetilde{h} = h$. Finally since for any $j \in \mathsf{J}$

$$\phi_j \circ \widetilde{h} = \pi_j \circ \mathrm{eq} \circ \widetilde{h} = \pi_j \circ h = \psi_j$$

we have that W factors uniquely through L and so L is the limit of X. □

Chapter 4

More on functors

4.1 Representable functors

Definition 4.1.1. A functor $F\colon \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists an object $X \in \mathsf{C}$ such that

$$F \cong \mathrm{Hom}_{\mathsf{C}}(X, -)$$

where $\mathrm{Hom}_{\mathsf{C}}(X, -)$ is the covariant Hom functor.

Example 4.1.1. The covariant Hom functor is representable.

Definition 4.1.2 (Concrete category)**.** A category C is **concrete** if there exists a faithful functor $U : \mathsf{C} \to \mathsf{Set}$. Such a functor is often called a **forgetful functor**.

Theorem 4.1.1 (Yoneda - [Rie, Theorem 2.2.4])**.** Let C be a category, $X \in \mathsf{C}$ an object, and $F\colon \mathsf{C} \to \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

4.2 Adjunctions

Definition 4.2.1 (Adjoint functors)**.** Let $L\colon \mathsf{C} \to \mathsf{D}$ and $R\colon \mathsf{D} \to \mathsf{C}$ be a pair of functors. We say that (L, R) is a pair of **adjoint functors** or that L is left-adjoint to R and R is right-adjoint to L if there exists a natural isomorphism of bifunctors

$$\mathrm{Hom}_{\mathsf{D}}(L(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(-, R(-)) \tag{4.2.1}$$

We represent this diagramatically using

$$\mathsf{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathsf{D}$$

Remark 4.2.1 (Unit and counit of an adjunction)**.** Let (L, R) be a pair of adjoint functors. The natural isomorphism (4.2.1) gives the natural isomorphisms

$$\mathrm{Hom}_{\mathsf{D}}(L(-), L(-)) \cong \mathrm{Hom}_{\mathsf{C}}(-, (G \circ L)(-))$$
$$\mathrm{Hom}_{\mathsf{D}}((L \circ R)(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(R(-), R(-))$$

In particular we get natural transformations

$$\eta\colon \mathbb{1}_{\mathsf{C}} \Rightarrow R \circ L \qquad \varepsilon\colon L \circ R \Rightarrow \mathbb{1}_{\mathsf{D}}$$

that satisfy the *triangular identities*

$$\begin{array}{ccccc} L & \xRightarrow{L\eta} & L\circ R\circ L & & \\ & \searrow^{\mathbb{1}_F} & \Downarrow{\scriptstyle \varepsilon F} & & \\ & & L & & \end{array} \qquad \begin{array}{ccccc} R & \xRightarrow{\eta R} & R\circ L\circ R & & \\ & \searrow^{\mathbb{1}_R} & \Downarrow{\scriptstyle F\varepsilon} & & \\ & & R & & \end{array}$$

Conversely, if L, R are two pair of functors and ε, η are natural transformations as above then (L, R) form an adjoint pair. For a proof of this see [**riehlCategoryTheory**].

Remark 4.2.2. Left and right adjoints are dual concepts.

Proposition 4.2.1 (Adjoints are unique)**.** The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L\colon \mathsf{C} \to \mathsf{D}$ be a functor.

Let $R_1, R_2\colon \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y}\colon \operatorname{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1}\colon \operatorname{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \operatorname{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism

$$\begin{array}{ccccc} & & \operatorname{Hom}_{\mathsf{C}}(-,R_1(Y)) & & \\ & \nearrow & \Downarrow{\scriptstyle \Theta_{-,Y}} & \searrow & \\ \mathsf{C} & & & & \mathsf{Set} \\ & \searrow & & \nearrow & \\ & & \operatorname{Hom}_{\mathsf{C}}(-,R_2(Y)) & & \end{array}$$

By the Yoneda lemma $\Theta_{-,Y}$ induces a unique isomorphism in C

$$\alpha_Y\colon R_1(Y) \xrightarrow{\sim} R_2(Y)$$

which ensemble into a natural isomorphism

$$\alpha\colon R_1 \Rightarrow R_2$$

□

Theorem 4.2.1 ([Rie, Theorem 4.5.2])**.** Right adjoints preserve all limits that exist in their codomain.
Dually left adjoints preserve all colimits that exist in their domain.

4.3 Functor categories

Definition 4.3.1 (Functor category). Let C and D be categories. The **functor category** $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has:

- **Objects**: functors $F : \mathsf{C} \to \mathsf{D}$.
- **Morphisms**: natural transformations $\eta : F \implies G$.

Composition and identities are defined componentwise.

Proposition 4.3.1. Let C and D be two categories and $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ be its functor category. If D has limits (resp. colimits) of shape J then $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Proof. Consider a J-shaped diagram $X : \mathsf{J} \Rightarrow [\mathsf{C}, \mathsf{D}]$. We wish to construct a limit object $L \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ together with a limit cone $\pi_j : L \to X_j$ for all $j \in \mathsf{J}$.

For any object $c \in \mathsf{C}$ the *evaluation* functor

$$\begin{array}{ccc} \mathrm{ev}_c : [\mathsf{C}, \mathsf{D}] & \longrightarrow & \mathsf{D} \\ F & & F(c) \\ \eta \Downarrow & \longmapsto & \downarrow \eta_c \\ G & & G(c) \end{array}$$

allows us to define a diagram $X^c := \mathrm{ev}_c \circ X : J \Rightarrow \mathsf{D}$. By hypothesis there exists

$$\lim_{j \in \mathsf{J}} X^c(j) =: L(c) \qquad \pi^c_j : L(c) \to X^c(j)$$

To make L into a functor we need to define its action on morphisms. Consider a morphism $f : c \to d$ in C. Since X_j is a functor for all $j \in \mathsf{J}$ we have an induced morphism $X_j f : X_j(c) \to X_j(d)$. Let's consider the family of morphism

$$\left\{ X_j f \circ \pi^c_j : L(c) \to X_j(d) \right\}_{j \in \mathsf{J}}$$

We want to show that this forms a cone over $L(d)$ with apex $L(c)$

Let $u : j \to k$ be a morphism in J. Since $Xu : X_j \Rightarrow X_k$ is a natural transformation we have a commutative diagram

$$\begin{array}{ccc} X_j(c) & \xrightarrow{(Xu)_c} & X_k(x) \\ {\scriptstyle X_j f}\downarrow & & \downarrow{\scriptstyle X_k f} \\ X_j(d) & \xrightarrow[(Xu)_d]{} & X_k(d) \end{array}$$

Since π^c is a limit over X^c we also now that $(Xu)_c \circ \pi^c_j = \pi^c_k$. Therefore we have

$$(Xu)_d \circ (X_j f \circ \pi^c_j) = ((Xu)_d \circ X_j f) \circ \pi^c_j = X_k f \circ (Xu)_c \circ \pi^c_j = X_k f \circ \pi^c_k$$

i.e: it is a valid cone over $L(d)$. By the universal property of $L(d) = \lim_{j \in \mathsf{J}} X^d(j)$ there exists a unique morphism $Lf : L(c) \to L(d)$ such that for all $j \in \mathsf{J}$ the following diagram commutes

$$\begin{array}{ccc} L(c) & \xrightarrow{Lf} & L(d) \\ {\scriptstyle \pi_j^c}\downarrow & & \downarrow{\scriptstyle \pi_j^d} \\ X^c(j) & \xrightarrow[X_j f]{} & X^d(j) \end{array}$$

The functiorality of L follows directly from the uniqueness of this induced map into $L(d)$. Moreover the above commutative square is the naturallity condition for π_j with respect to f. Therefore, for each $j \in \mathsf{J}$ the collection $\{\pi_j^c\}_{c\in\mathsf{C}}$ is a natural transformation $\pi_j \colon L \Rightarrow X_j$. It is an easy check to see that this collection forms a cone over X in the functor category $\mathsf{Fct}(\mathsf{C},\mathsf{D})$.

To finish the proof, we need to check the universal property. Consider another functor $M \in \mathsf{Fct}(\mathsf{C},\mathsf{D})$ equipped with a cone $\tau_j \colon M \Rightarrow X_j$.

For each $c \in \mathsf{C}$ the evaluation $\tau_j^c := \mathrm{ev}_c \circ M : M(c) \to X^c(j)$ is a cone over X^c. This induces a unique morphism $\phi_c : M(c) \to L(c)$ such that $\pi_j^c \circ \phi_c = \tau_j^c$.

To show that $\phi \equiv \{\phi_c\}_{c\in\mathsf{C}}$ is a natural transformation let $f : c \to d$ be a morphism in C. We want to see that the following diagram commutes

$$\begin{array}{ccc} M(c) & \xrightarrow{\phi_c} & L(c) \\ {\scriptstyle Mf}\downarrow & & \downarrow{\scriptstyle Lf} \\ M(d) & \xrightarrow[\phi_d]{} & L(d) \end{array}$$

Postcompossing with π_j^d we get, on the one hand

$$\pi_j^d \circ (Lf \circ \phi_c) = (\pi_j^d \circ Lf) \circ \phi_c = X_j f \circ \pi_j^c \circ \phi_c = X_j f \circ \tau_j^c$$

And on the other

$$\pi_j^d \circ (\phi_d \circ Mf) = (\pi_j^d \circ \phi_d) \circ Mf = \tau_j^d \circ Mf$$

Since τ_j is a natural transformation we have

$$X_j f \circ \tau_j^c = \tau_j^d \circ Mf$$

Therefore since the projections agree for all $j \in \mathsf{J}$ by uniqueness of the maps π_j we get the commutativity of the diagram. Its uniqueness as a morphism of cones follows directly from the objectwise uniqueness of each ϕ_c. □

PART II

Homological algebra

Chapter 5

Additive categories

Definition 5.0.1 (Additive category). A category C is **additive** if it satisfies

Ad.1) For every pair of objects $X, Y \in \mathsf{C}$ the set $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

Ad.2) There exists a zero object $0 \in \mathsf{C}$.

Ad.3) C admits finite coproducts.

Ad.4) C admits finite products.

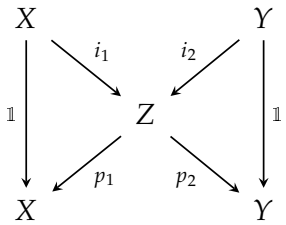
Example 5.0.1.

- The category of abelian groups Ab is additive. More generally, for any ring R the category of R-modules $\mathsf{Mod}(R)$ is additive.
- The category of groups Grp is not additive, since given two group homomorphisms f, g its sum is not necessarilly a homomorphism.

Notation. If $X, Y \in \mathsf{C}$ we denote by $X \oplus Y$ its coproduct and call it their **direct sum**.

Lemma 5.0.1. Let C be a category satisfying Ad.1 and Ad.2.
Suppose that C verifies

Ad.5) For any pair of objects $X, Y \in \mathsf{C}$ there exists $Z \in \mathsf{C}$ and morphisms



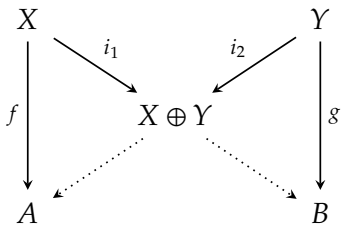
such that

$$p_1 \circ i_1 = \mathbb{1}_X \quad p_1 \circ i_2 = 0$$
$$p_2 \circ i_2 = \mathbb{1}_Y \quad p_2 \circ i_1 = 0$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathbb{1}_Z$$

Then conditions Ad.4, Ad.3 and Ad.5 are equivalent and the objects $X \times Y$, $X \oplus Y$ and Z are naturally isomorphic.

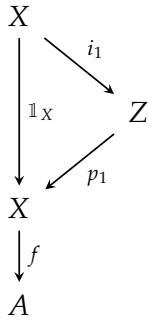
Proof. We will show that conditions Ad.3 and Ad.5 are equivalent and that $X \oplus Y \cong Z$. A dual argument shows that Ad.4 and Ad.5 are equivalent and that $X \times Y \cong Z$.

Ad.3 $\implies$ Ad.5 Let $X \oplus Y$ be the coproduct of X and Y, that comes with morphisms $i_1 \colon X \to X \oplus Y$ and $i_2 \colon Y \to X \oplus Y$.
By definition, for all morphisms $f \colon X \to A$, $g \colon Y \to B$ there exist unique morphisms $X \oplus Y \to A$, $X \oplus Y \to B$ such that



In particular if $f = \mathbb{1}_X$ and $g = \mathbb{1}_Y$

Ad.5 $\implies$ Ad.3 We are going to see that Z verifies the universal property of the coproduct.
Let $f : X \to A$ be a morphism. We want to find a morphism $\widetilde{f} \colon Z \to A$ such that $f = \widetilde{f} \circ i_1$. We have the following



Therefore if $\widetilde{f} \coloneqq f \circ p_1$ since $i_1 \circ p_1 = \mathbb{1}_X$ we have $\widetilde{f} \circ i_1 = f$.
Analogously we see that any morphism $g : Y \to B$ can can be factored through Z. Therefore since Z verifies the universal property of the coproduct, $Z \cong X \oplus Y$. □

Remark 5.0.1. Given objects $A_1, \dots, A_n$ and $B_1, \dots, B_m$ in an additive category C, we can represent morphisms $f \colon A_1 \oplus \cdots \oplus A_n \to B_1 \oplus \cdots \oplus B_m$ as $n \times m$ matrices

$$\begin{pmatrix} f_{11} & \cdots & f_{1n} \\ \vdots & \ddots & \vdots \\ f_{m1} & \cdots & f_{mn} \end{pmatrix} \text{ where } f_{k\ell} \coloneqq p_k \circ f \circ i_\ell \colon A_\ell \to B_k$$

Using that $\sum_k i_k \circ p_k = 1$, it follows that addition and composition of matrices obey the usual rules of matrix addition and multiplication respectively.

Lemma 5.0.2. A morphism $f \colon X \to Y$ in an additive category is a monomorphism if and only if for every morphism $g \colon Z \to X$

$$f \circ g = 0 \implies g = 0$$

Dually f is an epimorphism if and only if for every morphism $g \colon Y \to Z$

$$g \circ f = 0 \implies g = 0$$

Proof. We will check the monomorphism caracterization.

$\Longrightarrow$ Let $f : X \rightarrowtail Y$ be a monomorphism. Consider a morphism $g : Z \to X$ such that $f \circ g = 0$. Since $f \circ 0 = 0$ applying left-cancellativity we have that $g = 0$.

$\Longleftarrow$ Let $g_1, g_2 : Z \rightrightarrows X$ be two morphisms such that $f \circ g_1 = f \circ g_2$.
Since $\mathrm{Hom}(Z, Y)$ is an abelian group we have

$$0 = f \circ g_1 - f \circ g_2 = f \circ (g_1 - g_2)$$

Therefore by hypothesis $g_1 - g_2 = 0 \iff g_1 = g_2$, proving that f is a monomorphism. □

Lemma 5.0.3. Let I be a small category. If C is additive then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Proof.

Ad.1 For any two natural transformations $\phi, \psi : \mathsf{I} \Rightarrow \mathsf{C}$ we can define their addition componentwise: $(\phi + \psi)_i = \phi_i + \psi_i$ for all $i \in \mathsf{I}$. Since $\phi_i, \psi_i \in \mathrm{Hom}_{\mathsf{C}}(-,-)$ this is well defined.

Ad.2 Consider the zero functor

$$\begin{array}{ccc} 0 : \mathsf{I} & \longrightarrow & \mathsf{C} \\ X & & 0 \\ f \downarrow & \longmapsto & \downarrow 0 \\ Y & & 0 \end{array}$$

This is clearly the zero object in $\mathsf{Fct}(\mathsf{I}, \mathsf{C},)$.

Ad.3, Ad.4 Direct from Proposition 4.3.1. □

Definition 5.0.2 (Additive functor). Let C, D be additive categories. A functor $F : \mathsf{C} \to \mathsf{D}$ is **additive** if for all $X, Y \in \mathsf{C}$

$$\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$$

is a morphism of abelian groups.

Example 5.0.2. For a fixed R-module M the functor $\mathrm{Hom}_R(M, -) : \mathsf{Mod}(R) \to \mathsf{Ab}$ is additive.

Proposition 5.0.1. Let C, D be additive categories. A functor $F : \mathsf{C} \to \mathsf{D}$ is additive if and only if it commutes with finite direct sums, i.e: if $F(0) \cong 0$ and for all $X, Y \in \mathsf{C}$

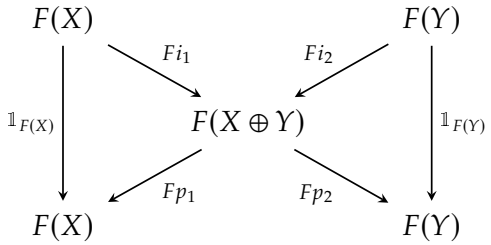
$$F(X \oplus Y) = F(X) \oplus F(Y)$$

Proof.

$\Longrightarrow$ Since F is a functor $F(\mathbb{1}_0) = \mathbb{1}_{F(0)}$. Furthermore since F is additive if $0_{X,Y}$ is the zero morphism between $X, Y \in \mathsf{C}$ then $F(0_{X,Y}) = 0_{F(X),F(Y)}$ because group homomorphisms preserve the identity. In particular taking $X = Y = 0$ since $0_{0,0} = \mathbb{1}_0$ we have

$$\mathbb{1}_{F(0)} = F(\mathbb{1}_0) = F(0_{0,0}) = 0_{F(0),F(0)}$$

which means that $F(0)$ is the zero object in D.
Let $X, Y \in \mathsf{C}$ be two objects and $X \oplus Y$ be its direct sum. Applying F to the diagram in Lemma 5.0.1 defining $Z \cong X \oplus Y$ we get



Note that

$$Fp_1 \circ Fi_1 = F(p_1 \circ i_1) = F(\mathbb{1}_X) = \mathbb{1}_{F(X)} \quad Fp_1 \circ Fi_2 = F(p_1 \circ i_2) = F(0) = 0$$
$$Fp_2 \circ Fi_2 = F(p_2 \circ i_2) = F(\mathbb{1}_Y) = \mathbb{1}_{F(Y)} \quad Fp_2 \circ Fi_1 = F(p_2 \circ i_1) = F(0) = 0$$

Furthermore since F is additive

$$Fp \circ Fi + Fj \circ Fq = F(p \circ i + j \circ q) = F\mathbb{1}_{X \oplus Y} = \mathbb{1}_{F(X \oplus Y)}$$

Therefore by Lemma 5.0.1 $F(X \oplus Y) \cong F(X) \oplus F(Y)$.

$\Longleftarrow$ Applying Lemma 5.0.1 to $X \oplus X$ and $Y \oplus Y$ we get maps $\Delta_X : X \to X \oplus X$ and $\nabla_Y : Y \oplus Y \to Y$ such that

$$p_1 \circ \Delta_X = \mathbb{1}_X = p_2 \circ \Delta_X$$
$$\nabla_Y \circ i_1 = \mathbb{1}_Y = \nabla_Y \circ i_2$$

Then for any pair of morphisms $f, g : X \to Y$ in D we have

$$f + g = \nabla_Y \circ (f \oplus g) \circ \Delta_X$$

where $f \oplus g$ is the unique morphism such that

$$p_k \circ (f \oplus g) \circ i_j = \begin{cases} f & j = k = 1 \\ g & j = k = 2 \\ 0 & \text{otherwise} \end{cases}$$

Applying F to this expression we get

$$F(f + g) = F(\nabla_Y \circ (f \oplus g) \circ \Delta_X) = F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X$$

Since by hypothesis F preserves finit direct sums

$$F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X = \nabla_{F(Y)} \circ (Ff \oplus Fg) \oplus \Delta_{F(X)} = Ff + Fg$$

Therefore $F(f + g) = Ff + Fg$, which means that F is additive. □

Proposition 5.0.2. Let $L \dashv R : \mathsf{C} \to \mathsf{D}$ be a pair of adjoint functors between additive categories. Then L is additive if and only if R is additive.

Proof. Since left adjoints preserve colimits and right adjoints preserve limits we have

$$L(X \oplus Y) = L(X) \oplus L(Y)$$
$$R(X \oplus Y) = R(X \times Y) = R(X) \times R(Y) = R(X) \oplus R(Y)$$

Note that if R is additive then $L(0) = 0$ since

$$\mathrm{Hom}_{\mathsf{C}}(L(0), 0) = \mathrm{Hom}_{\mathsf{D}}(0, R(0)) = \mathrm{Hom}_{\mathsf{D}}(0, 0)$$

and viceversa.

Therefore by Proposition 5.0.1 we have the result. □

Definition 5.0.3 (Full additive subcategory). Let C be an additive category.
A full subcategory D of C is a **full additive subcategory** if the inclussion functor $\iota\colon D \hookrightarrow \mathsf{C}$ is additive.

Remark 5.0.2. By Proposition 5.0.1 a full subcategory D of C is a full additive subcategory if and only if $0 \in \mathsf{D}$ and D is closed under finite direct sums.

5.1 Complexes

From now until the end of the chapter, let C be an additive category.

Definition 5.1.1 (Differential object). A **differential object** $(X^\bullet, \mathrm{d}_X^\bullet)$ in C is a sequence of objects $X^k \in \mathsf{C}$ and morphisms d_X^k for every $k \in \mathbb{Z}$:

$$\cdots \longrightarrow X^{k-1} \xrightarrow{\mathrm{d}_X^{k-1}} X^k \xrightarrow{\mathrm{d}_X^k} X^{k+1} \longrightarrow \cdots$$

A morphism of differential objects $f^\bullet\colon (X^\bullet, \mathrm{d}_X^\bullet) \to (Y^\bullet, \mathrm{d}_Y^\bullet)$ is a collection of morphisms $\{f^k\colon X^k \to Y^k\}_{k\in\mathbb{Z}}$ such that, for all $k \in \mathbb{Z}$, the following diagram commutes

$$\begin{array}{ccccccc} \cdots & \longrightarrow & X^k & \xrightarrow{\mathrm{d}_X^k} & X^{k+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^k} & & \downarrow{\scriptstyle f^{k+1}} & & \\ \cdots & \longrightarrow & Y^k & \xrightarrow[\mathrm{d}_Y^k]{} & Y^{k+1} & \longrightarrow & \cdots \end{array}$$

Therefore we get a category $\mathsf{Diff}(\mathsf{C})$ of differential objects in C. By abuse of notation we will often denote differential objects $(X^\bullet, \mathrm{d}_X^\bullet)$ as $X^\bullet$.

Remark 5.1.1. We can view the category of differential objects $\mathsf{Diff}(\mathsf{C})$ as the functor category $\mathsf{Fct}(\mathbb{Z}, \mathsf{C})$. Therefore by Lemma 5.0.3 $\mathsf{Diff}(\mathsf{C})$ is an additive category.

Remark 5.1.2. For all $n \in \mathbb{Z}$ there exists a fully faithful functor

$$\iota_n\colon \mathsf{C} \to \mathsf{Diff}(\mathsf{C})$$

which sends an object $X \in \mathsf{C}$ to the differential object

$$\cdots \longrightarrow 0 \longrightarrow X \longrightarrow 0 \longrightarrow \cdots$$

and a morphism $f\colon X \to Y$ in C to

$$\begin{array}{ccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle f} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & Y & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

Therefore we can identify an object $X \in \mathsf{C}$ as a differential object using ι_0. When the context is clear, we will directly write $X \in \mathsf{Diff}(\mathsf{C})$ instead of $\iota_0(X) \in \mathsf{Diff}(\mathsf{C})$.

Definition 5.1.2 (Complex). A **complex** in C is a differential object $(X^\bullet, \mathrm{d}_X^\bullet)$ such that for all $k \in \mathbb{Z}$

$$\mathrm{d}_X^{k+1} \circ \mathrm{d}_X^k = 0$$

We denote by $\mathsf{Comp}(\mathsf{C})$ the full additive subcategory of $\mathsf{Diff}(\mathsf{C})$ consisting of complexes in C.

Definition 5.1.3. A complex $X^\bullet$ is

1. **Bounded above** if $X^n = 0$ of $n \gg 0$.
2. **Bounded below** if $X^n = 0$ of $n \ll 0$.
3. **Bounded** if it is bounded above and below.

We denote by $\mathsf{Comp}^*(\mathsf{C})$ ($* = +, -, \mathrm{b}$) the full additive subcategory of $\mathsf{Comp}(\mathsf{C})$ consisting of bounded above (resp. bounded below, bounded) complexes. To emphasize that we are working with unbounded complexes we may write $\mathsf{Comp}^{\mathrm{ub}}(\mathsf{C}) := \mathsf{Comp}(\mathsf{C})$.

Remark 5.1.3. Note that for any $n \in \mathbb{N}$, the inclussion functor ι_n can be viewed as a functor

$$\iota_n : \mathsf{C} \to \mathsf{Comp}^*(\mathsf{C})$$

Therefore we can identify $X \in \mathsf{C}$ with $\iota_0(X) \in \mathsf{Comp}^*(\mathsf{C})$.

5.2 Homotopy of complexes

Definition 5.2.1 (Homotopy).

1. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is **homotopic to zero** if for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ in C such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

   Visually

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & X^{p-1} & \longrightarrow & X^p & \longrightarrow & X^{p+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^{p-1}} & \swarrow{\scriptstyle s^p} & \downarrow{\scriptstyle f^p} & \swarrow{\scriptstyle s^{p+1}} & \downarrow{\scriptstyle f^{p+1}} & & \\ \cdots & \longrightarrow & Y^{p-1} & \longrightarrow & Y^p & \longrightarrow & Y^{p+1} & \longrightarrow & \cdots \end{array}$$

2. Two morphisms $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are **homotopic** if $f^\bullet - g^\bullet$ is homotopic to zero. We denote this by $f^\bullet \sim g^\bullet$.
3. A complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$ is **homotopic to zero** if $\mathbb{1}_{X^\bullet} \sim 0$.
4. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is a **homotopy equivalence** if there exists $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **homotopy equivalent** and write $X^\bullet \sim Y^\bullet$.

Lemma 5.2.1. Let $F : \mathsf{C} \to \mathsf{D}$ be an additive functor. If $f^\bullet : X^\bullet \to Y^\bullet$ is homotopic to zero then $F f^\bullet$ is also homotopic to zero.

Proof. Since $f^\bullet$ is homotopic to zero for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ such that

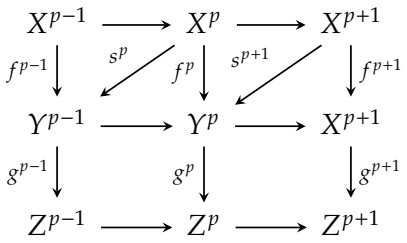
$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$F f^p = F(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p) = F s^{p+1} \circ \mathrm{d}_{F(X)}^p + \mathrm{d}_{F(Y)}^{p-1} \circ F s^p \qquad \square$$

Lemma 5.2.2. Let $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to Z^\bullet$ be two morphisms of complexes in $\mathsf{Comp}(\mathsf{C})$. If either f or g is homotopic to 0 then $g \circ f$ is homotopic to 0.

Proof. Assume that f is homotopic to 0. The proof consists on chasing the following diagram:



We have that for all $p \in \mathbb{Z}$ there exists $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$\begin{aligned} g^p \circ f^p &= g^p \circ \left(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p \right) \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \left(g^p \circ \mathrm{d}_Y^{p-1} \right) \circ s^p \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \mathrm{d}_Z^{p-1} \circ \left(g^{p-1} \circ s^p \right) \end{aligned}$$

□

Chapter 6

Abelian categories

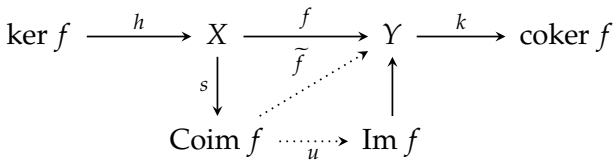
Definition 6.0.1. Let $f : X \to Y$ be a morphism in an additive category C that admits kernels and cokernels. Consider the following

$$\ker f \xrightarrow{h} X \xrightarrow{f} Y \xrightarrow{k} \operatorname{coker} f$$

We denote

$$\operatorname{Im} f := \ker k \qquad \operatorname{Coim} f := \operatorname{coker} h$$

I.e: we have the following diagram



Note that, since $f \circ h = 0$, f factors uniquely through $\operatorname{Coim} f$ producing the morphism $\widetilde{f}$. Furthermore since $k \circ f = k \circ \widetilde{f} = 0$ we have that $\widetilde{f}$ factors uniquely through $\operatorname{Im} f$. Therefore we have a canonical morphism

$$u : \operatorname{Coim} f \to \operatorname{Im} f$$

Definition 6.0.2 (Abelian category)**.** An additive category C is **abelian** if it verifies

Ab.1) Any morphism in C admits a kernel and a cokernel.

Ab.2) For any morphism f in C the canonical morphism $\operatorname{Coim} f \to \operatorname{Im} f$ is an isomorphism.

Lemma 6.0.1. An abelian category admits all finite limits and colimits.

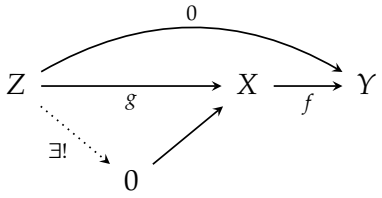
Proof. Direct from Theorem 3.2.1. □

Lemma 6.0.2. Let $f : X \to Y$ be a morphism in an abelian category. Then f is a monomorphism if and only if $0 \to X$ is its kernel.
Dually f is an epimorphism if and only if $Y \to 0$ is its cokernel.

Proof. We will do the kernel one.

$\Longrightarrow$ Suppose f is a monomorphism.
If $g : Z \to X$ is a morphism such that $f \circ g = 0$ then by Lemma 5.0.2 $g = 0$. In particular g factors uniquely through $0 \to X$, i.e: $0 \to X$ is the kernel of f.

$\Longleftarrow$ Suppose now that $0 \to X$ is the kernel of f.
Consider a morphism $g\colon Z \to X$ such that $f \circ g = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $g = 0$, making f a monomorphism by Lemma 5.0.2. □

Proposition 6.0.1. Let I be a small category and C an abelian category. Then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is abelian.

Proof. By Lemma 5.0.3 we have that $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Furthermore since C has kernels and cokernels, which are no more than special cases of limits and colimits, by Proposition 4.3.1 $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ has kernels and cokernels and they are computed elementwise.

Finally since $\operatorname{Coim} f(i) \xrightarrow{\sim} \operatorname{Im} f(i)$ for all $i \in \mathsf{I}$ we have that $\operatorname{Coim} f \xrightarrow{\sim} \operatorname{Im} f$. □

Corollary 6.0.1. If C is abelian then the categories of complexes $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ are abelian.

Proof. Direct since by Proposition 6.0.1 the category of differential objects $\mathsf{Diff}(\mathsf{C})$ is abelian. □

In what remains of the chapter we will assume that C is an abelian category.

Definition 6.0.3.

1. A complex $X \xrightarrow{f} Y \xrightarrow{g} Z$ is **exact** if $\operatorname{Im} f \xrightarrow{\sim} \ker g$. Note that this is equivalent to $\operatorname{coker} f \xrightarrow{\sim} \operatorname{Im} g$.
2. More generally, a sequence of morphisms $X^p \xrightarrow{\mathrm{d}^p} \cdots \to X^n$ with $\mathrm{d}^{i+1} \circ \mathrm{d}^i = 0$ for all $i \in \{p, \dots, n-1\}$ is **exact** if $\operatorname{Im} \mathrm{d}^i \xrightarrow{\sim} \ker \mathrm{d}^{i+1}$ for all $i \in \{p, \dots, n-1\}$.
3. A **short exact sequence** is an exact sequence of the form

$$0 \to X \to Y \to Z \to 0$$

Remark 6.0.1. A short sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ is short exact if and only if f is a monomorphism, g is an epimorphism and $\operatorname{Im} f = \ker g$.
Indeed, it is exact at X iff $\ker f = \operatorname{Im} 0 = 0$ and is exact at Z iff $\operatorname{coker} g = \operatorname{Im} 0 = 0$.

Example 6.0.1. Any morphism $f : X \to Y$ may be decomposed in short exact sequences

$$0 \longrightarrow \ker f \rightarrowtail X \twoheadrightarrow \operatorname{Coim} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Im} f \rightarrowtail Y \twoheadrightarrow \operatorname{coker} f \longrightarrow 0$$

with $\operatorname{Coim} f \cong \operatorname{Im} f$.
Indeed, by definition we have $\operatorname{Coim} f = \operatorname{coker}\left(\ker f \to X\right)$ and $\operatorname{Im} = \ker(Y \to$

coker f).

Proposition 6.0.2. Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

be a short exact sequence in an abelian category C. We say that the sequence **splits** if any of the following equivalent conditions are satisfied

1. There exists $h\colon Z \to Y$ such that $g \circ h = \mathbb{1}_Z$.
2. There exists $k\colon Y \to X$ such that $k \circ f = \mathbb{1}_X$.
3. There exists $\varphi = (k, h)$ and $\psi = \binom{f}{g}$ such that $Y \xrightarrow{\varphi} X \oplus Z$ and $X \otimes Z \xrightarrow{\psi} Y$ are inverses to each other.
4. The sequence is homotopic to 0.
5. The sequence is isomorphic to $0 \to X \to X \oplus Z \to Z \to 0$.

Proof.

$1 \implies 3$ Since $g = g \circ h \circ g$ we have that $g \circ (\mathbb{1}_Y - h \circ g) = 0$, i.e: $\mathbb{1}_Y - h \circ g$ factors through $\ker g$ and so it factors through X. Therefore there exists $h\colon Y \to Z$ such that $\mathbb{1}_Y - h \circ g = f \circ k$.

$3 \implies 1$ Since $g \circ f = 0$ we have $g = g \circ h \circ g$, i.e: $(g \circ h - \mathbb{1}_Z) \circ g$. Therefore since g is an epimorphism we have $g \circ h - \mathbb{1}_Z = 0 \iff g \circ h = \mathbb{1}_Z$.

$2 \iff 3$ Analogous to $1 \iff 3$.

$3 \iff 4$ By definition the sequence is homopic to zero if and only if there exists

$$\begin{array}{ccccccccc} 0 & \longrightarrow & X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathbb{1}_X} & \swarrow{\scriptstyle k} & \downarrow{\scriptstyle \mathbb{1}_Y} & \swarrow{\scriptstyle h} & \downarrow{\scriptstyle \mathbb{1}_Z} & & \\ 0 & \longrightarrow & X & \xrightarrow[f]{} & Y & \xrightarrow[g]{} & Z & \longrightarrow & 0 \end{array}$$

such that $\mathbb{1}_X = k \circ f$, $\mathbb{1}_Y = h \circ g$ and $\mathbb{1}_Z = h \circ g + f \circ k$.

$3 \iff 5$ Trivial. □

6.1 Exact functors

Definition 6.1.1. Let C, D be two abelian categories and

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in an C.
An additive functor $F\colon \mathsf{C} \to \mathsf{D}$ is **left-exact** if

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

is exact in D.
Dually, it is **right-exact** if

$$F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$

is exact in D.
If F is both left and right exact we say that it is **exact**.

We state the following characterization of exact functors, which we will not prove.

Lemma 6.1.1 ([Sch, Lemma 5.2.2]). Let $F\colon \mathsf{C} \to \mathsf{D}$ be an additive functor.

1. The following are equivalent
   (a) F is left exact (resp. right exact).

   (b) F commutes with kernels: for any morphism $f: X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F f$ (resp. with cokernels).
   (c) F commutes with finite limits (resp. finite colimits).
   (d) For any short exact sequence $0 \to X \to Y \to Z \to 0$ in C the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in D (resp. $F(X) \to F(Y) \to F(Z) \to 0$ is exact).

2. The following are equivalent

   (a) F is exact.
   (b) For any exact sequence $X \to Y \to Z$ in C the sequence $F(X) \to F(Y) \to F(Z)$ is exact in D.
   (c) F commutes with finite limits and finite colimits.

As a direct application of Lemma 6.1.1.

Example 6.1.1. The Hom bifunctor $\mathrm{Hom}_{\mathsf{C}}: \mathsf{C}^{\mathrm{op}} \times \mathsf{C} \to \mathsf{Ab}$ is left exact with respect to each of its arguments.

Example 6.1.2. Recall Theorem 4.2.1.
Right adjoints are left exact. Dually, left adjoints are right exact.

Example 6.1.3. Let I be a small category. If C admits I-indexed limits then the functor $\lim: \mathsf{Fct}(\mathsf{I}^{\mathrm{op}}, \mathsf{C}) \to \mathsf{C}$ is left exact. Dually if C admits I-indexed colimits then the functor $\mathrm{colim}: \mathsf{Fct}(\mathsf{I}, \mathsf{C}) \to \mathsf{C}$ is right exact.

Example 6.1.4. Let R be a ring and $\mathcal{I}$ an index set. The two functors

$$\prod_{i \in \mathcal{I}}: \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R) \qquad \bigoplus_{i \in \mathcal{I}}: \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$$

are exact.

Example 6.1.5. Let R be a ring and $\mathcal{I}$ a small directed category. The functor $\varinjlim_{i \in \mathcal{I}}: \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$ is exact.

6.1.1 Diagram lemmas

Theorem 6.1.1 (Freyd-Mitchell embedding - [Wei94, Section 1.6]). Let C be a small abelian category. Then there exists a ring R (with unit, not necessarilly commutative) and a fully faithful exact functor $F: \mathsf{C} \to \mathsf{Mod}(R)$ which embeds C as a full subcategory of $\mathsf{Mod}(R)$.

Lemma 6.1.2 (Five lemma). Consider a commutative diagram

$$\begin{array}{ccccccccc}
X^0 & \xrightarrow{\alpha^0} & X^1 & \xrightarrow{\alpha^1} & X^2 & \xrightarrow{\alpha^2} & X^3 \\
\downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \downarrow{\scriptstyle f^2} & & \downarrow{\scriptstyle f^3} \\
Y^0 & \xrightarrow[\beta^0]{} & Y^1 & \xrightarrow[\beta^1]{} & Y^2 & \xrightarrow[\beta^2]{} & Y^3
\end{array}$$

where the rows are exact.
If f^0 is an epimorphism and f^1, f^3 are monomorphisms then f^2 is a monomorphism. Dually if f^3 is an monomorphism and f^0, f^2 are epimorphisms then f^1 is a epimorphism.

Proof. Let $x_2 \in X^2$ such that $f^2(x_2) = 0$. We want to see that $x_2 = 0$.

We have that $(f^3 \circ \alpha^2)(x_2) = 0$ which, since f^3 is a monomorphism, implies that $\alpha^2(x_2) = 0$. Since the first row is exact there exists $x_1 \in X^1$ such that $\alpha(x_1) = x_2$.

Let $y_1 := f^1(x_1)$. Since $(\beta^1 \circ f^1)(x_1) = 0$ and the second row is exact there exists $y_0 \in X^0$ such that $\beta^0(y_0) = f^1(x_1)$.

Since f^0 is an epimorphism there exists $x_0 \in X^0$ such that $y_0 = f^0(x_0)$.

Since $(f^1 \circ \alpha^0)(x_0) = f^1(x_1)$ and f^1 is a monomorphism we have $\alpha^0(x_0) = x_1$. Therefore $x_2 = \alpha^1(x_1) = 0$. □

Lemma 6.1.3 (Snake lemma). Consider a commutative diagram

$$\begin{array}{ccccccc} & & X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' \longrightarrow 0 \\ & & \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' \end{array}$$

where the rows are exact.
Then there exists a morphism $\delta\colon \ker\gamma \to \operatorname{coker}\alpha$ giving rise to an exact sequence

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \ker\alpha & \dashrightarrow & \ker\beta & \dashrightarrow & \ker\gamma & \dashrightarrow & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & \dashrightarrow & \operatorname{coker}\alpha & \dashrightarrow & \operatorname{coker}\beta & \dashrightarrow & \operatorname{coker}\gamma & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Proof. We will first check that $\ker\alpha \to \ker\beta \to \ker\gamma$ is exact. Let $x \in \ker\beta$ with $g(x) = 0$. Since the first row is exact there exists $x' \in X'$ with $f(x') = x$. Then $(f' \circ \alpha)(x') = (\beta \circ f)(x') = 0$. Since f' is a monomorphism we get $\alpha(x') = 0$ and thus $x' \in \ker\alpha$.

A dual argument shows that $\operatorname{coker}\alpha \to \operatorname{coker}\beta \to \operatorname{coker}\gamma$ is exact.

It only remains to construct δ. Let $x'' \in \ker\gamma$ and choose $x \in X$ with $g(x) = x''$. Set $y := \beta(y)$. Since $g'(y) = 0$ there exists $y' \in Y'$ such that $f'(y') = y$. Therefore we can define $\delta(x'')$ to be the image of y' in $\operatorname{coker}\alpha$. □

6.2 Complexes and cohomology

Recall that the categories $\mathsf{Comp}^*(\mathsf{C})$ are abelian for $* = +, -, \mathrm{b}, \mathrm{ub}$.

Definition 6.2.1 (Cohomology). Let

$$X^\bullet := \cdots \longrightarrow X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} X^n \xrightarrow{\mathrm{d}_X^{n}} X^{n+1} \longrightarrow \cdots$$

be a complex in C (i.e: $X^\bullet \in \mathsf{Comp}(\mathsf{C})$). We define

$$\begin{aligned} Z^n(X^\bullet) &:= \ker \mathrm{d}_X^n \\ B^n(X^\bullet) &= \operatorname{Im} \mathrm{d}_X^{n-1} \\ \mathrm{H}^n(X^\bullet) &:= {}^{Z^n(X^\bullet)}\!/_{B^n(X^\bullet)} = \operatorname{coker}(B^n(X^\bullet) \to Z^n(X^\bullet)) \end{aligned}$$

We call $\mathrm{H}^n(X^\bullet)$ the n-th **cohomology** of $X^\bullet$.

Remark 6.2.1. A complex $X^\bullet$ is exact at X^n if and only if $\mathrm{H}^i(X^\bullet) = 0$.

Lemma 6.2.1. Let C be an abelian category and $f^\bullet \colon X^\bullet \to Y^\bullet$ a morphism in $\mathsf{Comp}(\mathsf{C})$ homotopic to zero. Then $\mathrm{H}^n f^\bullet \colon \mathrm{H}^n(X^\bullet) \to \mathrm{H}^n(Y^\bullet)$ is the zero morphism for all $n \in \mathbb{Z}$.
In particular if $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are homotopic $\mathrm{H}^n f^\bullet = \mathrm{H}^n g^\bullet$ for all $n \in \mathbb{Z}$.

Proof. Let

$$f^n = s^{n+1} \circ \mathrm{d}_X^n + \mathrm{d}_Y^{n-1} \circ s^n$$

Then $\mathrm{d}_X^n = 0$ in $\ker \mathrm{d}_X^n$ and the image of d_Y^{n-1} is zero on ${}^{\ker \mathrm{d}_Y^n}\!/_{\operatorname{Im} \mathrm{d}_Y^{n-1}}$.

Therefore $\mathrm{H}^n f : {}^{Z^n(X^\bullet)}\!/_{B^n(X^\bullet)} \to {}^{Z^n(Y^\bullet)}\!/_{B^n(Y^\bullet)}$ is the zero morphism. □

Corollary 6.2.1. Let $X^\bullet \sim Y^\bullet$ be two homotopic complexes. Then $\mathrm{H}^n(X^\bullet) \cong \mathrm{H}^n(Y^\bullet)$.

Proof. Since $X^\bullet \sim Y^\bullet$ there exist maps $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$.

Therefore by Lemma 6.2.1 for all $n \in \mathbb{Z}$ we have that

$$\begin{aligned} \mathrm{H}^n(g^\bullet \circ f^\bullet) \cong \mathrm{H}^n \mathbb{1}_{X^\bullet} = \mathbb{1}_{\mathrm{H}^n(X^\bullet)} \\ \mathrm{H}^n(f^\bullet \circ g^\bullet) \cong \mathrm{H}^n \mathbb{1}_{Y^\bullet} = \mathbb{1}_{\mathrm{H}^n(Y^\bullet)} \end{aligned}$$

Furthermore, since H^n is a functor

$$\begin{aligned} \mathrm{H}^n(g^\bullet \circ f^\bullet) = \mathrm{H}^n g^\bullet \circ \mathrm{H}^n f^\bullet \\ \mathrm{H}^n(f^\bullet \circ g^\bullet) = \mathrm{H}^n f^\bullet \circ \mathrm{H}^n g^\bullet \end{aligned}$$

which means that $\mathrm{H}^n f^\bullet$ and $\mathrm{H}^n g^\bullet$ are inverses of each other. □

Definition 6.2.2 (Quasi-isomorphism)**.** Let C be an abelian category. A morphism $f^\bullet \colon X^\bullet \to Y^\bullet$ is a **quasi-isomorphism** (abbreviated q.is) if $\mathrm{H}^n f^\bullet$ is an isomorphism for all $n \in \mathbb{Z}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **quasi-isomorphic**.
In particular $X^\bullet$ is quasi-isomorphic to zero if and only if $X^\bullet$ is exact.

Definition 6.2.3 (Resolution)**.** Let C be an abelian category and $X \in \mathsf{C}$. A **right resolution** of X is a quasi-isomorphism $X \to Y^\bullet$ with $Y^\bullet \in \mathsf{Comp}^+(\mathsf{C})$.
Dually, a **left resolution** is a quasi-isomorphism $Z^\bullet \to X$ with $Z^\bullet \in \mathsf{Comp}^-(\mathsf{C})$.

Remark 6.2.2. Let $X \to Y^\bullet$ be a right resolution of X.
This means that we have a morphism of complexes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle \varepsilon} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & Y^0 & \longrightarrow & Y^1 & \longrightarrow & \cdots \end{array}$$

Remark 6.2.3. We have short exact sequences

$$X^{n-1} \xrightarrow{d_X^{n-1}} \ker d_X^n \longrightarrow H^n(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow H^n(X^\bullet) \longrightarrow \operatorname{coker} d_X^{n-1} \xrightarrow{d_X^n} X^{n+1}$$

We can glue this into an exact sequence

$$0 \longrightarrow H^n(X^\bullet) \longrightarrow \operatorname{coker} d_X^{n-1} \xrightarrow{d_X^n} \ker d_X^{n+1} \longrightarrow H^{n+1}(X^\bullet) \longrightarrow 0$$

Theorem 6.2.1. Let

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

be a short exact sequence in an abelian category $\mathsf{Comp}(\mathsf{C})$.

1. For all $k \in \mathbb{Z}$ the sequence $H^k(X^\bullet) \to H^k(Y^\bullet) \to H^k(Z^\bullet)$ is exact.
2. For all $k \in \mathbb{Z}$ there exists $\delta^k \colon H^k(Z^\bullet) \to H^{k+1}(X^\bullet)$ making the long sequence

$$\begin{array}{l} \cdots \longrightarrow H^{k-1}(Y^\bullet) \longrightarrow H^{k-1}(Z^\bullet) \xrightarrow{\delta^{k-1}} \\ H^k(X^\bullet) \longrightarrow H^k(Y^\bullet) \longrightarrow H^k(Z) \xrightarrow{\delta^k} \\ H^{k+1}(X^\bullet) \longrightarrow H^{k+1}(Y^\bullet) \longrightarrow \cdots \end{array}$$

exact. Furthermore δ^k can be constructed to be functorial with respect to short exact sequences of $\mathsf{Comp}(\mathsf{C})$.

Proof. Consider the diagram

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & H^k(X^\bullet) & & H^k(Y^\bullet) & & H^k(Z^\bullet) & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \operatorname{coker} d_X^{k-1} & \longrightarrow & \operatorname{coker} d_Y^{k-1} & \longrightarrow & \operatorname{coker} d_Z^{k-1} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker d_X^{k+1} & \longrightarrow & \ker d_Y^{k+1} & \longrightarrow & \ker d_Z^{k+1} & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & H^{k+1}(X^\bullet) & & H^{k+1}(Y^\bullet) & & H^{k+1}(Z^\bullet) & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

where the columns are exact by Remark 6.2.3 and the rows are exact by hypothesis.

Therefore the result follows from the Snake lemma. □

Chapter 7

Derived functors

7.1 Injective and projective objects

Definition 7.1.1. Let C be a category. An object $I \in \mathsf{C}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & & \\ Z & & \end{array}$$

where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & \nearrow_{h} & \\ Z & & \end{array}$$

Dually, an object $P \in \mathsf{C}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & & X \\ & \nearrow_{h} & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

Notation. If C is any category we denote by $\mathcal{I}_{\mathsf{C}}$ its full subcategory made of injective objects and by $\mathcal{P}_{\mathsf{C}}$ its full subcategory of projective objects.

Proposition 7.1.1. Let C be an abelian category. An object $I \in \mathsf{C}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(-, I)$ is exact.
Dually, an object $P \in \mathsf{C}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(P, -)$ is exact.

Proof. By Example 6.1.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in \mathcal{I}_{\mathsf{C}}$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 6.1.1 (recall that Hom is a contravariant functor), we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.

Since by Example 6.1.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.

Consider a morphism $\alpha \colon X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta \colon Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f \colon X \rightarrowtail Y$ in C. We have the following short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.

Therefore if $\gamma \colon X \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(X, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects. □

Lemma 7.1.1. Let

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

be a short exact sequence in C with X injective. Then the sequence splits.

Proof. By definition taking $k = \mathbb{1}_X$ there exists $h : Y \to X$ such that $k \circ f = \mathbb{1}_X$. Then we just need to apply Proposition 6.0.2. □

Remark 7.1.1. If $F \colon \mathsf{C} \to \mathsf{D}$ is an additive functor between abelian categories and the hypothesis of Lemma 7.1.1 are satisfied the sequence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$

splits and in particular it is exact.

Lemma 7.1.2. Let $X, Z \in \mathsf{C}$. Then $X \oplus Z$ is injective if and only if X and Z are injective.

Proof. This follows from the fact that if F, G are two additive functors between abelian categories then $F \oplus G$ is exact if and only if F and G are exact and $\mathrm{Hom}(-, I \oplus J) \cong \mathrm{Hom}(-, I) \oplus \mathrm{Hom}(-, J)$. □

Proposition 7.1.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in C with X and Z injective. Then Y is injective.

Proof. By Lemma 7.1.1 since X is injective the sequence splits. Therefore the sequence is isomorphic to

$$0 \longrightarrow X \longrightarrow X \oplus Z \longrightarrow Z \longrightarrow 0$$

Finally since X and Z are injective by Lemma 7.1.2 $X \oplus Z \cong Y$ is injective. □

Definition 7.1.2. A category has **enough injectives** if for every object $X \in \mathsf{X}$ there exists an injective object $I \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $X \rightarrowtail I$.
Dually, C has **enough projectives** if for every object $X \in \mathsf{A}$ there exists a projective object $P \in \mathcal{P}_{\mathsf{C}}$ and an epimorphism $P \twoheadrightarrow X$.

Theorem 7.1.1. Let C be an abelian category with enough injectives. Then every object $X \in \mathsf{C}$ admits a right resolution $X \to I^\bullet$ where $I^n \in \mathcal{I}_{\mathsf{C}}$ is injective for all $n \in \mathbb{Z}$.
Dually, if C has enough projectives every object $X \in \mathsf{C}$ admits a left resolution $P^\bullet \to X$ where $P^n \in \mathcal{P}_{\mathsf{C}}$ is projective for all $n \in \mathbb{Z}$.

Proof. Let $X \in \mathsf{C}$.

We want to construct an exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^n \longrightarrow \cdots$$

We will do it by induction over n.

For the base case, since C has enough injectives there exists an injective object $I_0 \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $X \rightarrowtail I_0$.

Suppose now that we have a exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^{n-1} \longrightarrow I^n$$

Since C is abelian there exists

$$\operatorname{coker}(I^{n-1} \to I^n) \colon I^n \twoheadrightarrow B^n$$

Note that

$$I^{n-1} \longrightarrow I^n \twoheadrightarrow B^n \longrightarrow 0$$

is exact. Since C has enough injectives there exists $I^{n+1} \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $B^n \rightarrowtail I^{n+1}$ Therefore

$$\begin{array}{ccccc} & & B^n & & \\ & & \uparrow & \searrow & \\ I^{n-1} & \longrightarrow & I^n & \longrightarrow & I^{n+1} \end{array}$$

is exact and induction follows. □

Theorem 7.1.2. Let C be a category with enough injectives. Let $X \xrightarrow{\varepsilon} I^\bullet, Y \xrightarrow{\eta} J^\bullet$ be two right-resolutions of $X, Y \in \mathsf{C}$.

If $J^\bullet$ is made of injectives then any morphism $f: X \to Y$ can be extended to a morphism $f^\bullet: I^\bullet \to J^\bullet$ in the sense that $\eta \circ f^0 = f \circ \varepsilon$. Furthermore this extension is unique up to homotopy.
We call $f^\bullet$ a **lifting** of f.

Proof. We will first prove existence by constructing $f^\bullet$. We will do it by induction.

For the base case $n = 0$ we have:

$$\begin{array}{ccc} X & \xrightarrow{\varepsilon} & I^0 \\ {\scriptstyle f}\downarrow & & \\ Y & \xrightarrow[\eta]{} & J^0 \end{array}$$

Since ε is a monomorphism and J^0 is injective there exists a unique $f^0: I^0 \to J^0$ making the diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{\varepsilon} & I^0 \\ {\scriptstyle f}\downarrow & & \vdots{\scriptstyle f^0} \\ Y & \xrightarrow[\eta]{} & J^0 \end{array}$$

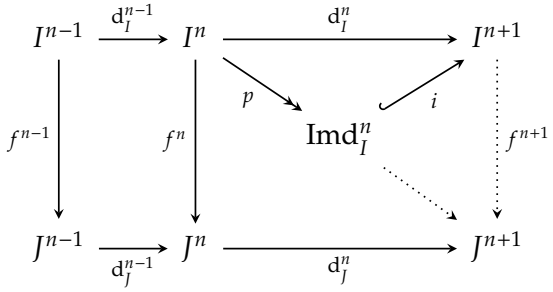
Suppose now that we have constructed $f^i: I^i \to J^i$ for $i \in \{0, \dots, n\}$ making the following diagram commute:

$$\begin{array}{ccccccccccc} X & \xrightarrow{\varepsilon} & I^0 & \xrightarrow{\mathrm{d}_I^0} & \cdots & \longrightarrow & I^{n-1} & \xrightarrow{\mathrm{d}_I^{n-1}} & I^n & \xrightarrow{\mathrm{d}_I^n} & I^{n+1} \\ {\scriptstyle f}\downarrow & & {\scriptstyle f^0}\downarrow & & & & {\scriptstyle f^{n-1}}\downarrow & & {\scriptstyle f^n}\downarrow & & \\ Y & \xrightarrow[\eta]{} & J^0 & \xrightarrow[\mathrm{d}_J^0]{} & \cdots & \longrightarrow & J^{n-1} & \xrightarrow[\mathrm{d}_J^{n-1}]{} & J^n & \xrightarrow[\mathrm{d}_J^n]{} & J^{n+1} \end{array}$$

Consider $g := \mathrm{d}_J^n \circ f^n: I^n \to J^{n+1}$. Since $\mathrm{d}_I^{n-1} \circ f^n = f^{n-1} \circ d_J^{n-1}$ and $\mathrm{d}_J^{n-1} \circ \mathrm{d}_J^n = 0$ we have that g vanishes on $\operatorname{Im} \mathrm{d}_I^{n-1}$. Therefore g induces a morphism

$$I^n \Big/ \operatorname{Im} \mathrm{d}_I^{n-1} \to J^{n+1}$$

Since $I^\bullet$ is exact we have $I^n \Big/ \operatorname{Im} \mathrm{d}_I^{n-1} \cong \operatorname{Im} \mathrm{d}_I^n \subseteq I^{n+1}$. Therefore since J^{n+1} is injective the map $\operatorname{Im} \mathrm{d}_I^{n-1} \to J^{n+1}$ can be extended to the whole I^{n+1} giving a morphism $f^{n+1}: I^{n+1} \to J^{n+1}$. Diagramatically:



This completes the existence part of the proof.

To see uniqueness, suppose that we have two liftings $f^\bullet, g^\bullet \colon I^\bullet \to J^\bullet$ of f. We want to find an homotopy $h^n \colon I^n \to J^{n-1}$ such that for all $n \in \mathbb{N}$

$$\phi^n := f^n - g^n = \mathrm{d}_J^{n-1} \circ h^n + h^{n+1} \circ \mathrm{d}_I^n$$

Again, we will do this by induction over n.

For the base case $n = 1$ note that $\phi^{-1} = f - f = 0$ (since both $f^\bullet$ and $g^\bullet$ are liftings of f). Therefore

$$\phi^0 \circ \varepsilon = \eta \circ (f - f) = 0$$

Since $\phi^0 \circ \varepsilon = 0$ we have that ϕ^0 factors through $\operatorname{coker} \varepsilon_1 \cong \operatorname{Im} \mathrm{d}_I^0$. Using that J^0 is injective we can construct $h^1 \colon I^1 \to J^0$ such that $h^1 \circ \mathrm{d}_I^0 = \phi^0$.

Assume now that we have constructed $h^i \colon I^i \to J^{i-1}$ for $i \in \{1, \dots, n\}$. We want to construct $h^{n+1} \colon I^{n+1} \to J^n$ such that

$$\psi := \phi^n - \mathrm{d}_J^{n-1} \circ h^n = h^{n+1} \circ \mathrm{d}_I^n$$

Note that

$$\begin{aligned}
\psi \circ \mathrm{d}_I^{n-1} &= \phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ h^n \circ \mathrm{d}_I^{n-1} \\
&= \phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ \left(\phi^{n-1} - \mathrm{d}_J^{n-2} \circ h^{n-1} \right) \\
&= \left(\phi^n \circ \mathrm{d}_I^{n-1} - \mathrm{d}_J^{n-1} \circ \phi^{n-1} \right) + \left(\mathrm{d}_J^{n-1} \circ \mathrm{d}_J^{n-2} \right) \circ h^{n-1} \\
&= 0
\end{aligned}$$

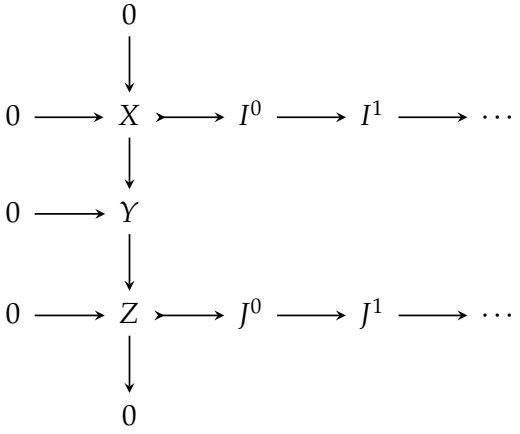
Therefore ψ vanishes on $\operatorname{Im} \mathrm{d}_I^{n-1}$ and thus it factors through $I^n / \operatorname{Im} \mathrm{d}_I^{n-1} \cong \operatorname{Im} \mathrm{d}_I^n \subseteq I^{n+1}$. Using that J^n is injective this map extends to $h^{n+1} \colon I^{n+1} \to J^n$. □

Corollary 7.1.1. Let $I^\bullet, J^\bullet$ be two injective resolutions of an object $X \in \mathsf{C}$. Then

$$\mathrm{H}^\bullet(I^\bullet) \cong \mathrm{H}^\bullet(J^\bullet)$$

Proof. Taking $\mathbb{1}_X$ in Theorem 7.1.2 we get that $I^\bullet$ and $J^\bullet$ are homotopic. Therefore the result follows from Corollary 6.2.1. □

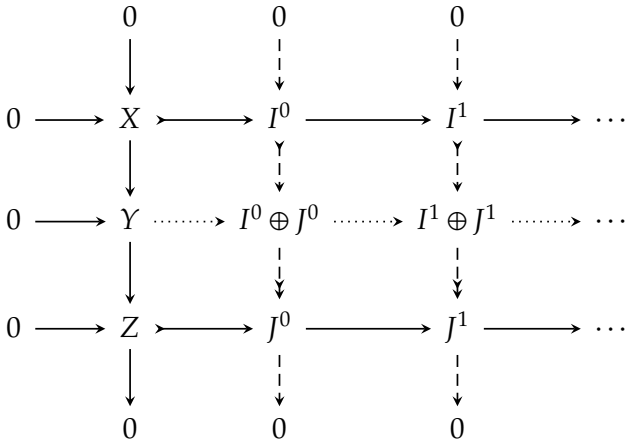
Theorem 7.1.3 (Horseshoe lemma). Consider a commutative diagram



where the column is exact and the rows are injective resolutions. Then $I^\bullet \oplus J^\bullet$ is an injective resolution of Y and the left-hand column lifts to an exact sequence of complexes

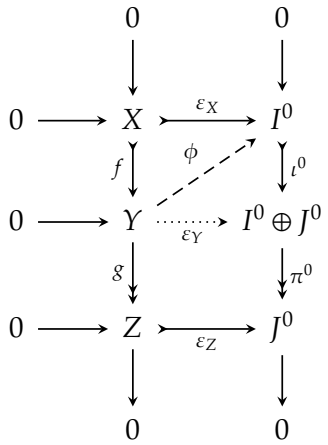
$$0 \longrightarrow I^\bullet \xrightarrow{\iota} I^\bullet \oplus J^\bullet \xrightarrow{\pi} J^\bullet \longrightarrow 0$$

where $\iota^n : I^n \rightarrowtail I^n \oplus J^n$ and $\pi^n : I^n \oplus J^n \twoheadrightarrow J^n$ are the natural inclussion and projection respectively. Diagramatically



Proof. We will do it by induction over n. Note that for all $n \in \mathbb{N}$ by Lemma 7.1.2 $I^n \oplus J^n$ is injective.

For the base case $n = 0$ we have



Since I^0 is injective and $f : X \rightarrowtail B$ is a monomorphism there exists a morphism $\phi : Y \to I^0$ such that $\phi \circ f = \varepsilon_X$. Define

$$\begin{aligned} \varepsilon_Y : Y &\to I^0 \oplus J^0 \\ y &\mapsto (\phi, \varepsilon_Z \circ g)(y) \end{aligned}$$

Let's see that the top and bottom square commute.

For the top one, let $x \in X$. On the one hand $\left(\iota^0 \circ \varepsilon_X\right)(x) = (\varepsilon(x), 0)$. On the other using that $\phi \circ f = \varepsilon_X$ and $g \circ f = 0$ we have

$$\left(\varepsilon_Y \circ f\right)(x) = \left(\phi \circ f, \varepsilon_Z \circ g \circ f\right)(y) = (\varepsilon(x), 0)$$

Now, to see the commutativity of the bottom square note that for all $y \in Y$

$$\left(\pi^0 \circ \varepsilon_Y\right)(y) = \left(\varepsilon_Z \circ g\right)(y)$$

It only remains to see that ε_Y is a monomorphism. Suppose that $\varepsilon_Y(y) = 0$. Then $\phi(y) = 0$ and $\left(\varepsilon_Z \circ g\right)(y) = 0$. Since ε_Z is a monomorphism $g(y) = 0$. Because $\ker g = \operatorname{Im} f$ there exists $x \in X$ such that $f(x) = y$. Then $\left(\phi \circ f\right)(x) = \varepsilon_X(x) = 0$. Using that ε_X is a monomorphism $x = 0$, which means that $y = f(0) = 0$ and thus ε_Y is a monomorphism.

This completes the base case.

We have the following diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \varepsilon_X} & & \downarrow{\scriptstyle \varepsilon_X} & & \downarrow{\scriptstyle \varepsilon_X} & & \\
0 & \longrightarrow & I^0 & \xrightarrow{\iota^0} & I^0 \oplus J^0 & \xrightarrow{\pi^0} & J^0 & \longrightarrow & 0
\end{array}$$

Since the vertical maps are monomorphisms their kernels are 0. If we denote by X^1, Y^1 and Z^1 their cokernels by the Snake lemma we get a short exact sequence:

$$0 \longrightarrow X^1 \longrightarrow Y^1 \longrightarrow Z^1 \longrightarrow 0$$

Furthermore the shifted complexes $I^{\geq 1}$ and $J^{\geq 1}$ are injective resolutions of X^1 and Z^1 respectively. Therefore we are in the exact same situation as in the base case and by induction we can construct $I^n \oplus J^n$ and short exact sequences

$$0 \longrightarrow X^n \longrightarrow Y^n \longrightarrow Z^n \longrightarrow 0$$

for all $n \geq 0$.

To finish the proof we need to define differentials $\mathrm{d}_Y^n : I^n \oplus J^n \to I^{n+1} \oplus J^{n+1}$. We define these as by composing the quotient map $I^n \oplus J^n \to Y^{n+1}$ with the monomorphism $Y^{n+1} \rightarrowtail I^{n+1} \oplus J^{n+1}$ constructed in the inductive step. In matrix form

$$\mathrm{d}_Y^n := \begin{pmatrix} \mathrm{d}_X^n & \delta^n \\ 0 & \mathrm{d}_Z^n \end{pmatrix}$$

where d_X and d_Z are the differentials of the resolutions $I^\bullet$ and $J^\bullet$ and δ^n is the connecting morphism from the Snake lemma. By construction we have $\ker \mathrm{d}_Y^n = \operatorname{Im} \mathrm{d}_Y^{n-1}$, making $I^\bullet \oplus J^\bullet$ exact. □

Remark 7.1.2. In the proof of Theorem 7.1.3 we have made implicit use of the Freyd-Mitchell embedding theorem. A diagramatic proof of (the dual of) this result can be seen at [Wei94, Lemma 2.8.8].

Proposition 7.1.3. Let $R\colon \mathsf{D} \to \mathsf{C}$ be a additive functor that is right adjoint to an exact functor $L\colon \mathsf{C} \to \mathsf{D}$. If $I \in \mathsf{D}$ is an injective object then $R(I)$ is also injective.
Dually, if $L\colon \mathsf{C} \to \mathsf{D}$ is an additive functor that is left adjoint to an exact functor $R\colon \mathsf{D} \to \mathsf{C}$ and $P \in \mathsf{C}$ is a projective object then $L(P)$ is also projective.

Proof. By Proposition 7.1.1 we want to show that $\mathrm{Hom}_{\mathsf{C}}(-, R(I))$ is exact.

Given a monomorphism $f\colon X \to Y$ in C we have

$$\begin{array}{ccc} \mathrm{Hom}_{\mathsf{D}}(L(Y), I) & \xrightarrow{Lf^*} & \mathrm{Hom}_{\mathsf{D}}(L(A), I) \\ \sim\downarrow & & \downarrow\sim \\ \mathrm{Hom}_{\mathsf{C}}(Y, R(I)) & \xrightarrow[f^*]{} & \mathrm{Hom}_{\mathsf{C}}(A, R(I)) \end{array}$$

Since L is exact and I is injective Lf^* is surjective. □

7.2 Derived functors

In this section C and D will be abelian categories, which we will assume that have enough injectives.

We will consider left-exact functors $F\colon \mathsf{C} \to \mathsf{D}$, but there are dual definitions and statements for right-exact functors assuming C and D have enough projectives.

Definition 7.2.1. Let C be an abelian category with enough injectives. Let $F\colon \mathsf{C} \to \mathsf{D}$ be a left-exact functor. We define the i-th **right derived functor** of F

$$\mathrm{R}^i F\colon \mathsf{C} \to \mathsf{D}$$

as, for any object $X \in \mathsf{C}$:

$$\mathrm{R}^i F(X) \coloneqq \mathrm{H}^i(F(I^\bullet))$$

where $I^\bullet$ is an injective resolution of X, and for any morphism $f\colon X \to Y$

$$\mathrm{R}^i F f \coloneqq \mathrm{H}^i(F f^\bullet)$$

where $f^\bullet$ is a lifting of f.

Remark 7.2.1. Since F is left exact the sequence $0 \to F(X) \to F(I^0) \to F(I^1)$ is exact. Therefore we always have

$$\mathrm{R}^0 F(X) \cong F(X)$$

Furthermore, under this identification if $f\colon X \to Y$ is a morphism in C then

$$\mathrm{R}^0 F f = F f$$

Remark 7.2.2. If $X \in \mathsf{C}$ is injective then for all $i > 0$

$$\mathrm{R}^i F(X) = 0$$

Indeed, one can resolve X by $0 \to X \to X \to 0$.

Proposition 7.2.1. For all $i \geq 0$ the right derived functors $\mathrm{R}^i F\colon \mathsf{C} \to \mathsf{D}$ are additive.

Proof. Let $f, g\colon X \to Y$ be two morphisms in C. Clearly, if $f^\bullet$ and $g^\bullet$ are liftings of f and g respectively then $f^\bullet + g^\bullet$ is a lifting of $f + g$. Therefore using that F and H^i are additive

$$\mathrm{R}^i F(f+g) = \mathrm{H}^i(F(f^\bullet + g^\bullet)) = \mathrm{H}^i\left(Ff^\bullet + Fg^\bullet\right) = \mathrm{H}^i(Ff^\bullet) + \mathrm{H}^i(Fg^\bullet) = \mathrm{R}^i F f + \mathrm{R}^i F g$$

□

Theorem 7.2.1. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be a short exact sequence in C. There exists a long exact sequence:

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \\ \quad\quad \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Y) \longrightarrow \mathrm{R}^1 F(Z) \\ \quad\quad \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Y) \longrightarrow \cdots \end{array}$$

Proof. Choose injective resolutions $I^\bullet$ and $J^\bullet$ of X and Z respectively. By the Horseshoe lemma $I^\bullet \oplus J^\bullet$ is an injective resolution of Y and we have a short exact sequence of injective complexes

$$0 \longrightarrow I^\bullet \longrightarrow I^\bullet \oplus J^\bullet \longrightarrow J^\bullet \longrightarrow 0$$

Since I^n is injective by Lemma 7.1.1 the sequences

$$0 \longrightarrow I^n \longrightarrow I^n \oplus J^n \longrightarrow J^n \longrightarrow 0$$

split. Therefore since F is additive we have a short exact sequence

$$0 \longrightarrow F(I^\bullet) \longrightarrow F(I^\bullet \oplus J^\bullet) \longrightarrow F(J^\bullet) \longrightarrow 0$$

Writing its corresponding long exact sequence in cohomology we get

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \\ \quad\quad \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Y) \longrightarrow \mathrm{R}^1 F(Z) \\ \quad\quad \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Y) \longrightarrow \cdots \end{array}$$

□

Definition 7.2.2. An object $Q \in \mathsf{C}$ is **F-acyclic** if for all $i > 0$

$$\mathrm{R}^i F(Q) = 0$$

An **F-acyclic resolution** of an object $X \in \mathsf{C}$ is a right resolution $X \to Q^\bullet$ made of F-acyclic objects.

Theorem 7.2.2. Let $Q^\bullet$ be a F-acylic resolution of $X \in \mathsf{C}$. Then for all $i \geq 0$

$$\mathrm{R}^i F(X) \cong \mathrm{H}^i(F(Q^\bullet))$$

Proof. Let $I^\bullet$ be an injective resolution of X. By Theorem 7.1.2 there exists a morphism $f^\bullet \colon Q^\bullet \to I^\bullet$ lifting $\mathbb{1}_X$:

$$\begin{array}{ccc} X & \longrightarrow & Q^\bullet \\ {\scriptstyle \mathbb{1}_X}\downarrow & & \downarrow{\scriptstyle f^\bullet} \\ X & \longrightarrow & I^\bullet \end{array}$$

This gives a morphism of complexes $Ff : F(Q^\bullet) \to F(I^\bullet)$, which induces maps on cohomology

$$\phi_i \colon \mathrm{H}^i(F(Q^\bullet)) \to \mathrm{H}^i(F(I^\bullet)) = \mathrm{R}^i F(X)$$

for all $i \geq 0$. We will prove that ϕ_i are isomorphisms by induction over i.

For the base case $i = 0$ since F is left exact the sequences $0 \to F(X) \to F(Q^0) \to F(Q^1)$ and $0 \to F(X) \to F(I^0) \to F(I^1)$ are exact. Therefore

$$\mathrm{H}^0(F(Q^\bullet)) \cong F(X) \cong \mathrm{H}^0(F(I^\bullet))$$

Suppose now that ϕ_i is an isomorphism for all $i \in \{0, \ldots, n\}$. Consider the short exact sequence

$$0 \longrightarrow X \longrightarrow Q^0 \longrightarrow Z^1 \longrightarrow 0$$

where $Z^1 := \mathrm{Im}(Q^0 \to Q^1)$. By Theorem 7.2.1 we have a long exact sequence

$$\begin{array}{l} 0 \longrightarrow F(X) \longrightarrow F(Q^0) \longrightarrow F(Z^1) \xrightarrow{\delta^0} \\ \mathrm{R}^1 F(X) \longrightarrow \mathrm{R}^1 F(Q^0) \longrightarrow \mathrm{R}^1 F(Z^1) \xrightarrow{\delta^1} \\ \mathrm{R}^2 F(X) \longrightarrow \mathrm{R}^2 F(Q^0) \longrightarrow \cdots \end{array}$$

Since Q^0 is F-acyclic for all $i > 0$ we get an isomorphism

$$\mathrm{H}^{i+1}(F(X)) = \mathrm{R}^{i+1} F(X) \cong \mathrm{R}^i F(Z^1) = \mathrm{H}^i(F(Z^1))$$

On the other hand, the truncated complex

$$0 \longrightarrow Z^1 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

is an F-acyclic resolution of Z^1. Applying the induction hypothesis for all $j \in \{1, \ldots, n\}$ we have

$$\mathrm{R}^j F(Z^1) \cong \mathrm{H}^j(F(Q^{\bullet, \geq 1})) = \mathrm{H}^{j+1}(F(Q^\bullet))$$

In particular for $j = n$ combining the two isomorphisms

$$\mathrm{R}^{n+1} F(X) \cong \mathrm{H}^{n+1} F(Q^\bullet)$$

It remains to check the case where $i = 1$. From the exact sequence

$$0 \longrightarrow Z^1 \longrightarrow Q^1 \longrightarrow Z^2 \longrightarrow 0$$

where $Z^2 := \operatorname{Im}\left(Q^1 \to Q^2\right)$ using that F is left exact we get

$$F(Z^1) \cong \ker\left(F(Q^1) \to F(Q^2)\right)$$

Therefore using the exact sequence

$$F(Q^0) \longrightarrow F(Z^1) \longrightarrow \mathrm{R}^1 F(X) \longrightarrow 0$$

we have

$$\begin{aligned}\mathrm{R}^1 F(X) &\cong {}^{F(Z^1)}\Big/_{\operatorname{Im}\left(F(Q^0) \to F(Q^1)\right)} \\ &\cong {}^{\ker\left(F(Q^1) \to F(Q^2)\right)}\Big/_{\operatorname{Im}\left(F(Q^0) \to F(Q^1)\right)} \\ &= \mathrm{H}^1 F(Q^\bullet)\end{aligned}$$

□

Theorem 7.2.3. Let $F\colon \mathsf{C} \to \mathsf{D}$ and $G\colon \mathsf{D} \to \mathsf{E}$ be two left exact functors of abelian categories. Suppose that C and D have enough injectives.

1. If G is exact then $\mathrm{R}^i(G \circ F) \cong G \circ \mathrm{R}^i F$.
2. If F is exact then there is a natural morphism $\mathrm{R}^i(G \circ F) \to \mathrm{R}^i G \circ F$.
3. Let $X \in \mathsf{C}$ be F-acyclic and suppose that F sends injective objects to G-acyclic ones. Then $\mathrm{R}^i(G \circ F)(X) \cong (\mathrm{R}^i G)(F(X))$.

Proof. For $X \in \mathsf{C}$ let $X \xrightarrow{\text{q.is}} I^\bullet$ be an injective resolution of X. Then

$$\mathrm{R}^i(G \circ F)(X) = \mathrm{H}^i\left((G \circ F)(I^\bullet)\right)$$

1. If G is exact then we have an isomorphism
$$\mathrm{H}^i\left((G \circ F)(I^\bullet)\right) \cong G(\mathrm{H}^i(F(I^\bullet)))$$
2. Consider an injective resolution $F(X) \xrightarrow{\text{q.is}} J^\bullet$ of $F(X)$. By Theorem 7.1.2 there exists a morphism $F(I^\bullet) \to J^\bullet$. Applying G we get a morphism
$$(G \circ F)(I^\bullet) \to G(J^\bullet)$$
Therefore since $\mathrm{H}^i\left((G \circ F)(I^\bullet)\right) = \mathrm{R}^i(G \circ F)(X)$ and $\mathrm{H}^i(G(J^\bullet)) = \mathrm{R}^i G(F(X))$ the result follows.
3. By hypothesis $F(I^\bullet)$ is quasi-isomorphic to $F(X)$. Therefore $(\mathrm{R}^i G)(F(X)) \cong \mathrm{H}^i(G(F(I^\bullet)))$ by Theorem 7.2.2 and $\mathrm{H}^i(G(F(I^\bullet))) \cong \mathrm{R}^i(G \circ F)(X)$. □

PART III

Sheaves

Chapter 8

Presheaves and sheaves

8.1 Presheaves and sheaves

Definition 8.1.1 (Presheaf). Let X be a topological space and let C be a category. The category of C-valued presheaves over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) := \mathsf{Fct}(\mathsf{Open}(X)^{\mathrm{op}}, \mathsf{C})$$

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of $\mathcal{F}$ over U*.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) \colon \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Open}(X)}$$

in C making the following diagram commute for all $V \subseteq U$

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Definition 8.1.2 (Constant presheaf). Let X be a topological space. For any object $A \in \mathsf{C}$ the **constant presheaf** with value A, denoted $\underline{A}_X^{\mathrm{pre}}$ is the C-valued presheaf associated to the constant functor

$$\begin{array}{ccc} \underline{A}_X^{\mathrm{pre}} : \mathsf{Open}(X)^{\mathrm{op}} & \longrightarrow & \mathsf{C} \\ U & & A \\ {\scriptstyle V \subseteq U}\downarrow & \longmapsto & \downarrow{\scriptstyle \mathbb{1}_A} \\ V & & A \end{array}$$

Note that this defines a functor

$$\underline{-}_X^{\mathrm{pre}} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$$

We now turn our attention to a particular class of presheaves.

Instead of working on a general category C, we will work with categories which are "sets with an algebraic structure" (for example Grp, Ab, Ring, ... but not Top).

Definition 8.1.3. A **type of algebraic structure** is a category C together with a functor $F\colon \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

Example 8.1.1. The following categories, together with their respective forgetful functors are types of algebraic structures:

- The category of pointed sets.
- The category of (abelian) groups.
- The category of monoids.
- The category of rings.
- The category of R-modules for a fixed ring R.
- The category of Lie algebras over a fixed field.

We state some properties of this types of categories without proof.

Lemma 8.1.1 ([Stacks, Tag 007O]). Let (C, F) be a type of algebraic structure. Then:

1. C has a terminal object 1 and $F(1) = \{*\}$.
2. C has products and $F\left(\prod X_i\right) = \prod F\left(X_i\right)$.
3. C has pullbacks and $F(X \times_Z Y) = F(X) \times_{F(Z)} F(Y)$.
4. C has equalizers and if $\mathrm{eq} \to X$ is the equalizer of $f, g\colon X \rightrightarrows Y$ then $F(\mathrm{eq}) \to F(X)$ is the equalizer of $Ff, Fg\colon F(X) \rightrightarrows F(Y)$.
5. $X \to Y$ is a monomorphism if and only if $F(X) \to F(Y)$ is injective.
6. If $Ff\colon F(X) \to F(Y)$ is surjective then f is an epimorphism.
7. Given $X_1 \to X_2 \to \cdots$, $\varinjlim X_i$ exists and $F(\varinjlim X_i) = \varinjlim F(X_i)$. And more generally for any direct limit.
8. Direct limits commute with limits.

In what follows C will be a type of algebraic structure.

Definition 8.1.4. Let X be a topological space. The category of **C-valued sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of $\mathsf{PSh}_X(\mathsf{C})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections $(\forall i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j})$ there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = s_i$.

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Notation. Given a ring R, we denote the category of sheaves of R-modules over a topological space X by $\mathsf{Mod}(\underline{R}_X)$.

Remark 8.1.1. Axioms SH.1 and SH.2 can be interpreted as the following equaliser sequence:

$$\mathscr{F}(U) \longrightarrow \prod_{i\in I} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in I\times I} \mathscr{F}(U_i \cap U_j)$$

where $\{U_i\}_{i\in I}$ is an open cover of U .
In particular taking the empty cover of $\varnothing \in \mathsf{Open}(X)$, since the empty product is terminal we have

$$\mathscr{F}(\varnothing) \longrightarrow 1 \rightrightarrows 1$$

i.e: $\mathscr{F}(\varnothing)$ is a terminal object of C

Example 8.1.2 (Sheaves on a point)**.** Let $\{*\}$ be the one-point topological space. We are going see that there is an equivalence of categories

$$\mathsf{Sh}_{\{*\}}(\mathsf{C}) \simeq \mathsf{C}$$

Consider the functors

$$F\colon \mathsf{Sh}_{\{*\}}(\mathsf{C}) \to \mathsf{C} \qquad G\colon \mathsf{C} \to \mathsf{Sh}_{\{*\}}(\mathsf{C})$$

where $F(\mathscr{F}) \coloneqq \mathscr{F}(\{*\})$ for all $\mathscr{F} \in \mathsf{Sh}_{\{*\}}(\mathsf{C})$ and for all $X \in \mathsf{C}$, $G(X)$ is the sheaf defined as

$$G(X)(U) \coloneqq \begin{cases} G & U = \{*\} \\ 1 & U = \varnothing \end{cases}$$

with the obvious restriction maps. We have

$$(F \circ G)(X) = G \qquad (G \circ F)(\mathscr{F}) \cong \mathscr{F}$$

where the isomorphism comes from $\mathscr{F}(\varnothing)$ being terminal (but not necessarilly 1).

8.1.1 Stalks

Definition 8.1.5. Let X be a topological space and let $\mathcal{F}$ be a C-valued presheaf. For all points $p \in X$ we define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p \coloneqq \varinjlim_{\substack{U\ni p \\ U\in\mathsf{Open}(X)}} \mathcal{F}(U)$$

Note that this comes with maps $\mathcal{F}(U) \to \mathcal{F}_p$ for all $U \subseteq X$ open such that $p \in U$. Since this is a direct limit following Example 3.2.5 we may describe $\mathcal{F}_p$ as

$$\mathcal{F}_p = \{(U,s) \mid p \in U, s \in \mathcal{F}(U)\}\,/_{\sim}$$

where

$$(U,s) \sim (V,t) \iff \exists W \subseteq U \cap V \mid p \in W \wedge s|_W = t|_W$$

Using this characterization we may define the stalk of a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ as

$$\begin{aligned} \phi_x\colon \mathcal{F}_x &\to \mathcal{G}_x \\ (U,s) &\mapsto (U, \phi(s)) \end{aligned}$$

Therefore we have a stalk functor

$$-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$$

By using the same definition we can extend this to a functor

$$-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$$

Remark 8.1.2. By the properties of algebraic structures, stalks commute with finite limits.

Lemma 8.1.2. Let $\mathscr{F}$ be a C-valued sheaf over a topological space X. For all open subsets $U \subseteq X$ the map $\mathscr{F}(U) \to \prod_{p \in U} \mathscr{F}_p$ is injective.
In other words, a sheaf is completely determined by its germs.

Proof. Let $s, t \in \mathscr{F}(U)$ such that for all $s_p = t_p$ in $\mathscr{F}_p$ for all $p \in U$.

By definition of stalk there exists an open neighbourhood $V_p \subseteq U$ of p such that $s|_{V_p} = t|_{V_p}$. Since this works for all $p \in U$, we have an open cover $\{V_p\}_{p \in U}$ of U.

Therefore since $\mathscr{F}$ is a sheaf by SH.1 we have $s = t$. □

Theorem 8.1.1. Let $\phi, \psi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves where $\mathcal{F}$ is a C-valued presheaf and $\mathcal{G}$ is a C-valued sheaf. Then $\phi = \psi$ if and only if for all $p \in X$ the stalks at p are equal, i.e: $\phi_p = \psi_p$.

Proof. The direction [$\implies$] is trivial.

For the [$\impliedby$] direction suppose that $\phi_p = \psi_p$ for all $p \in X$ and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \overset{\phi}{\underset{\psi}{\rightrightarrows}} & \mathcal{G}(U) \\ {\scriptstyle i_{\mathcal{F}}}\downarrow & & \downarrow{\scriptstyle i_{\mathcal{G}}} \\ \prod_{p \in U} \mathcal{F}_p & \underset{\prod \psi_p}{\overset{\prod \phi_p}{\rightrightarrows}} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

For all $s \in \mathcal{F}(U)$ we have

$$i_{\mathcal{G}}(\phi(U)(s)) = \big(\phi_p(s_p)\big)_{p \in U} = \big(\psi_p(s_p)\big)_{p \in U} = i_{\mathcal{G}}(\psi(U)(s))$$

Since $i_{\mathcal{G}}$ is injective we have

$$\phi(U)(s) = \psi(U)(s)$$

Therefore since this holds for all $s \in \mathcal{F}(U)$ and $U \subseteq X$ open we have that $\phi = \psi$. □

Proposition 8.1.1. Let $\phi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then ϕ is a isomorphism if and only if for all $p \in X$ the induced stalk map $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism.

Proof.

$\Longrightarrow$ Suppose that $\phi : \mathscr{F} \to \mathscr{G}$ is an isomorphism in $\mathsf{Sh}_X(\mathsf{C})$. Then there exists $\psi : \mathscr{G} \to \mathscr{F}$ such that

$$\psi \circ \phi = \mathbb{1}_{\mathscr{F}} \qquad \phi \circ \psi = \mathbb{1}_{\mathscr{G}}$$

Therefore since taking stalks is functorial for all $p \in X$ we have

$$\psi_p \circ \phi_p = \mathbb{1}_{\mathscr{F}_p} \qquad \phi_p \circ \psi_p = \mathbb{1}_{\mathscr{G}_p}$$

i.e: $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism with inverse ψ_p.

$\Longleftarrow$ Note that we only need to check the case where $\mathsf{C} = \mathsf{Set}$ since the functor $F \colon \mathsf{C} \to \mathsf{Set}$ reflects isomorphisms. To do this, we will see that for all open subsets $U \subseteq X$ the map $\phi(U) \colon \mathscr{F}(U) \to \mathscr{G}(U)$ is injective and surjective.
To see injectivity, let $s, t \in \mathscr{F}(U)$ such that $\phi(U)(s) = \phi(U)(t)$. For all $p \in U$ we have

$$\phi_p(s_p) = \big(\phi(U)(s)\big)_p = \big(\phi(U)(t)\big)_p = \phi_p(t_p)$$

Therefore since ϕ_p is injective $s_p = t_p$. Using sheaves are determined by its germs we get $s = t$ and thus $\phi(U)$ is injective.
To see surjectivity consider $t \in \mathscr{G}(U)$. For all $p \in U$ since ϕ_p is surjective there exists $\sigma_p \in \mathscr{F}_p$ such that $\phi_p(\sigma_p) = t_p$. By definition of stalk there exists an open neighbourhood $V_p \subseteq U$ of p and a section $s^{(p)} \in \mathscr{F}(V_p)$ such that $s^{(p)}_p = \sigma_p$.
By shrinking V_p if necessary we can assume that

$$\phi(V_p)(s^{(p)}) = t|_{V_p}$$

Now consider two such neighbourhoods V_p and V_q. On the intersection we have

$$\phi(V_p \cap V_q)(s^{(p)}|_{V_p \cap V_q}) = t|_{V_p \cap V_q} = \phi(V_p \cap V_q)(s^{(p)}|_{V_p \cap V_q})$$

Therefore since $\phi(V_p \cap V_q)$ is injective $s^{(p)}|_{V_p \cap V_q} = s^{(p)}|_{V_p \cap V_q}$. By the gluability axiom there exists a unique section $s \in \mathscr{F}(U)$ such that $s|_{V_p} = s^{(p)}$ for all $p \in U$.
Finally since $\phi(U)(s)|_{V_p} = \phi(V_p)(s^{(p)}) = t|_{V_p}$ for all $p \in U$ the uniqueness of the gluability axiom now on $\mathscr{G}$ implies that $\phi(U)(s) = t$ and thus $\phi(U)$ is surjective. □

Definition 8.1.6. Let X be a topological space.

1. A presheaf $\mathcal{F}$ is a **subpresheaf** of a presheaf $\mathcal{G}$ if for all open sets $U \subseteq X$ we have $\mathcal{F}(U) \subseteq \mathcal{G}(U)$ and the restriction maps of $\mathcal{G}$ induce the restriction maps of $\mathcal{F}$.
   If $\mathscr{F}$ and $\mathscr{G}$ are sheaves we say that $\mathscr{F}$ is a **subsheaf** of $\mathscr{G}$.
2. A morphism of presheaves $\phi \colon \mathcal{F} \to \mathcal{G}$ is **injective** if for all open sets $U \subseteq X$ the map $\mathcal{F}(U) \to \mathcal{G}(U)$ is injective.
3. A morphism of presheaves $\phi \colon \mathcal{F} \to \mathcal{G}$ is **surjective** if for all open sets $U \subseteq X$ the map $\mathcal{F}(U) \to \mathcal{G}(U)$ is surjective.
4. A morphism of sheaves $\varphi \colon \mathscr{F} \to \mathscr{G}$ is **injective** if for all open sets $U \subseteq X$ the map $\mathscr{F}(U) \to \mathscr{G}(U)$ is injective.
5. A morphism of sheaves $\varphi \colon \mathscr{F} \to \mathscr{G}$ is **surjective** if for all open sets $U \subseteq X$ and every section $s \in \mathscr{G}(U)$ there exists an open covering $U = \bigcup_{i \in I} U_i$ such that $s|_{U_i}$ is in the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$ for all i.

8.2 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a "forgetful" functor:

$$\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

We want to associate to each presheaf a sheaf.

Definition 8.2.1 (Sheafification). Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#} : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.
That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathcal{G})$$

The unit of this adjunction gives a canonical morphism $\theta : \mathcal{F} \mapsto \mathcal{F}^{\#}$ in $\mathsf{PSh}_X(\mathsf{C})$ which verifies that for any presheaf morphism $\phi : \mathcal{F} \to \mathcal{G}$ where $\mathcal{F}$ is a presheaf and $\mathcal{G}$ is a sheaf there exists a unique morphism $\psi : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi \circ \theta = \phi$.

Construction (Godement construction of sheafification). Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) := \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V} : \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a C-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.
Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i \in \mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i \in \mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$.
Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta : \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the subpresheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p\in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal propery of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi\colon \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi\colon \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$. Diagramatically

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\phi} & \mathcal{G} \\ {\scriptstyle\theta}\downarrow & \nearrow_{\psi} & \\ \mathcal{F}^\# & & \end{array}$$

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^\#(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $s|_{U_i} = \theta(t_i)$ for all $i \in \mathcal{I}$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define

$$\psi(\theta(t_i)) \coloneqq \phi(t_i) \in \mathcal{G}(U_i)$$

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y\colon \mathcal{F}_y \to \mathcal{G}_y$ we have

$$\phi_y((t_i)_y) = \phi_y((t_j)_y)$$

Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that

$$\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$$

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $t|_{U_i} = \phi(t_i)$ for all $i \in \mathcal{I}$. Therefore we can define $\psi_U(s) \coloneqq t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\psi_x = \phi_x$ for all $x \in X$, making it unique.

Lemma 8.2.1. Let $\mathscr{F}$ be a sheaf. Then $(\iota\mathscr{F})^\# \cong \mathscr{F}$, i.e: $-^\# \circ \iota \cong \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$.

Proof. Consider the identity map $\mathbb{1} : \iota\mathscr{F} \to \mathscr{F}$. Since $\iota\mathscr{F}$ is a presheaf and $\mathscr{F}$ is a sheaf by definition of sheafification there exists a unique morphism $\psi : (\iota\mathscr{F})^\# \to \mathscr{F}$ such that $\psi \circ \theta = \mathbb{1}$. Note that

$$(\theta \circ \psi) \circ \theta = \theta \circ (\psi \circ \theta) = \theta$$

Applying now the definition of sheafification to $\theta : \mathscr{F} \to (\iota\mathscr{F})^\#$ we get a unique map $h : (\iota\mathscr{F})^\# \to (\iota\mathscr{F})^\#$ such that $h \circ \theta = \theta$. Therefore since $\mathbb{1} \circ \theta = \theta$ and $(\theta \circ \psi) \circ \theta = \theta$ we have that $\theta \circ \psi = \mathbb{1}$, which means that θ is an isomorphism. □

Lemma 8.2.2. Let X be a topological space and $\mathcal{F}$ a C-valued presheaf over X. Then for all $p \in X$ there is bijection of stalks

$$\mathcal{F}_p \cong (\mathcal{F}^{\#})_p$$

Proof. Direct from the construction of sheafification. □

Remark 8.2.1. We will later give another proof of Lemma 8.2.2 that does not use the constructions of sheafification.

8.3 Sheaves in an abelian category

In this section, X will be a fixed topological space and C will be an abelian category for which sheafification exists.

Note that by Proposition 6.0.1 $\mathsf{PSh}_X(\mathsf{C})$ is an abelian category.

We are going to see that $\mathsf{Sh}_X(\mathsf{C})$ is an abelian category.

Consider the following category-theoretic auxiliary lemma.

Lemma 8.3.1. Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two functors. Suppose that

1. A and B are additive categories, a, b are additive functors and there is an adjunction $b \dashv a$.
2. B is an abelian category and b is left-exact.
3. $b \circ a \cong \mathbb{1}_{\mathsf{A}}$.

Then A is abelian.

Proof. Since B is an abelian by Lemma 6.0.1 all finite limits and colimits exist en B.

Since b is left-adjoint by Example 6.1.2 it is right-exact and therefore, exact. Analogously we have that a is left-exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. Since B is abelian there exist $\ker_{\mathsf{B}} \varphi$ and $\operatorname{coker}_{\mathsf{B}} \varphi$.

Since exact functors preserve kernels and cokernels we have that

$$b(\ker_{\mathsf{B}} \varphi) \cong \ker_{\mathsf{A}}(b\varphi) \qquad b(\operatorname{coker}_{\mathsf{B}} \varphi) \cong \operatorname{coker}_{\mathsf{A}}(b\varphi)$$

Therefore we have that every morphism of the form $b\varphi$ in A admits a kernel and a cokernel.

Note that, since $b \circ a \cong \mathbb{1}_{\mathsf{A}}$, every morphism ψ in A is of the form $\psi = b\varphi$. Thus all kernels and cokernels exist in A and

$$\ker_{\mathsf{A}} \psi \cong b \ker_{\mathsf{B}}(a\psi) \qquad \operatorname{coker}_{\mathsf{A}} \psi \cong b \operatorname{coker}_{\mathsf{B}}(a\psi) \tag{8.3.1}$$

It only remains to show that $\operatorname{Coim}_{\mathsf{A}} \psi \cong \operatorname{Im}_{\mathsf{A}} \psi$. Since A has all finite limits and colimits a is left exact.

By Lemma 6.1.1 we have that a transforms kernels into kernels. Therefore

$$\begin{aligned}
\operatorname{coim}_{\mathsf{A}}\psi &= \operatorname{coker}_{\mathsf{A}}\left(\ker_{\mathsf{A}}\psi \to A_1\right) && \text{(definition)}\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(a(\ker_{\mathsf{A}}\psi \to A_1)\right) && (8.3.1)\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(\ker_{\mathsf{B}}(a\psi) \to a(A_1)\right) && (a \text{ preserves kernels})\\
&= b\operatorname{Coim}_{\mathsf{B}}(a\psi) && \text{(definition)}\\
&\cong b\operatorname{Im}_{\mathsf{B}}(a\psi) && (\mathsf{B} \text{ is abelian})\\
&= b(\ker_{\mathsf{B}}\left(a(A_2) \to \operatorname{coker}_{\mathsf{B}}(a\psi)\right)) && \text{(definition)}\\
&\cong \ker_{\mathsf{A}}\left((b\circ a)(A_2) \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && (b \text{ preserves kernels})\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && b\circ a \cong \mathbb{1}_{\mathsf{A}}\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to \operatorname{coker}_{\mathsf{A}}\psi\right) && (8.3.1)\\
&= \operatorname{Im}_{\mathsf{A}}\psi && \text{(definition)}
\end{aligned}$$

□

Lemma 8.3.2. $\mathsf{Sh}_X(\mathsf{C})$ is an additive category and the forgetful functor $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is additive.

Proof. Let's check first that $\mathsf{Sh}_X(\mathsf{C})$ satisfies additive category axioms. Recall that Ad.3 is equivalent to Ad.4 by Lemma 5.0.1.

We will use the characterization of sheaves in terms of equalisers.

Ad.1 Since $\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G}) = \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G})$ and $\mathsf{PSh}_X(\mathsf{C})$ is additive, $\operatorname{Hom}_{\mathsf{Sh}_X}$ admits the structure of an abelian group.

Ad.2 Note that the zero object in $\mathsf{PSh}_X(\mathsf{C})$ is given by the constant presheaf $\underline{0}_X^{\mathrm{pre}}$. Note that the sequence

$$\underline{0}_X^{\mathrm{pre}}(U) = 0 \longrightarrow \prod_{i\in I}\underline{0}_X^{\mathrm{pre}}(U_i) = 0 \rightrightarrows \prod_{(i,j)\in I\times I}\underline{0}_X^{\mathrm{pre}}(U_i\cap U_j) = 0$$

is trivially an equaliser for all open covers $U = \bigcup_{i\in I} U_i$. Therefore $\underline{0}_X^{\mathrm{pre}}$ is a sheaf and thus $\mathsf{Sh}_X(\mathsf{C})$ admits a zero object.

Ad.4 Let $\mathscr{F},\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$. We define their product elementwise as

$$(\mathscr{F}\times\mathscr{G})(U) = \mathscr{F}(U)\times\mathscr{G}(U)$$

Since this coincides with the product in $\mathsf{PSh}_X(\mathsf{C})$ it is a presheaf. Furthermore for any open cover $U = \bigcup_{i\in I} U_i$ we have that

$$\begin{aligned}
(\mathscr{F}\times\mathscr{G})(U) = \mathscr{F}(U)\times\mathscr{G}(U) \to \prod_{i\in I}(\mathscr{F}\times\mathscr{G})(U_i) &= \prod_{i\in I}(\mathscr{F}(U_i))\times\prod_{i\in I}(\mathscr{G}(U_i))\\
\rightrightarrows \prod_{(i,j)\in I\times I}(\mathscr{F}\times\mathscr{G})(U_i\cap U_j) &= \prod_{i\in I}\left(\mathscr{F}(U_i\cap U_j)\right)\times\prod_{i\in I}\left(\mathscr{G}(U_i\cap U_j)\right)
\end{aligned}$$

is an equaliser since it can be seen as the "product" of two equalisers. Therefore $\mathscr{F}\times\mathscr{G}$ is a sheaf.

By the construction of the zero object and products in $\mathsf{Sh}_X(\mathsf{C})$ we have that ι preserves both. Therefore by Proposition 5.0.1 ι is additive. □

Proposition 8.3.1. Let C be an abelian category. The sheafification functor $-^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

Proof. Since $-^{\#}$ is left adjoint by Example 6.1.2 it is right exact.

To see that it is left exact we will use Lemma 6.1.1. In particular we are going to see that sheafification commutes with finite limits.

Let $\mathcal{F} : \mathsf{I} \to \mathsf{PSh}_X(\mathsf{C})$ be a diagram in $\mathsf{PSh}_X(\mathsf{C})$ with I finite.

We need to see that

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#} \cong \lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}$$

Note that since $\iota : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is right adjoint it preserves limits. Therefore limits of sheaves are computed exactly as limits of presheaves.

By definition of limit we have a morphism

$$\phi : \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#} \to \lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}$$

To see that this is an isomorphism we will see that the induced stalk maps ϕ_p are isomorphisms for all $p \in X$, i.e: we want to show that

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p$$

On the one hand since stalks commute with sheafification we have

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)_p$$

Furthermore since stalks commute with finite limits

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p$$

On the other hand since stalks commute with finite limits

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i^{\#})_p$$

Furthermore since stalks commute with sheafification we have

$$\lim_{i\in\mathsf{I}} \left((\mathcal{F}_i)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p$$

Therefore

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p$$

□

Theorem 8.3.1. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{C})$.

1. The category $\mathsf{Sh}_X(\mathsf{C})$ is abelian.
2. The kernel of φ is the same as the kernel of φ in $\mathsf{PSh}_X(\mathsf{Ab})$.

3. φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.
4. The cokernel of φ is the sheafification of the cokernel of φ in $\mathsf{PSh}_X(\mathsf{Ab})$.
5. φ is surjective if and only if φ is surjective as a map of sheaves.
6. A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in I}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$ if and only if for all $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact.

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 4, but these come directly from Lemma 8.3.1 with

$$\mathsf{A} := \mathsf{Sh}_X(\mathsf{C}) \quad \mathsf{B} := \mathsf{PSh}_X(\mathsf{B})$$
$$a := \iota \colon \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C}) \text{ forgetful} \quad b := -^{\#} \colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C}) \text{ sheafification}$$

We can indeed apply this lemma since

1. $\mathsf{Sh}_X(\mathsf{C})$ is additive by Lemma 8.3.2 and $\mathsf{PSh}_X(\mathsf{C})$ is additive by Lemma 5.0.3, ι is additive by Lemma 8.3.2 and $-^{\#}$ is additive by Proposition 5.0.2, and the adjunction $-^{\#} \dashv \iota$ is given by definition of sheafification.
2. $\mathsf{PSh}_X(\mathsf{C})$ is abelian by Proposition 6.0.1 and $-^{\#}$ is left exact by Proposition 8.3.1.
3. $-^{\#} \circ \iota = \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$ comes from Lemma 8.2.1. □

Chapter 9

Operations on sheaves

9.1 Continuous maps and (pre)sheaves

Consider a continuous map between topological spaces $f : X \to Y$.

We want to find a way of translating presheaves and sheaves from X to Y along f and viceversa.

We will start in the presheaf case and move to the sheaf case using sheafification.

Definition 9.1.1 (Direct image presheaf). Let $f : X \to Y$ be a continuous map. For any C-valued presheaf over X we define the **direct image** (or pushforward) of $\mathcal{F}$ via f as the C-valued presheaf $f_*\mathcal{F}$ over Y which, for any open subset $V \subseteq Y$

$$\left(f_*\mathcal{F}\right)(V) \coloneqq \mathcal{F}\left(f^{-1}(V)\right)$$

and for any $V_1 \subseteq V_2 \subseteq Y$ the restriction is given by

$$\begin{array}{ccc} f_*(\mathcal{F}(V_2)) & = & \mathcal{F}(f^{-1}(V_2)) \\ \vdots & & \downarrow \\ f_*(\mathcal{F}(V_1)) & = & \mathcal{F}(f^{-1}(V_1)) \end{array}$$

Note that this defines a functor

$$f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$$

Lemma 9.1.1. Let $f : X \to Y$ and $g : Y \to Z$ be two continuous maps. Then the direct images $(g \circ f)_*$ and $g_* \circ f_*$ are equal.

Proof. Let $\mathcal{F}$ be a presheaf over X and $W \subseteq Z$ an open subset. On the one hand

$$(g \circ f)_*(\mathcal{F}(W)) = \mathcal{F}((g \circ f)^{-1}(W)) = \mathcal{F}((f^{-1} \circ g^{-1})(W))$$

On the other

$$(g_* \circ f_*)(\mathcal{F}(W)) = f_*(\mathcal{F}(g^{-1}(W))) = \mathcal{F}((f^{-1} \circ g^{-1})(W)) \qquad \square$$

Lemma 9.1.2. If $\mathscr{F}$ is a sheaf then $f_*\mathscr{F}$ is a sheaf.

Proof. This follows since for any open cover $V = \bigcup_i V_i$ we have

$$f^{-1}\left(\bigcup_i V_i\right) = \bigcup_i f^{-1}(V_i)$$ □

This means that we have a functor

$$f_* \colon \mathsf{Sh}_X(\mathsf{C}) \longrightarrow \mathsf{Sh}_Y(\mathsf{C})$$

which verifies the presheaf composition law.

Definition 9.1.2 (Inverse image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. For any C-valued presheaf $\mathcal{G}$ over Y we define the **inverse image** (or pullback) of $\mathcal{G}$ via f as the C-valued presheaf $f_{\mathrm{pre}}^{-1}\mathcal{G}$ over X which, for any open subset $U \subseteq X$

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \coloneqq \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V)$$

Since this is a direct limit following Example 3.2.5 we may describe $\mathcal{F}_p$ as

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) = \{(s, V) \mid f(U) \subseteq V \in \mathsf{Open}(X) \wedge s \in \mathcal{G}(V)\} \Big/ \sim$$

where

$$(s, V) \sim (t, W) \iff \exists Z \subseteq V \cap W \mid f(U) \subseteq Z \in \mathsf{Open}(Y) \wedge s|_Z = t|_Z$$

In this way, if $U_1 \subseteq U_2 \subseteq X$ are open subsets of X the restriction is given by

$$\mathrm{res}_{U_2,U_1}\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)([s, V]) = [(s, V)]$$

Note that this defines a functor $f_{\mathrm{pre}}^{-1} \colon \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.

Theorem 9.1.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category. Then there exists an adjunction

$$f_{\mathrm{pre}}^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f_{\mathrm{pre}}^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. We want to construct natural transformations

$$\eta \colon \mathbb{1}_{\mathsf{PSh}_Y(\mathsf{C})} \implies f_* \circ f_{\mathrm{pre}}^{-1}$$
$$\varepsilon \colon f_{\mathrm{pre}}^{-1} \circ f_* \implies \mathbb{1}_{\mathsf{PSh}_X(\mathsf{C})}$$

making for all $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ the following diagrams commute

$$\begin{array}{ccc} f_{\mathrm{pre}}^{-1}(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{G}}} & f_{\mathrm{pre}}^{-1}(\mathcal{G}) \\ {\scriptstyle f_{\mathrm{pre}}^{-1}(\eta_{\mathcal{G}})}\downarrow & \nearrow_{\varepsilon_{f_{\mathrm{pre}}^{-1}(\mathcal{G})}} & \\ \left(f_{\mathrm{pre}}^{-1} \circ f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G}) & & \end{array} \qquad \begin{array}{ccc} f_*(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{F}}} & f_*(\mathcal{G}) \\ {\scriptstyle \eta_{f_*(\mathcal{F})}}\downarrow & \nearrow_{f_*(\varepsilon_{\mathcal{F}})} & \\ \left(f_* \circ f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F}) & & \end{array}$$

Note that, by definition of the direct image presheaf and inverse image presheaf we have, for $U \in \mathsf{Open}(X)$ and $V \in \mathsf{Open}(Y)$

$$\left(f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F})(U) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} (f_*\mathcal{F})(W) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} \mathcal{F}(f^{-1}(W))$$

$$\left(f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G})(V) = (f_{\mathrm{pre}}^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Open}(Y)}} \mathcal{G}(W)$$

Since we have $(f \circ f^{-1})(V) \subseteq V$, we can define $\eta_{\mathcal{G}}(V)$ as the canonical injection into the direct limit

$$\eta_{\mathcal{G}}(V) \colon \mathcal{G}(V) \to \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Open}(Y)}} \mathcal{G}(W)$$

On the other hand, since for every $W \subseteq f(U)$ open we have $f^{-1}(W) \subseteq (f^{-1} \circ f)(U) \subseteq U$ there exists a restriction morphism

$$\mathrm{res}_{f^{-1}(W),U} \colon \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

Using these maps we obtain a morphism

$$\varepsilon_{\mathcal{F}}(U) \colon \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Open}(Y)}} \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U) \qquad \square$$

Lemma 9.1.3. Let $f : X \to Y$ be a continuous map. Let $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ be a C-valued presheaf over Y. Then for all $p \in X$ there is a bijection of stalks.

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)_p \cong \mathcal{G}_{f(p)}$$

Proof.

$$\begin{aligned}
\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)_p &= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Open}(X)}} \left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \\
&= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Open}(X)}} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V) \right) \\
&= \varinjlim_{\substack{V \ni f(p) \\ V \in \mathsf{Open}(Y)}} \mathcal{G}(V) \\
&= \mathcal{G}_{f(p)}
\end{aligned}$$

$\square$

Corollary 9.1.1. Let C be a abelian category. For all continuous maps $f : X \to Y$ the presheaf inverse image $f_{\mathrm{pre}}^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is exact.

Lemma 9.1.4. Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps. Then the inverse images $(g \circ f)^{-1}_{\text{pre}}$ and $f^{-1}_{\text{pre}} \circ g^{-1}_{\text{pre}}$ coincide.

Proof. Using the adjunction $f^{-1}_{\text{pre}} \dashv f_*$ and Lemma 9.1.1 we have, on the one hand

$$(g \circ f)^{-1}_{\text{pre}} \dashv (g \circ f)_* = g_* \circ f_*$$

On the other hand note that

$$f^{-1}_{\text{pre}} \circ g^{-1}_{\text{pre}} \dashv g_* \circ f_*$$

Indeed

$$\begin{aligned} \text{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(f^{-1}_{\text{pre}} \circ g^{-1}_{\text{pre}}\right)(\mathcal{H}), \mathcal{F}\right) &\cong \text{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}\left(g^{-1}_{\text{pre}}(\mathcal{H}), f_*(\mathcal{F})\right) \\ &\cong \text{Hom}_{\mathsf{PSh}_Z(\mathsf{C})}\left(\mathcal{H}, \left(g_* \circ f_*\right)(\mathcal{F})\right) \end{aligned}$$

Therefore since adjoints are unique we have $(g \circ f)^{-1}_{\text{pre}} = f^{-1}_{\text{pre}} \circ g^{-1}_{\text{pre}}$. □

Definition 9.1.3. Let $f : X \to Y$ be a continuous map. The **inverse image** (or pullback) sheaf via f is the functor

$$f^{-1} := -^{\#} \circ f^{-1}_{\text{pre}} \circ \iota : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

where $-^{\#}$ is the sheafification functor, f^{-1}_{pre} is the inverse image of presheaves and $\iota : \mathsf{Sh}_Y(\mathsf{C}) \hookrightarrow \mathsf{PSh}_Y(\mathsf{C})$ is the forgetful functor.

Theorem 9.1.2. Let $f : X \to Y$ be a continuous map There exists an adjunction

$$f^{-1} \dashv f_* : \mathsf{Sh}_X(\mathsf{C}) \underset{f_*}{\overset{f^{-1}}{\leftrightarrows}} \mathsf{Sh}_Y(\mathsf{C})$$

between the direct and inverse images of sheaves.

Proof. Using the adjunctions $-^{\#} \dashv \iota$ and $f^{-1}_{\text{pre}} \dashv f_*$ we have

$$\begin{aligned} \text{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}(\mathscr{G}), \mathscr{F}) &= \text{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(\left(-^{\#} \circ f^{-1}_{\text{pre}} \circ \iota\right)(\mathscr{G}), \mathscr{F}\right) \\ &\cong \text{Hom}_{\mathsf{PSh}_X(\mathsf{C})}((f^{-1}_{\text{pre}} \circ \iota)(\mathscr{G}), \iota(\mathscr{F})) \\ &= \text{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}(\iota(\mathscr{G}), (f_* \circ \iota)(\mathscr{F})) \\ &= \text{Hom}_{\mathsf{PSh}_Y(\mathsf{C})}(\iota(\mathscr{G}), (\iota \circ f_*)(\mathscr{F})) \\ &= \text{Hom}_{\mathsf{Sh}_Y(\mathsf{C})}(\mathscr{G}, f_*(\mathscr{F})) \end{aligned}$$

□

Lemma 9.1.5. Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps. Then the inverse images $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ coincide.

Proof. Analogous to Lemma 9.1.4 □

Lemma 9.1.6. Let $f : X \to Y$ be a continuous map. Let $\mathscr{G} \in \mathsf{Sh}_Y(\mathsf{C})$ be a C-valued sheaf over Y. Then for all $p \in X$ there is a bijection of stalks.

$$\left(f^{-1}\mathscr{G}\right)_p \cong \mathscr{G}_{f(p)}$$

Proof.

$$\begin{aligned}
\left(f^{-1}\mathscr{G}\right)_p &= \left(\left(f^{-1}_{\mathrm{pre}}(\iota\mathscr{G})\right)^{\#}\right)_p && \text{(definition)}\\
&= \left(f^{-1}_{\mathrm{pre}}(\iota\mathscr{G})\right)_p && \text{(Lemma 8.2.2)}\\
&= (\iota\mathscr{G})_{f(p)} && \text{(Lemma 9.1.3)}\\
&= \mathscr{G}_{f(p)} && \text{(Lemma 8.2.1 and Lemma 8.2.2)}
\end{aligned}$$

□

Corollary 9.1.2. Let C be a abelian category. For all continuous maps $f : X \to Y$ the sheaf inverse image $f^{-1}\colon \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

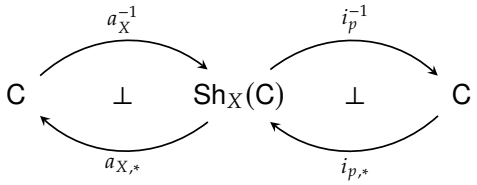
9.1.1 Particular cases

Let X be a topological space and $\{*\}$ the one-point topological space.

Given $p \in X$ we can construct continuous maps

$$\begin{aligned} a_X\colon X &\to \{*\} & i_p\colon \{*\} &\to X\\ x &\mapsto * & * &\mapsto p \end{aligned}$$

Since $\mathsf{Sh}_{\{*\}}(\mathsf{C}) \simeq \mathsf{C}$ we get two pairs of adjoint functors



Definition 9.1.4 (Global sections). The **global sections** functor is

$$\Gamma(-;X) \coloneqq a_{X,*}\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$$

Less abstractly, it sends a C-valued sheaf $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ to $\Gamma(\mathscr{F};X) = \mathscr{F}(X)$ and a morphism of sheaves $\phi : \mathscr{F} \to \mathscr{G}$ to $\Gamma(\phi, X) = \mathscr{F}(X) \xrightarrow{\phi(X)} \mathscr{G}(X)$.

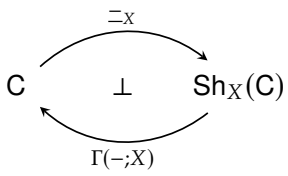
Definition 9.1.5 (Constant sheaf). The **constant sheaf** functor is

$$\underline{\ }_X \coloneqq a_X^{-1}\colon \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$$

Less abstractly, it sends an object $A \in \mathsf{C}$ to the sheaf

Complete. Should it be a remark/lemma that this is the sheafification of the constant presheaf?

Corollary 9.1.3. There is an adjunction



Proof. Direct from the adjunction $f^{-1} \dashv f_*$. □

Definition 9.1.6 (Skyscraper sheaf). The **skyscraper sheaf** functor is

$$\mathrm{sk}_p := i_{p,*} \colon \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$$

Less abstractly, it sends an object $A \in \mathsf{C}$ to the sheaf

$$(\mathrm{sk}_p\, A)(U) = \begin{cases} A & p \in U \\ 1 & p \notin U \end{cases}$$

for all open subsets $U \subseteq X$ with restriction maps for $V \subseteq U$

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_A & p \in V \\ \mathbb{1}_A & p \notin U \\ A \to 1 & p \in U \wedge p \notin V \end{cases}$$

The last functor i_p^{-1} is one we have already seen.

Lemma 9.1.7. For all points $p \in X$ we have

$$-_p \cong i_p^{-1}$$

Proof. Identifying $\mathsf{Sh}_{\{*\}}(\mathsf{C}) \simeq \mathsf{C}$ if $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ is a C-valued presheaf we have

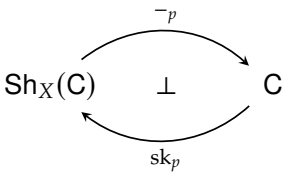
$$\left(i_{p,\mathrm{pre}}^{-1}\mathcal{F}\right)(\{*\}) = \varinjlim_{\substack{V \supseteq i_p(\{*\}) \\ V \in \mathsf{Open}(X)}} \mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(X)}} \mathcal{F}(V) = \mathcal{F}_p$$

Therefore if $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ is a C-valued sheaf we have

$$\begin{aligned} \left(i_p^{-1}\mathscr{F}\right)(\{*\}) &= \left(-^{\#} \circ i_{p,\mathrm{pre}}^{-1} \circ \iota\right)(\mathscr{F})(\{*\}) \\ &= \left((\iota(\mathscr{F}))^{\#}\right)_p \\ &= (\iota(\mathscr{F}))_p \\ &= \mathscr{F}_p \end{aligned}$$

□

Corollary 9.1.4. There is an adjunction

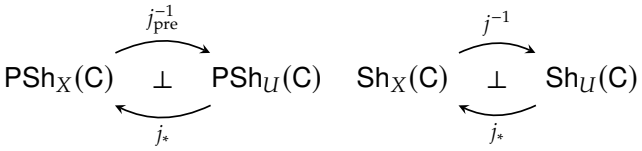


Proof. Direct from the adjunction $f^{-1} \dashv f_*$. □

9.2 Open immersions and (pre)sheaves

Let's consider now an open immersion $j \colon U \hookrightarrow X$, where X is a topological space and $U \subseteq X$ is an open subset.

In section 9.1 we have constructed two pairs of adjoint functors



Namely, the direct and inverse images of (pre)sheaves.

Lemma 9.2.1. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion

1. Let $\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$ be a C-valued presheaf over X.
   Then $j_{\mathrm{pre}}^{-1}\mathcal{G} \in \mathsf{PSh}_U(\mathsf{C})$ is given by, for any open subset $V \subseteq U$
   $$j_{\mathrm{pre}}^{-1}\mathcal{G}(V) = \mathcal{G}(V)$$
   Furthermore $j_{\mathrm{pre}}^{-1} \circ j_* = \mathbb{1}_{\mathsf{PSh}_U(\mathsf{C})}$.
2. Let $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$ be a C-valued sheaf over X.
   Then $j^{-1}\mathcal{G} \in \mathsf{Sh}_U(\mathsf{C})$ is given by, for any subset $V \subseteq U$
   $$j^{-1}\mathcal{G}(V) = \mathcal{G}(V)$$
   Furthermore $j^{-1} \circ j_* = \mathbb{1}_{\mathsf{Sh}_U(\mathsf{C})}$.

Proof. By definition we have

$$j_{\mathrm{pre}}^{-1}\mathcal{G}(V) = \varinjlim_{\substack{W \supseteq j(V) \\ W \in \mathsf{Open}(U)}} \mathcal{G}(W) = \varinjlim_{\substack{W \supseteq V \\ W \in \mathsf{Open}(U)}} \mathcal{G}(W)$$

Therefore since V is open this last direct limit is exactly $\mathcal{G}(V)$.

Furthermore if $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ for any open subset $V \subseteq U$ we have

$$\left(j_{\mathrm{pre}}^{-1} \circ j_*\right)\mathcal{F}(V) = j_{\mathrm{pre}}^{-1}\mathcal{F}(j^{-1}(V)) = \mathcal{F}(j^{-1}(V)) = \mathcal{F}(V)$$

The statement about sheaves follows directly since $V \mapsto \mathcal{G}(V)$ is clearly a sheaf if $\mathcal{G}$ is a sheaf. □

Lemma 9.2.2. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion
Let $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$ be a C-valued sheaf over X. For all $p \in U$ we have

$$\left(j^{-1}\mathcal{G}\right)_p = \mathcal{G}_p$$

Proof.

$$\left(j^{-1}\mathcal{G}\right)_p = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(U)}} j^{-1}\mathcal{G}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(U)}} \mathcal{G}(V) = \varinjlim_{\substack{V \ni p \\ V \in \mathsf{Open}(X)}} \mathcal{G}(V) = \mathcal{G}_p$$ □

Definition 9.2.1. Let X be a topological space and $j\colon U \hookrightarrow X$ an open immersion.

1. If $\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$ is a C-valued presheaf over X we call $j_{\mathrm{pre}}^{-1}\mathcal{G}$ the **restriction** of $\mathcal{G}$ to U, and denote it by $\mathcal{G}|_U$.
2. If $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$ is a C-valued sheaf over X we call $j^{-1}\mathcal{G}$ the **restriction** of $\mathcal{G}$ to U, and denote it by $\mathcal{G}|_U$.

Definition 9.2.2. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion. Let $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ be a C-valued presheaf over U. Suppose that C has an initial object $e \in \mathsf{C}$.
The **extension** $j^{\mathrm{pre}}_! \mathcal{F}$ of $\mathcal{F}$ by e is the C-valued presheaf over X given by, for any open subset $V \subseteq X$

$$j^{\mathrm{pre}}_! \mathcal{F}(V) \coloneqq \begin{cases} e & V \not\subseteq U \\ \mathcal{F}(V) & V \subseteq U \end{cases}$$

The restriction maps are given by, for any $W \subseteq V \subseteq X$ open subsets

$$\mathrm{res}_{V,W} \coloneqq \begin{cases} \mathrm{res}^U_{V,W} & V \subseteq U \\ \mathbb{1}_e & V \not\subseteq U \\ e \to \mathcal{F}(W) & W \subseteq U \subseteq V \end{cases}$$

where $\mathrm{res}^U_{V,W}$ is the restriction in $\mathsf{PSh}_U(\mathsf{C})$ and $e \to \mathcal{F}(W)$ the unique map from the initial object e.
Note that this defines a functor

$$j^{\mathrm{pre}}_! \colon \mathsf{PSh}_U(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$$

Lemma 9.2.3. Let $j\colon U \hookrightarrow X$ be an open immersion. For any C-valued presheaf $\mathcal{F}$ the stalk at a point $p \in X$ of the extension $j^{\mathrm{pre}}_! \mathcal{F}$ is

$$\left(j^{\mathrm{pre}}_! \mathcal{F}\right)_p = \begin{cases} e & p \notin U \\ \mathcal{F}_p & p \in U \end{cases}$$

Proof. By definition of stalk

$$\left(j^{\mathrm{pre}}_! \mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j^{\mathrm{pre}}_! \mathcal{F}(V)$$

Therefore if $p \notin U$

$$\left(j^{\mathrm{pre}}_! \mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j^{\mathrm{pre}}_! \mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} e = e$$

On the other hand if $p \in U$

$$\left(j^{\mathrm{pre}}_! \mathcal{F}\right)_p = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} j^{\mathrm{pre}}_! \mathcal{F}(V) = \varinjlim_{\substack{V \ni p \\ U \in \mathsf{Open}(U)}} \mathcal{F}(V) = \mathcal{F}_p$$

□

Corollary 9.2.1. If C is abelian then the extension by zero functor of presheaves $j^{\mathrm{pre}}_! \colon \mathsf{PSh}_U(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is exact.

Theorem 9.2.1. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.
There is a adjunction

$$\mathsf{PSh}_U(\mathsf{C}) \underset{j_{\mathrm{pre}}^{-1}}{\overset{j_!^{\mathrm{pre}}}{\rightleftarrows}} \mathsf{PSh}_X(\mathsf{C}) \quad \bot$$

Proof. Let $\mathcal{F} \in \mathsf{PSh}_U(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$. We want to construct an isomorphism

$$\Phi\colon \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(j_!^{\mathrm{pre}}(\mathcal{F}), \mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}(\mathcal{F}, j_{\mathrm{pre}}^{-1}(\mathcal{G}))$$

Let $\phi\colon j_!^{\mathrm{pre}}(\mathcal{F}) \to \mathcal{G}$ be a morphism of presheaves over X. We want to define a morphism of presheaves $\Phi(\phi)\colon \mathcal{F} \to j_{\mathrm{pre}}^{-1}(\mathcal{G})$ over U. Since j is an open map by Lemma 9.2.1 we have $j_{\mathrm{pre}}^{-1}\mathcal{F}(V) = \mathcal{F}(V)$ for all open subsets $V \subseteq X$. Therefore since ϕ provides a component map $\phi_V\colon j_!\mathcal{F}(V) \to \mathcal{G}(V)$ we can define

$$\Phi(\phi)_V \coloneqq \phi_V$$

Since ϕ is a presheaf morphism over X it commutes with the restriction maps for all open subsets of X. In particular, it commutes with the restriction maps for all open subsets of U and thus $\Phi(\phi)$ is a valid morphism of presheaves over U.

Now let $\psi\colon \mathcal{F} \to j_{\mathrm{pre}}^{-1}(\mathcal{G})$ be a morphism of presheaves over U. We want to define a morphism of presheaves $\Psi(\psi)\colon j_!^{\mathrm{pre}}(\mathcal{F}) \to \mathcal{G}$ over X. For any open set $W \subseteq X$ we define the component $\Psi(\psi)_W\colon j_!^{\mathrm{pre}}\mathcal{F}(W) \to \mathcal{G}(W)$ componentwise as

$$\Psi(\psi)_W \coloneqq \begin{cases} \psi_W & W \subseteq U \\ e \to \mathcal{G}(W) & W \nsubseteq U \end{cases}$$

Since this commutes with the restriction maps it is a valid morphism of presheaves over X.

It only remains to see that Φ and Ψ are inverses. If $V \subseteq U$ is an open subset then

$$(\Phi \circ \psi)(\psi)_V = \Phi(\psi_V) = \psi_V$$

On the other hand if $W \subseteq X$ then we distinguish two cases. If $W \subseteq U$ then

$$(\Psi \circ \Phi)(\phi)_W = \Psi(\phi_W) = \phi_W$$

If $W \nsubseteq U$ then both $(\Psi \circ \Phi)(\phi)_W$ and ϕ_W are maps from the initial object e to $\mathcal{G}(W)$ and therefore coincide. □

Definition 9.2.3. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.
The sheaf **restriction** by e functor is

$$j_! \coloneqq -^{\#} \circ j_!^{\mathrm{pre}} \circ \iota : \mathsf{Sh}_U(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

Lemma 9.2.4. Let $j\colon U \hookrightarrow X$ be an open immersion. For any C-valued sheaf $\mathscr{F}$ over U the stalk at a point $p \in X$ of the extension $j_!\mathscr{F}$ is

$$(j_!\mathscr{F})_p = \begin{cases} e & p \notin U \\ \mathscr{F}_p & p \in U \end{cases}$$

Proof. Analogous to Lemma 9.2.3 □

Corollary 9.2.2. If C is abelian then the extension by zero functor of sheaves $j_!\colon \mathsf{Sh}_U(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

Theorem 9.2.2. Let X be a topological space and $j\colon U \hookrightarrow X$ be an open immersion. Suppose that C has an initial object $e \in \mathsf{C}$.
There is a adjunction

$$\mathsf{Sh}_U(\mathsf{C}) \underset{j^{-1}}{\overset{j_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathsf{Sh}_X(\mathsf{C})$$

Proof. For all $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ and $\mathscr{G} \in \mathsf{Sh}_U(\mathsf{C})$ we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(j_!(\mathscr{F}), \mathscr{G}\right) &\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(-^{\#} \circ j_!^{\mathrm{pre}} \circ \iota\right)(\mathscr{F}), \mathscr{G}\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\left(j_!^{\mathrm{pre}} \circ \iota\right)(\mathscr{F}), \iota(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}\left(\iota(\mathscr{F}), \left(j_{\mathrm{pre}}^{-1} \circ \iota\right)(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{PSh}_U(\mathsf{C})}\left(\iota(\mathscr{F}), \left(\iota \circ j^{-1}\right)(\mathscr{G})\right) \\
&\cong \mathrm{Hom}_{\mathsf{Sh}_U(\mathsf{C})}\left(\mathscr{F}, j^{-1}(\mathscr{G})\right)
\end{aligned}$$

□

9.3 Closed immersions and (pre)sheaves

Let's consider now a closed immersion $i\colon Z \hookrightarrow X$, where X is a topological space and $Z \subseteq X$ is an closed subset.

In section 9.1 we have constructed two pairs of adjoint functors

$$\mathsf{PSh}_X(\mathsf{C}) \underset{i_*}{\overset{i_{\mathrm{pre}}^{-1}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathsf{PSh}_Z(\mathsf{C}) \qquad \mathsf{Sh}_X(\mathsf{C}) \underset{i_*}{\overset{i^{-1}}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} \mathsf{Sh}_Z(\mathsf{C})$$

Namely, the direct and inverse images of (pre)sheaves.

Lemma 9.3.1. Let $\mathscr{F}$ be a C-valued sheaf. The stalks of $i_*\mathscr{F}$ at a point $p \in X$ is

$$(i_*\mathscr{F})_p = \begin{cases} 1 & p \notin Z \\ \mathscr{F}_p & p \in Z \end{cases}$$

Proof. If $p \notin Z$ then there exist arbitrarilly small open neighbourhoods U of p that do not intersect Z. Therefore since $\mathscr{F}$ is a sheaf $\mathscr{F}(i^{-1}(U)) = 1$, proving the first case.

If $p \in Z$ then any neighbourhood of p is of the form $U \cap Z$ with U a neighbourhood of p in X. □

Corollary 9.3.1. If C is abelian then the direct image sheaf functor

$$i_*\colon \mathsf{Sh}_Z(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

is exact.

Definition 9.3.1. Let $\mathscr{F}$ be a C-valued sheaf over X. The **support** of $\mathscr{F}$ is

$$\operatorname{supp}\mathscr{F} := \{p \in X \mid \mathscr{F}_p \neq 1\}$$

If $s \in \mathscr{F}(X)$ is a section then its support is

$$\operatorname{supp} s := \{p \in X \mid s_x \neq 1 \text{ in } \mathscr{F}_x\}$$

Proposition 9.3.1. The direct image sheaf functor

$$i_*\colon \mathsf{Sh}_Z(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$$

is fully faithfull. Furthermore, its essential image consists of those sheaves whose support is contained in Z.

Proof. Let $\mathscr{F} \in \mathsf{Sh}_Z(\mathsf{C})$. Consider the map

$$i^{-1}i_*\mathscr{F} \to \mathscr{F}$$

Since by Lemma 9.3.1 this is an isomorphism for all $p \in Z$ by Proposition 8.1.1 it is a isomorphism. Therefore $i^{-1}i_* \cong \mathbb{1}_{\mathsf{Sh}_Z(\mathsf{C})}$, which means that i_* is fully faithfull.

To see the essential image part, we are going to see that $\mathscr{G} \in \operatorname{Im} i_*$ if and only if $\mathscr{G}_p = 1$ for all $p \in X \setminus Z$. Note that the direction $\Longrightarrow$ is direct from Lemma 9.3.1. To see $\Longleftarrow$, suppose that $\mathscr{G}$ verifies $\mathscr{G}_p = 1$ for all $p \in X \setminus Z$. Then the map

$$\mathscr{G} \to i_*i^{-1}\mathscr{G}$$

is an isomorphism on all stalks and thus by Proposition 8.1.1 it is an isomorphism. □

Definition 9.3.2. Let X be a topological space and $Z \subseteq X$ a closed subset. For any C-valued sheaf $\mathscr{F}$ over X, the **extraordinary inverse image** is the subsheaf $i^!\mathscr{F}$ of $\mathscr{F}$ defined as, for any open subset $U \subseteq X$

$$\left(i^!\mathscr{F}\right)(U) := \{s \in \mathscr{F}(U) \mid \operatorname{supp} s \subseteq Z \cap U\}$$

Note that this is the largest subsheaf of $\mathscr{F}$ whose support is contained in Z.
By Proposition 9.3.1 we can view this as a functor

$$i^!\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Z(\mathsf{C})$$

Theorem 9.3.1. Let X be a topological space and $i\colon Z \hookrightarrow X$ be a closed immersion. If C is abelian, there is a adjunction

$$\mathsf{Sh}_Z(\mathsf{C}) \underset{i^!}{\overset{i_*}{\rightleftarrows}} \mathsf{Sh}_X(\mathsf{C}) \qquad (\perp)$$

Proof. We want to show that, for any C-valued sheaf $\mathscr{G} \in \mathsf{Sh}_Z(\mathsf{C})$ and $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(i_*\mathscr{G}, \mathscr{F}\right) \cong \operatorname{Hom}_{\mathsf{Sh}_Z(\mathsf{C})}\left(\mathscr{G}, i^!\mathscr{F}\right)$$

Consider a morphism $\phi\colon i_*\mathscr{G} \to \mathscr{F}$. We want to construct a morphism $\psi\colon \mathscr{G} \to i^!\mathscr{F}$.

For any open subset $U \subseteq X$ we have a morphism in C

$$\phi_U : \mathscr{G}(U \cap Z) \to \mathscr{F}(U)$$

Let $W \subseteq Z$ be an open set and write $W = U \cap Z$ for some $U \subseteq X$ open. We want to show that the image of ϕ_U lands on $i^!\mathscr{F}(U)$. Let $s \in \mathscr{G}(U \cap Z)$ and consider its restriction to the open set $U \setminus Z$ of X. By the commutativity of the restriction maps with ϕ we get

$$\big(\phi_U(s)\big)|_{U\setminus Z} = \phi_{U\setminus Z}(s|_{(U\setminus Z)\cap Z})$$

Since $(U \setminus Z) \cap Z = \varnothing$ we have

$$\big(\phi_U(s)\big)|_{U\setminus Z} = \phi_{U\setminus Z}(s|_{(U\setminus Z)\cap Z}) = \phi_{U\setminus Z}(0) = 0$$

which means that

$$\operatorname{supp}(\phi_U(s)) \subseteq U \cap Z = W$$

Therefore since the image of ϕ_U is contained in $i^!\mathscr{F}(U)$ it naturally induces a map

$$\psi_W : \mathscr{G}(W) \to i^!\mathscr{F}(W)$$

Since by construction these maps commute with the restrictions, these maps define a morphism $\psi : \mathscr{G} \to i^!\mathscr{F}$.

Conversely, suppose now that we have a morphism $\psi : \mathscr{G} \to i^!\mathscr{F}$. We now want to construct a morphism $\phi : i_*\mathscr{G} \to \mathscr{F}$.

For any open subset $V \subseteq X$ let $H = V \cap Z$. We have a morphism in C

$$\psi_H : \mathscr{G}(H) \to i^!\mathscr{F}(H)$$

By composing ψ_H with the inclussion $i^!\mathscr{F}(H) \hookrightarrow \mathscr{F}(V)$ we obtain a map $\phi_V : \mathscr{G}(V \cap Z) \to \mathscr{F}(V)$. Therefore since these maps commute with the restrictions they assemble into a morphism $\phi : i_*\mathscr{G} \to \mathscr{F}$. □

Theorem 9.3.2. Let X be a topological space and $U \subseteq X$ be an open subset. Consider the open and closed immersions $j : U \hookrightarrow X$ and $i : X \setminus U \hookrightarrow X$.
If C is abelian for all C-valued sheaves $\mathscr{F} \in \mathsf{Sh}_X(\mathsf{C})$ there exists a short exact sequence

$$0 \longrightarrow j_!j^{-1}\mathscr{F} \longrightarrow \mathscr{F} \longrightarrow i_*i^{-1}\mathscr{F} \longrightarrow 0$$

Furthermore for any morphism of sheaves $\mathscr{F} \to \mathscr{G}$ we get a morphism of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & j_!j^{-1}\mathscr{F} & \longrightarrow & \mathscr{F} & \longrightarrow & i_*i^{-1}\mathscr{F} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & j_!j^{-1}\mathscr{G} & \longrightarrow & \mathscr{G} & \longrightarrow & i_*i^{-1}\mathscr{G} & \longrightarrow & 0 \end{array}$$

i.e: this construction is functorial.

Proof. Using the characterization of the stalks at Lemma 9.1.6, Lemma 9.2.2, Lemma 9.2.3 and Lemma 9.3.1 we have

$$\big(j_!j^{-1}\mathscr{F}\big)_p = \begin{cases} 0 & p \in X \setminus U \\ \big(j^{-1}\mathscr{F}\big)_p & p \in U \end{cases} = \begin{cases} 0 & p \in X \setminus U \\ \mathscr{F}_p & p \in U \end{cases}$$

$$\big(i_*i^{-1}\mathscr{F}\big)_p = \begin{cases} 0 & p \in X \setminus U \\ \big(i^{-1}\mathscr{F}\big)_p & p \in U \end{cases} = \begin{cases} 0 & p \in X \setminus U \\ \mathscr{F}_p & p \in U \end{cases}$$

Therefore for all $p \in X$ the sequence

$$0 \longrightarrow \left(j_! j^{-1}\mathscr{F}\right)_p \longrightarrow \mathscr{F}_p \longrightarrow \left(i_* i^{-1}\mathscr{F}\right)_p \longrightarrow 0$$

is exact and therefore so is

$$0 \longrightarrow j_! j^{-1}\mathscr{F} \longrightarrow \mathscr{F} \longrightarrow i_* i^{-1}\mathscr{F} \longrightarrow 0$$

The functoriallity comes from the naturality of the adjunctions $j_! \dashv j^{-1} \dashv j_*$ and $i^{-1} \dashv i_* \dashv i^!$. □

PART IV

Sheaf cohomology

Chapter 10

Cohomology of sheaves

In what follows, let $\mathbf{k}$ be a fixed commutative ring.

We want to define the cohomology of X with coefficients in a sheaf of $\mathbf{k}$-modules as the right derived functor of the global sections functor

$$\Gamma(X, -)\colon \mathsf{Mod}(\underline{\mathbf{k}}_X) \to \mathsf{Mod}(\mathbf{k})$$

To do this, we need to check the following:

1. $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ and $\mathsf{Mod}(\mathbf{k})$ are abelian categories.
2. The functor $\Gamma(X, -)$ is left exact.
3. $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ has enough injectives.

Note that $\mathsf{Mod}(\mathbf{k})$ is an abelian category. Therefore by Theorem 8.3.1 $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ is also abelian.

Recall that by Corollary 9.1.3 $\Gamma(X, -)$ is right adjoint. Therefore by Example 6.1.2 $\Gamma(X, -)$ is left exact.

It only remains to show that $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ has enough injectives.

Lemma 10.0.1. The category $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ of sheaves of $\mathbf{k}$-modules over X has enough injectives.

Proof. We will use the following claim

> The category $\mathsf{Mod}(R)$ of modules over a ring R has enough injectives.

For a proof, see [Alu09, Corollary VIII.6.4].

Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ and consider its stalk $\mathscr{F}_p \in \mathsf{Mod}(\mathbf{k})$ at a point $p \in X$.

Since $\mathsf{Mod}(\mathbf{k})$ has enough injectives we can choose an object I_p and a monomorphism

$$i_p\colon \mathscr{F}_p \rightarrowtail I_p$$

Consider the sheaf

$$\mathscr{I} := \bigoplus_{p \in X} \mathrm{sk}_p \, \underline{I_p}$$

where $\mathrm{sk}_p \underline{I_p}$ is the skyscraper sheaf supported at the constant sheaf $\underline{I_p}$. More explicitly, for every open set $U \subseteq X$ the sheaf $\mathscr{I}$ is defined as

$$\mathscr{I}(U) = \bigoplus_{p \in U} I_p$$

Recall that by Corollary 9.1.4 the skyscraper functor is left-adjoint to the stalk functor, which is exact. Therefore by Proposition 7.1.3 $\mathscr{I}$ is an injective object of $\mathsf{Mod}(\underline{\mathbf{k}}_X)$. □

This means that we can derive all right-exact functors from the category of sheaves of $\mathbf{k}$-modules.

Having seen this, we now can finally define sheaf cohomology.

Definition 10.0.1 (Sheaf Cohomology). Let $\mathscr{F}$ be a sheaf of $\mathbf{k}$-modules over a topological space X. The i-th **cohomology** of X with coefficients in $\mathscr{F}$ is the i-th right derived functor of the global sections functor:

$$\mathrm{H}^i(X; \mathscr{F}) := R^i\Gamma(X, \mathscr{F})$$

Therefore, given a sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ to compute its cohomology we *only* need to construct an injective (or $\Gamma(X, -)$ acyclic by Theorem 7.2.2) resolution of $\mathscr{F}$

$$0 \longrightarrow \mathscr{F}^{(0)} \longrightarrow \mathscr{F}^{(1)} \longrightarrow \cdots$$

and compute the cohomology of the $\mathbf{k}$-modules

$$0 \longrightarrow \Gamma(X, \mathscr{F}^{(0)}) \longrightarrow \Gamma(X, \mathscr{F}^{(1)}) \longrightarrow \cdots$$

Since injective resolutions are usually hard to find (note how Lemma 10.0.1 doesn't give an *easy* construction) we will mostly work with $\Gamma(X, -)$-acyclic resolutions.

Lemma 10.0.2. Let X be a topological space and $U \subseteq X$ an open subset.

1. If $\mathscr{I}$ is an injective sheaf over X then $\mathscr{I}|_U$ is injective over U.
2. For any sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ we have $\mathrm{H}^i(U; \mathscr{F}) = \mathrm{H}^i(U; \mathscr{F}|_U)$.

Proof. Let $j \colon U \hookrightarrow X$ be the open immersion. There is an adjunction $j_! \dashv j^{-1}$. Furthermore, $j_!$ is exact. Therefore 1 follows from Proposition 7.1.3.

To see 2, by definition $\mathrm{H}^i(U; \mathscr{F}) = \mathrm{H}^i(\Gamma(U, \mathscr{I}^\bullet))$ where $\mathscr{F} \to \mathscr{I}^\bullet$ is an injective resolution. By 1 $\mathscr{F}|_U \to \mathscr{I}^\bullet|_U$ is an injective resolution of $\mathscr{F}|_U$ in $\mathsf{Mod}(\underline{\mathbf{k}}_U)$. Therefore

$$\mathrm{H}^i(U; \mathscr{F}|_U) = \mathrm{H}^i(\Gamma(U, \mathscr{I}^\bullet|_U)) = \mathrm{H}^i(U; \mathscr{F})$$ □

Lemma 10.0.3. Let X be a topological space and $Z \subseteq X$ a closed subset.

1. If $\mathscr{I}$ is an injective sheaf over X then $\mathscr{I}_Z$ is injective over Z.
2. For any sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ we have $\mathrm{H}^i(X; \mathscr{F}_Z) = \mathrm{H}^i(Z; \mathscr{F}|_Z)$.

Proof. Analogous to Lemma 10.0.2 using the closed immersion $i \colon Z \hookrightarrow X$. □

Lemma 10.0.4. Let X be a topological space and $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$. If $K \subseteq X$ is compact

and relatively Hausdorff then

$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i(U; \mathscr{F}) \cong \mathrm{H}^i(K; \mathscr{F})$$

Proof. Suppose first that $i = 0$. In this case we want to show that there is an isomorphism induced by the restriction

$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U, \mathscr{F}) \xrightarrow{\sim} \Gamma(K, \mathscr{F})$$

We will do this by showing that this map is both injective and surjective:

Injectivity: Let $s \in \Gamma(U, \mathscr{F})$ for some $U \supset K$ open such that its restriction to K is zero, i.e: $s_x = 0$ for all $x \in K$. We have that for all $x \in K$ there exists an open neighbourhood $V_x \subseteq U$ such that $s|_{V_x} = 0$.
Since K is compact we can extract a finite subcover $\{V_{x_1}, \dots, V_{x_n}\}$ from $\{V_x\}_{x \in K}$. Note that $V \coloneqq \bigcup_{i=1}^n V_{x_i}$ is an open neighbourhood of K contained in U such that $s|_V = 0$. Therefore in the colimit s becomes 0, meaning that the map is injective.

Surjectivity: Let $s \in \Gamma(K, \mathscr{F})$. By definition there exists an open cover $\{U_j\}_{j=1}^n$ of K and local sections $s_j \in \Gamma(U_j, \mathscr{F})$ such that s_j represents s in $U_j \cap K$.
On the overlaps $K \cap U_j \cap U_k$ the sections s_j and s_k have the same germs. Therefore by the injectivity argument applied to the difference $s_j - s_k$ they must agree on an open neighbourhood $W_{j,k} \subseteq U_j \cap U_k$ of $K \cap U_j \cap U_k$.
Since K is compact and relatively Hausdorff it is normal.

Now, consider two functors $F, G \colon \mathsf{Mod}(\underline{\mathbf{k}}_X) \to \mathsf{Mod}(\mathbf{k})$

$$F(\mathscr{F}) \coloneqq \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U, \mathscr{F}) \qquad G(\mathscr{F}) \coloneqq \Gamma(K, \mathscr{F})$$

From the first step we have $F \cong G$. Therefore for all $i \geq 0$ we have an isomorphism

$$\mathrm{R}^i F(\mathscr{F}) \cong \mathrm{R}^i G(\mathscr{F})$$

To compute these functors, choose an injective resolution $\mathscr{F} \to \mathscr{I}^\bullet$ of $\mathscr{F}$. On the one hand

$$\mathrm{R}^i G(\mathscr{F}) = \mathrm{H}^i \Gamma(K; I^\bullet) = \mathrm{H}^i(K; \mathscr{F})$$

On the other

$$\mathrm{R}^i F(\mathscr{F}) = \mathrm{H}^i \left(\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \Gamma(U, \mathscr{I}^\bullet) \right) \cong \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i \left(\Gamma(U, \mathscr{I}^\bullet) \right) = \varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i \left(U; \mathscr{F} \right)$$

Therefore

$$\varinjlim_{\substack{U \supset K \\ U \in \mathsf{Open}(X)}} \mathrm{H}^i \left(U; \mathscr{F} \right) \cong \mathrm{R}^i F(\mathscr{F}) \cong \mathrm{R}^i G(\mathscr{F}) = \mathrm{H}^i(K; \mathscr{F}) \qquad \square$$

Theorem 10.0.1 (Open Mayer-Vietoris). Let X be a topological space. Suppose that $X = U \cup V$ for some open sets U, V. Then for every sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ there exists a long exact sequence in cohomology:

$$\begin{array}{l} 0 \longrightarrow \mathrm{H}^0(X;\mathscr{F}) \longrightarrow \mathrm{H}^0(U;\mathscr{F}) \oplus \mathrm{H}^0(V;\mathscr{F}) \longrightarrow \mathrm{H}^0(U\cap V;\mathscr{F}) \\ \quad \hookrightarrow \mathrm{H}^1(X;\mathscr{F}) \longrightarrow \mathrm{H}^1(U;\mathscr{F}) \oplus \mathrm{H}^1(V;\mathscr{F}) \longrightarrow \mathrm{H}^1(U\cap V;\mathscr{F}) \\ \quad \hookrightarrow \mathrm{H}^2(X;\mathscr{F}) \longrightarrow \mathrm{H}^2(U;\mathscr{F}) \oplus \mathrm{H}^2(V;\mathscr{F}) \longrightarrow \cdots \end{array}$$

Proof. Consider the sequence of sheaves of $\mathbf{k}$-modules

$$0 \longrightarrow \mathscr{F} \xrightarrow{s\mapsto(s|_U, s|_V)} \mathscr{F}|_U \oplus \mathscr{F}|_V \xrightarrow{(s,t)\mapsto s|_{U\cap V} - t|_{U\cap V}} \mathscr{F}|_{U\cap V} \longrightarrow 0$$

Let $p \in X = U \cup V$. If $p \in U$ but $p \notin V$ the induced sequence on the stalks is

$$0 \longrightarrow \mathscr{F}_p \longrightarrow \mathscr{F}_p \oplus 0 \longrightarrow 0 \longrightarrow 0$$

which is exact. Analogously if $p \in V$ but $p \notin U$ we also get an exact sequence.

Now, if $p \in U \cap V$ we have

$$0 \longrightarrow \mathscr{F}_p \xrightarrow{s\mapsto(s_x, s_x)} \mathscr{F}_p \oplus \mathscr{F}_p \xrightarrow{(s,t)\mapsto s-t} \mathscr{F}_p \longrightarrow 0$$

which is also exact.

Therefore

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}|_U \oplus \mathscr{F}|_V \longrightarrow \mathscr{F}|_{U\cap V} \longrightarrow 0$$

is exact in $\mathsf{Mod}(\underline{\mathbf{k}}_X)$. Applying Theorem 7.2.1 with $\Gamma(X, -)$ and using Lemma 10.0.2 we get the result. □

Theorem 10.0.2 (Closed Mayer-Vietoris). Let X be a topological space. Suppose that $X = Z \cup W$ for some closed sets Z, W. Then for every sheaf of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ there exists a long exact sequence in cohomology:

$$\begin{array}{l} 0 \longrightarrow \mathrm{H}^0(X;\mathscr{F}) \longrightarrow \mathrm{H}^0(X;\mathscr{F}_Z) \oplus \mathrm{H}^0(X;\mathscr{F}_W) \longrightarrow \mathrm{H}^0(X;\mathscr{F}_{Z\cap W}) \\ \quad \hookrightarrow \mathrm{H}^1(X;\mathscr{F}_Z) \longrightarrow \mathrm{H}^1(X;\mathscr{F}_Z) \oplus \mathrm{H}^1(X;\mathscr{F}_W) \longrightarrow \mathrm{H}^1(X;\mathscr{F}_{Z\cap W}) \\ \quad \hookrightarrow \mathrm{H}^2(X;\mathscr{F}) \longrightarrow \mathrm{H}^2(X;\mathscr{F}_U) \oplus \mathrm{H}^2(X;\mathscr{F}_V) \longrightarrow \cdots \end{array}$$

Proof. Analogous to the open version with

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_Z \oplus \mathscr{F}_W \longrightarrow \mathscr{F}_{Z\cap W} \longrightarrow 0 \quad \square$$

10.1 Acyclic resolutions

Definition 10.1.1 (Flasque sheaf)**.** A sheaf of $\mathbf{k}$-modules $\mathscr{F}$ is **flasque** (or **flabby**) if for any inclussion of open subsets $V \subseteq U$ the restriction map $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ is surjective.

Proposition 10.1.1. Flasque sheaves are $\Gamma(X, -)$-acyclic.

Proof. Let $\mathscr{F}$ be a flasque sheaf. Since $\mathsf{Mod}(\underline{\mathbf{k}}_X)$ has enough injectives we have a short exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{I}$ is an injective object. We will first show that, for any open subset $U \subseteq X$, the map $\mathscr{I}(U) \to \mathscr{G}(U)$ is surjective.

Let $s \in \mathscr{G}(U)$. Let $V, W \subseteq U$ be two open subsets and $t_V \in \mathscr{I}(V), t_W \in \mathscr{I}(W)$ be two sections lifting s to $\mathscr{I}$. Consider

$$t_V|_{V\cap U} - t_W|_{V\cap W} \in \mathscr{F}(V \cap W)$$

Since $\mathscr{F}$ is flasque there exists a section $x_V \in \mathscr{F}(V)$ such that $x_V|_{V\cap W} = t_V - t_W$. Therefore if $t'_V = t_V - x_V$ we have $t'_V|_{V\cap W} = t_W|_{V\cap W}$, i.e: there exists a section $t \in \mathscr{I}(V \cup W)$ such that $t|_V = t'|_V$ and $t'|_W = t|_W$. Clearly t is sent to $s|_{V\cap W}$.

Consider now a pair (W, t) which is maximal (for the obvious order relation), where W is an open subset of U and t is a section of $\mathscr{I}$ lifting s to W. Since s lifts locally on $\mathscr{I}$ and using the previous result we see that $W = U$.

But then using the long exact sequence in cohomology of our sequence and the fact that for $i > 0 \quad \mathrm{H}^i(X, \mathscr{I}) = 0$ imply that $\mathrm{H}^1(X, \mathscr{F}) = 0$ we get

$$\forall k - 1 \geq 1 \quad \mathrm{H}^k(X, \mathscr{F}) = \mathrm{H}^{k-1}(X, \mathscr{G})$$

Furthermore surjectivity of the maps $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ implies that $\mathscr{G}$ is flasque.

It follows by induction that

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$ □

Lemma 10.1.1. Injective sheaves are flasque.

Proof. Let $U' \subseteq U \subseteq X$ be open subsets and $\mathscr{I} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ be an injective sheaf. We want to see that the map $\mathscr{I}(U) \to \mathscr{I}(U')$ is surjective.

Consider the open immersions $j : U \hookrightarrow X$ and $j' : U' \hookrightarrow X$. Since $j_!$ is left adjoint to the restriction for any □

Using flasque sheaves we can construct a canonical (and functorial) resolution of a sheaf of $\mathscr{O}_X$-modules to compute its cohomology.

Construction (Godement resolution)**.** Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. The **Godement resolution** of $\mathscr{F}$ is constructed by considering the inclussion of $\mathscr{F}$ into the *Godement sheaf* $\mathscr{F}_{\text{God}}$

$$U \mapsto \bigoplus_{p \in U} \mathscr{F}_p =: \mathscr{F}_{\text{God}}(U)$$

Note that $\mathscr{F}_{\text{God}}$ is flasque.

We then inject the quotient ${}^{\mathscr{F}_{\text{God}}}\!/_{\mathscr{F}}$ into $\left({}^{\mathscr{F}_{\text{God}}}\!/_{\mathscr{F}}\right)_{\text{God}}$ and so on.

Example 10.1.1. Let's try to compute sheaf cohomology using the Godement resolution.
Let $X = \{p, q\}$ with the *Sierpiński* topology $\mathcal{T} = \{\varnothing, \{p\}, X\}$. Consider the constant sheaf supported at the integers $\mathscr{F} = \underline{\mathbb{Z}}$.
To compute the cohomology of $\mathscr{F}$, we are going to construct the complex of global sections:

$$0 \longrightarrow \Gamma(X, \mathscr{F}_{\text{God}}) \xrightarrow{\mathrm{d}^0} \Gamma\left(X, \left({}^{\mathscr{F}_{\text{God}}}\!/_{\mathscr{F}}\right)_{\text{God}}\right) \xrightarrow{\mathrm{d}^1} \cdots$$

Note that the stalks of $\mathscr{F}$ are:

$$\mathscr{F}_p = \mathscr{F}(\{p\}) = \mathbb{Z}, \qquad \mathscr{F}_q = \mathscr{F}(X) = \mathbb{Z}.$$

Consider the Godement sheaf associated to $\mathscr{F}$:

$$\mathscr{F}_{\text{God}}(U) = \bigoplus_{x \in U} \mathscr{F}_x.$$

We have that:

$$\Gamma(X, \mathscr{F}_{\text{God}}) = \mathscr{F}_{\text{God}}(X) = \mathscr{F}_p \oplus \mathscr{F}_q = \mathbb{Z} \oplus \mathbb{Z}.$$

This means that we have a map d^{-1} as follows:

$$\mathrm{d}^{-1} : \underbrace{\Gamma(X, \mathscr{F})}_{\mathbb{Z}} \to \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}}, \quad \mathrm{d}^{-1}(k) = (k, k).$$

Let $\mathscr{Q}^1 = {}^{\mathscr{F}_{\text{God}}}\!/_{\mathscr{F}}$ be the cokernel sheaf. If we compute its stalks, we have that:

$$(\mathscr{Q}^1)_p = {}^{\mathscr{F}_{\text{God},p}}\!/_{\mathscr{F}_p} = {}^{\mathbb{Z}}\!/_{\mathbb{Z}} = 0,$$
$$(\mathscr{Q}^1)_q = {}^{\mathscr{F}_{\text{God},q}}\!/_{\mathscr{F}_q} = {}^{\mathbb{Z}\oplus\mathbb{Z}}\!/_{\langle (k,k) \rangle} \cong \mathbb{Z}.$$

Therefore, the global sections of the Godement sheaf of $\mathscr{Q}^1$ are:

$$\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right) = (\mathscr{Q}^1)_p \oplus (\mathscr{Q}^1)_q = 0 \oplus \mathbb{Z} = \mathbb{Z}.$$

This produces a differential map:

$$\mathrm{d}^0 : \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}} \to \underbrace{\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right)}_{\mathbb{Z}}, \quad \mathrm{d}^0(m, n) = m - n.$$

Let $\mathscr{Q}^2$ be the cokernel of the map $\mathscr{F}_{\text{God}} \to (\mathscr{Q}^1)_{\text{God}}$. If we compute its stalks, we have:

$$(\mathscr{Q}^2)_p = {}^{0}\!/_{0} = 0,$$
$$(\mathscr{Q}^2)_q = {}^{\mathbb{Z}}\!/_{\mathbb{Z}} = 0.$$

Therefore, we have that:

$$\Gamma\left(X, (\mathscr{Q}^2)_{\text{God}}\right) = 0 \oplus 0 = 0.$$

This means that our resolution of global sections is:

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow[(m,n)\mapsto m-n]{d^0} \mathbb{Z} \longrightarrow 0$$

Therefore, the cohomology groups of X with coefficients in $\underline{\mathbb{Z}}$ are:

$$\begin{aligned}
&\mathrm{H}^0(X, \underline{\mathbb{Z}}) = \ker(d^0) = \{(m,n) \in \mathbb{Z} \oplus \mathbb{Z} \mid m - n = 0\} \cong \mathbb{Z}, \\
&\mathrm{H}^1(X, \underline{\mathbb{Z}}) = \operatorname{coker}(d^0) = {}^{\mathbb{Z}}\!/_{\operatorname{im}(d^0)} = {}^{\mathbb{Z}}\!/_{\mathbb{Z}} = 0, \\
&\mathrm{H}^i(X, \underline{\mathbb{Z}}) = 0 \quad \forall i > 1.
\end{aligned}$$

Chapter 11

Cohomology and homotopy

Our goal in this chapter will be to prove that sheaf cohomology is an homotopy invariant. We adapt the results of [KS90, Section 2.7].

Notation. We will denote $I := [0,1]$ the unit interval and $I_x := I \times \{x\}$.

Definition 11.0.1. Let X and Y be topological spaces. A map $f : X \to Y$ is continuous over a topological space S if it is continuous and there exists continuous maps $p_X : X \to S$ and $p_Y : Y \to S$ making the following diagram commute

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p_X}\searrow \quad \swarrow{\scriptstyle p_Y} & \\ & S & \end{array}$$

Definition 11.0.2. Let X, Y and S be topological spaces.

1. Let $f_0, f_1 : X \rightrightarrows Y$ be two continuous maps over S. We say that f_0 and f_1 are **homotopic** over S if there exists a continuous map $h : X \times I \to Y$ over S such that $f_0 = h(-,0)$ and $f_1 = h(-,1)$. We denote this by $f_0 \sim_S f_1$.
2. A continuous maps over S $f : X \to Y$ is a **homotopy equivalence** over S if there exists a continuous maps over S $g : Y \to X$ such that $g \circ f \sim_S \mathbb{1}_X$ and $f \circ g \sim_S \mathbb{1}_Y$. In this case we write $X \sim_S Y$.
3. A topological space Z is **contractible** over S if the map $p_Z : Z \to S$ is a homotopy equivalence.

Example 11.0.1. If $S = \{*\}$ is the one-point topological space we recover the usual definition of homotopy of topological spaces.

Remark 11.0.1. If $f_0, f_1 : X \rightrightarrows Y$ are homotopic over S we have a commutative diagram

$$\begin{array}{ccc} X & & \\ {\scriptstyle j_i}\downarrow & \searrow^{f_t} & \\ X \times I & \xrightarrow{h} & Y \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p_Y} \\ X & \xrightarrow[p_X]{} & S \end{array}$$

where $j_t(z) = (z,t)$ for $t \in \{0,1\}$.

Lemma 11.0.1. Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_I)$. Then

1. For $i > 1$ we have $\mathrm{H}^i(I;\mathscr{F}) = 0$.
2. If $\mathscr{F}(I) \to \mathscr{F}_t$ is an epimorphism for all $t \in I$ then $\mathrm{H}^1(I;\mathscr{F}) = 0$.

Proof. Let $j \geq 1$ and $s \in \mathrm{H}^j(I;\mathscr{F})$. We want to show that $s = 0$.

For $0 \leq t_1 < t_2 \leq 1$ consider the morphism

$$f_{t_1,t_2} : \mathrm{H}^i(I;\mathscr{F}) \to \mathrm{H}^i([t_1,t_2];\mathscr{F}|_{[t_1,t_2]})$$

and let

$$J := \{t \in [0,1] \mid f_{0,t}(s) = 0\}$$

Since $\mathrm{H}^j(\{0\},\mathscr{F}) = 0$ we have that $0 \in J$. Furthermore if $t \in J$ then for any $t' \leq t$ the map $f_{0,t'}$ factors as

$$\begin{array}{ccc} \mathrm{H}^i(I;\mathscr{F}) & \xrightarrow{f_{0,t'}} & \mathrm{H}^i([0,t'];\mathscr{F}) \\ {\scriptstyle f_{0,t}}\downarrow & \nearrow & \\ \mathrm{H}^i([0,t];\mathscr{F}) & & \end{array}$$

Therefore since $f_{0,t}(s) = 0$ we have that $f_{0,t'}(s) = 0$, i.e: $[0,t] \subseteq J$ and thus J is a sub-interval of $[0,1]$ starting at 0.

To prove that $J = [0,1]$ we will see that it is both open and closed.

Open Suppose $t_0 \in S$. By Lemma 10.0.4 we have that

$$\mathrm{H}^j([0,t_0];\mathscr{F}) \cong \varinjlim_{t>t_0} \mathrm{H}^j([0,t];\mathscr{F})$$

Therefore there exists $[0, t_0 + \varepsilon[$ such that $f_{0,t}(s) = 0$ for all $t \in [0, t_0 + \varepsilon[$, and thus J is open.

Closed Let $t_0 := \sup J$. We want to see that $t_0 \in J$. For $t \in [0,t_0]$ consider the Mayer-Vietoris sequence associated with $[0,t_0] = [0,t] \cup [t,t_0]$:

$$\cdots \longrightarrow \mathrm{H}^i([0,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^i([0,t];\mathscr{F}) \oplus \mathrm{H}^i([t,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^i(\{t\},\mathscr{F}) \longrightarrow \cdots$$

If $i > 1$ then since $\mathrm{H}^i(\{t\};\mathscr{F}) = \mathrm{H}^{i-1}(\{t\};\mathscr{F}) = 0$ we have an isomorphism

$$\mathrm{H}^i([0,t_0];\mathscr{F}) \cong \mathrm{H}^i([0,t];\mathscr{F}) \oplus \mathrm{H}^i([t,t_0];\mathscr{F})$$

Similarly, if $\mathscr{F}(I) \twoheadrightarrow \mathscr{F}_t$ is an epimorphism we have

$$\mathrm{H}^0([0,t_0];\mathscr{F}) \twoheadrightarrow \mathrm{H}^0(\{t\};\mathscr{F}) \longrightarrow \mathrm{H}^1([0,t_0];\mathscr{F}) \xrightarrow{\sim} \mathrm{H}^1([0,t];\mathscr{F}) \oplus \mathrm{H}^1([t,t_0];\mathscr{F})$$

In both cases for $f_{0,t_0}(s)$ to be zero it suffices for its components in $\mathrm{H}^i([0,t];\mathscr{F})$ and $\mathrm{H}^i([t,t_0];\mathscr{F})$ to vanish.
On the one hand since $t < t_0 = \sup J$ we can choose $t \in J$, so $f_{0,t}(s) = 0$. On the other by Lemma 10.0.4 we have

$$\varinjlim_{t<t_0} \mathrm{H}^i([t,t_0];\mathscr{F}) \cong \mathrm{H}^i(\{t_0\};\mathscr{F}) = 0$$

Therefore $f_{0,t_0}(s) = 0$ and thus $t_0 \in J$.

Since J is non-empty, open and closed and $[0,1]$ is connected necessarily $J = [0,1]$. Therefore $s = f_{0,1}(s) = 0$. □

Lemma 11.0.2. Let $f : X \to Y$ be a continuous map over S.
For all $i \geq 0$ there exists a natural transformation

$$f^{\#,i} := \mathrm{R}^i p_{Y,*} \circ p_Y^{-1} \to \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$

Proof. Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ be a sheaf over S.

Consider an injective resolution $p_Y^{-1}(\mathscr{F}) \to \mathscr{I}^\bullet$ of the inverse image $p_Y^{-1}(\mathscr{F}) \in \mathsf{Mod}(\underline{\mathbf{k}}_Y)$ Using the unit of the adjunction $f_* \dashv f^{-1}$ we get a natural map of complexes in Y

$$\mathscr{I}^\bullet \to \left(f_* \circ f^{-1}\right)(\mathscr{I}^\bullet)$$

Applying the direct image $p_{Y,*}$ to this, since $p_Y \circ f = p_X$ we get a morphism of complexes in S

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet)$$

Note that, since f^{-1} is exact, $f^{-1}(\mathscr{I}^\bullet)$ is an exact resolution of $p_X^{-1}(\mathscr{F})$. However, it need not be made of injectives. To fix this, consider a proper injective resolution $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$. By Theorem 7.1.2 there exists a morphism (unique up to homotopy of complexes) lifting the identity

$$f^{-1}(\mathscr{I}^\bullet) \to \mathscr{J}^\bullet$$

Applying $p_{X,*}$ we get a morphism in S

$$\left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Therefore composing with the previous map we get

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{X,*} \circ f^{-1}\right)(\mathscr{I}^\bullet) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Applying the i-th right derived functor we get a map

$$\left(\mathrm{R}^i p_{Y,*} \circ p_Y^{-1}\right)(\mathscr{F}) \to \left(\mathrm{R}^i p_{X,*} \circ p_X^{-1}\right)(\mathscr{F})$$ □

Remark 11.0.2. If $S = \{*\}$ then $\mathsf{Sh}_*(\mathsf{C}) \simeq \mathsf{C}$ and $p_Z^{-1}\underline{M} = \underline{M}$ for all $M \in \mathsf{C}$ and $Z \in \{X, Y\}$.
In this case the morphism of Lemma 11.0.2 is

$$f^{\#,i}(\underline{M}) \colon \mathrm{H}^i(Y; \underline{M}) \to \mathrm{H}^i(X; \underline{M})$$

We denote this map by f^*.

Theorem 11.0.1. Let $f_0, f_1 : X \rightrightarrows Y$ be two homotopic maps over S. Then $f_0^{\#,i} = f_1^{\#,i}$ for all $i \geq 0$.

Proof. Let $\mathscr{F} \in \mathsf{Sh}_S(\mathsf{C})$. Choose injective resolutions $p_Y^{-1}(\mathscr{F}) \to \mathscr{I}^\bullet$ and $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$.

Since $f_t = h \circ j_t$ we can factor $\mathscr{I}^\bullet \to \left(f_{t,*} \circ f_t^{-1}\right)(\mathscr{I}^\bullet)$ through

$$\mathscr{I}^\bullet \to \left(h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \to \left(\left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)(\mathscr{I}^\bullet)$$

Pushing this factorization via $p_{Y,*}$ we get

$$p_{Y,*}(\mathscr{I}^\bullet) \to \left(p_{Y,*} \circ h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \to \left(p_{Y,*} \circ \left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)(\mathscr{I}^\bullet)$$

On the one hand since $p_Y \circ h = p_X \circ p$ the middle term becomes

$$\left(p_{Y,*} \circ h_* \circ h^{-1}\right)(\mathscr{I}^\bullet) = \left(\left(p_Y \circ h\right)_* \circ h^{-1}\right)(\mathscr{I}^\bullet) = \left(\left(p_X \circ p\right)_* \circ h^{-1}\right)(\mathscr{I}^\bullet)$$

On the other since $f_t = h \circ j_t$ and $p_Y \circ f_t = p_X$ we have

$$\left(p_{Y,*} \circ \left(h \circ j_t\right)_* \circ \left(h \circ j_t\right)^{-1}\right)(\mathscr{I}^\bullet) = \left(\left(p_Y \circ f_t\right)_* \circ f_t^{-1}\right)(\mathscr{I}^\bullet) = \left(p_{X,*} \circ f_t^{-1}\right)(\mathscr{I}^\bullet)$$

As in Lemma 11.0.2 since $f_t^{-1}(\mathscr{I}^\bullet)$ is an exact (but not necessarily injective) resolution of $p_X^{-1}(\mathscr{F})$ there exists a unique (up to homotopy) lifting of the identity

$$f_t^{-1}(\mathscr{I}^\bullet) \to \mathscr{J}^\bullet$$

where $p_X^{-1}(\mathscr{F}) \to \mathscr{J}^\bullet$ is an injective resolution of $p_X^{-1}(\mathscr{F})$. Applying $p_{X,*}$ we obtain a map

$$\left(p_{X,*} \circ f_t^{-1}\right)(\mathscr{I}^\bullet) \to p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Therefore we obtain a factorization of $f^{\#,t}$

$$p_{Y,*}(\mathscr{I}^\bullet) \xrightarrow{g_1} \left(p_{X,*} \circ p_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \xrightarrow{g_2} p_{X,*}\left(\mathscr{J}^\bullet\right)$$

Denote by

$$\alpha_t \coloneqq \mathrm{H}^\bullet(g_2)\colon\ \mathrm{H}^\bullet\left(\left(p_{X,*} \circ p_* \circ h^{-1}\right)(\mathscr{I}^\bullet)\right) \to \mathrm{H}^\bullet\left(p_{X,*}\left(\mathscr{J}^\bullet\right)\right)$$

the induced map on cohomology of g_2. Since g_1 does not depend on t, we only need to see that $\alpha_0 = \alpha_1$.

The map α_t is induced by

$$\left(p_* \circ h^{-1}\right)(\mathscr{I}^\bullet) \to \underbrace{\left(j_t^{-1} \circ h^{-1}\right)}_{f_t^{-1}}(\mathscr{I}^\bullet) \to \mathscr{J}^\bullet$$

Note that $h^{-1}(\mathscr{I}^\bullet)$ is an exact sequence resolving $\left(h^{-1} \circ p_Y^{-1}\right)(\mathscr{F}) \cong \left(p^{-1} \circ p_X^{-1}\right)(\mathscr{F})$. Since by Lemma 11.0.1 higher cohomology ($k > 0$) vanishes, the cohomology of α_t is concentrated in degree zero, which is

$$\left(p_* \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \to \left(j_t^{-1} \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F})$$

However, note that we have isomorphisms

$$p_X^{-1}(\mathscr{F}) \xrightarrow{\sim} \left(p_* \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \xrightarrow{\sim} \left(j_t^{-1} \circ p^{-1} \circ p_X^{-1}\right)(\mathscr{F}) \xrightarrow{\sim} p_X^{-1}(\mathscr{F})$$

whose composition is the identity on p_X^{-1}, which means that α_t is induced by the identity. Indeed:

– The first one comes from the unit of the adjunction $p_* \dashv p^{-1}$.

– Applying p_* to the unit of the adjunction $j_{t,*} \dashv j^{-1}$ we get

$$p_* \xrightarrow{\sim} p_* \circ j_{t,*} \circ j_t^{-1} = \left(p \circ j_t\right)_* \circ j_t^{-1} = \mathbb{1}_* \circ j_t^{-1} = j_t^{-1}$$

– The last one is direct from the composition of the inverse image

$$j_t^{-1} \circ p^{-1} = \left(p \circ j_t\right)^{-1} = \mathbb{1}^{-1} = \mathbb{1}$$

Therefore since α_t does not depend on t we get $\alpha_0 = \alpha_1$ and thus $f_0^{\#,i} = f_1^{\#,i}$. □

Corollary 11.0.1. Let $f : X \to Y$ be a homotopy equivalence over S. Then $f^{\#,i}$ is a natural isomorphism for all $i \geq 0$.

Proof. By definition there exists a continuous map over S $g : Y \to X$ such that $g \circ f \sim_S \mathbb{1}_X$ and $f \circ g \sim_S \mathbb{1}_Y$.

Since $f^{\#,i}$ is a natural transformation we have

$$\left(g \circ f\right)^{\#,i} = f^{\#,i} \circ g^{\#,i} \qquad \left(f \circ g\right)^{\#,i} = g^{\#,i} \circ f^{\#,i}$$

Therefore since homotopic maps induce the same transformation

$$f^{\#,i} \circ g^{\#,i} = (\mathbb{1}_X)^{\#,i} = \mathbb{1} \qquad g^{\#,i} \circ f^{\#,i} = (\mathbb{1}_Y)^{\#,i} = \mathbb{1}$$ □

Corollary 11.0.2. Let X be contractible over S. Then for all $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ the adjunction unit map

$$\mathscr{F} \to \left(\mathrm{R}^0 p_{X,*} \circ p_X^{-1}\right)(\mathscr{F})$$

is an isomorphism and for all $i > 0$

$$\left(\mathrm{R}^i p_{X,*} \circ p_X^{-1}\right)(\mathscr{F}) = 0$$

Proof. We can view S as a space over itself via the identity map $\mathbb{1}_S : S \to S$. By hypothesis the map $p_X : X \to S$ is a homotopy equivalence over S. Therefore by Corollary 11.0.1 $p_X^{\#,i}$ is a natural isomorphism

$$\mathrm{R}^i \mathbb{1}_{S,*} \circ \mathbb{1}_S^{-1} \xrightarrow{\sim} \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$

For $i = 0$ we have $\mathrm{R}^0 \mathbb{1}_{S,*} = \mathbb{1}$. Therefore

$$\mathbb{1}_S \xrightarrow{\sim} \mathrm{R}^0 p_{X,*} \circ p_X^{-1}$$

On the other hand since $\mathbb{1}_{S,*}$ is exact $\mathrm{R}^i \mathbb{1}_{S,*} = 0$ for all $i > 0$. Therefore

$$0 \xrightarrow{\sim} \mathrm{R}^i p_{X,*} \circ p_X^{-1}$$ □

Definition 11.0.3. Let X be a topological space over S with $p_X : X \to S$. Let $i : A \hookrightarrow X$ be a subspace with $p_A := p_X|_A$.
We say that A is a **deformation retract** of X over S if there exists a continuous map over S $r : X \to A$ with $r \circ i = \mathbb{1}_A$ and a homotopy over S $F : X \times I \to A$ between $\mathbb{1}_X$ and $i \circ r$. The map r is called the **retraction** of X into A.
If $F(a,t) = a$ for all $t \in I$ and $a \in A$, we say that A is a **strong deformation retract**

of X over S.

Corollary 11.0.3. Let $i\colon A \hookrightarrow X$ be a strong deformation retract of X over S. Then the induced map

$$i^{\#,n}\colon \mathrm{R}^n p_{X,*} \circ p_X^{-1} \to \mathrm{R}^n p_{A,*} \circ p_A^{-1}$$

is a natural isomorphism for all $n \geq 0$.

Proof. Let $r\colon X \to A$ be the retraction of X into A. Since $r \circ i = \mathbb{1}_A$ we have for all $n \geq 0$

$$i^{\#,n} \circ r^{\#,n} = (i \circ r)^{\#,n} = \mathbb{1}_A^{\#,n} = \mathbb{1}_{\mathrm{R}^i p_{A,*} \circ p_A^{-1}}$$

On the other hand since $i \circ r \sim_S \mathrm{id}_X$ by Theorem 11.0.1 we have

$$r^{\#,n} \circ i^{\#,n} = (r \circ i)^{\#,n} = \mathbb{1}_X^{\#,n} = \mathbb{1}_{\mathrm{R}^i p_{X,*} \circ p_X^{-1}}$$

Therefore $r^{\#,n}$ and $i^{\#,n}$ are inverses of each other and thus natural isomorphisms. □

Chapter 12

Some examples of cohomology

We focus our attention now to the computation of the cohomology of some topological spaces. Our main tools will be the Mayer-Vietoris sequence and the homotopy invariance of cohomology.

Example 12.0.1 ($\mathbb{S}^1$). Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ be a sheaf of $\mathbf{k}$-modules over S and $X := S \times \mathbb{S}^1$. Consider the open cover of $\mathbb{S}^1 = H^+ \cup H^-$ where H^+ is the upper hemisphere and H^- the lower one, both extended slightly past the equator. Let $U = S \times H^+$ and $V = S \times H^-$. Since H^+ and H^- are homeomorphic to open intervals, which are contractible, both U and V are contractible over S. Furthermore $U \cap V = (S \times W_1) \coprod (S \times W_2)$ consists of two disjoint pieces, each of these contractible over S.
Therefore since cohomology of contractible spaces is concentrated in degree 0 we only have

$$\begin{aligned} \mathrm{R}^0 p_{U,*}\mathscr{F}|_U &\cong \mathscr{F} \\ \mathrm{R}^0 p_{V,*}\mathscr{F}|_V &\cong \mathscr{F} \\ \mathrm{R}^0 p_{U\cap V,*}\mathscr{F}|_{U\cap V} &\cong \mathscr{F} \oplus \mathscr{F} \end{aligned}$$

Writing the Mayer-Vietoris sequence since the higher terms vanish we end up with

$$0 \longrightarrow \mathrm{R}^0 p_{X,*}\mathscr{F}_X \longrightarrow \mathscr{F} \oplus \mathscr{F} \xrightarrow{\alpha} \mathscr{F} \oplus \mathscr{F} \xrightarrow{\delta} \mathrm{R}^1 p_{X,*}\mathscr{F}|_X \longrightarrow 0$$

Therefore directly we get $\mathrm{R}^i p_{X,*}\mathscr{F}_X = 0$ for all $i \geq 2$.
Note that $\alpha \colon \mathscr{F} \oplus \mathscr{F} \to \mathscr{F} \oplus \mathscr{F}$ maps a pair of sections (s_U, s_V) into their difference on the overlap. Since the section s_U is defined globally on U its restriction to W_1 and to W_2 is identical and analogously for s_V. Therefore we have

$$\alpha(s_U, s_V) = (s_U - s_V, s_U - s_V)$$

and thus $\ker \alpha = \{(s, s) \mid s \in \mathscr{F}\}$ is the diagonal which means that

$$\mathrm{R}^0 p_{X,*}\mathscr{F}_X \cong \mathscr{F}$$

On the other hand by exactness δ is surjective, meaning that

$$\mathrm{R}^1 p_{X,*}\mathscr{F}|_X \cong \operatorname{coker} \alpha \cong {}^{\mathscr{F} \oplus \mathscr{F}}\big/_{\operatorname{Im} \alpha} = {}^{\mathscr{F} \oplus \mathscr{F}}\big/_{\langle (s,s) \rangle} \cong \mathscr{F}$$

Therefore

$$\mathrm{R}^i p_{S\times\mathbb{S}^1,*}\mathscr{F}_X = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = 1 \\ 0 & i \geq 2 \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{S}^1; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = 1 \\ 0 & i \geq 2 \end{cases}$$

Example 12.0.2 ($\mathbb{S}^n$). Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$. We are going to see that cohomology of $X = S \times \mathbb{S}^n$ over S by induction over n is

$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n \\ \mathscr{F} & i \notin \{0, n\} \end{cases}$$

We have already computed the base case $n = 1$ in Example 12.0.1, so we can suppose $n \geq 2$.
Consider the open cover $\mathbb{S}^n = H^+ \cup H^-$ where H^+ is the open upper hemisphere and H^- is the lower open hemisphere, both extending slightly past the equator. We have that $U \coloneqq S \times H^+$ and $V \coloneqq S \times H^-$ are contractible over S. Furthermore since $H^+ \cap H^-$ is a thickened equatorial band it can be *squeezed* into the equator itself, which it is a $(n-1)$-sphere. This means that $H^+ \cap H^-$ strong-deformation retracts onto $\mathbb{S}^{n-1}$ and thus, $U \cap V$ strong-deformation retracts onto $S \times \mathbb{S}^{n-1}$.
Since $\mathbb{S}^{n-1}$ is connected the first terms of the Mayer-Vietoris sequence are

$$0 \longrightarrow \mathrm{R}^0 p_{X,*}\mathscr{F}_X \longrightarrow \mathscr{F} \oplus \mathscr{F} \xrightarrow{\alpha} \mathscr{F} \xrightarrow{\delta} \mathrm{R}^1 p_{X,*}\mathscr{F}|_X \longrightarrow 0 \oplus 0 \longrightarrow \cdots$$

In this case we have $\alpha(s_U, s_V) = s_U - s_V$ and thus $\mathrm{R}^0 p_{X,*}\mathscr{F}|_X \cong \mathscr{F}$. Furthermore since α is clearly surjective and δ is surjective because the sequence is exact we have $\mathrm{R}^1 p_{X,*}\mathscr{F}|_X \cong 0$.
If we focus now on the higher terms of the Mayer-Vietoris sequence we have, for $i \geq 1$

$$\cdots \longrightarrow \mathrm{R}^i p_{U,*} \oplus \mathrm{R}^i p_{V,*} \longrightarrow \mathrm{R}^i p_{U\cap V,*} \xrightarrow{\delta} \mathrm{R}^{i+1} p_{X,*} \longrightarrow \mathrm{R}^{i+1} p_{U,*} \oplus \mathrm{R}^{i+1} p_{V,*} \longrightarrow \cdots$$

Since U and V are contractible over S these terms are zero by Corollary 11.0.2 and thus we get an isomorphism

$$\mathrm{R}^i p_{U\cap V,*} \xrightarrow{\sim} \mathrm{R}^{i+1} p_{X,*}$$

Substituting $U \cap V$ we get

$$\mathrm{R}^{i+1} p_{X,*}\mathscr{F}|_{S\times\mathbb{S}^n} \cong \mathrm{R}^i p_{X,*}\mathscr{F}|_{S\times\mathbb{S}^{n-1}} \cong \begin{cases} \mathscr{F} & i \in \{0, n-1\} \\ \mathscr{F} & i \notin \{0, n-1\} \end{cases}$$

Therefore

$$\mathrm{R}^i p_{S\times\mathbb{S}^n,*}\mathscr{F}|_{S\times\mathbb{S}^n} = \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n \\ 0 & i \notin \{0, n\} \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{S}^n; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = n \\ 0 & i \notin \{0, n\} \end{cases}$$

Example 12.0.3 ($\mathbb{R}^n \setminus \{0\}$)**.** We are going to see first that $X \coloneqq S \times (\mathbb{R}^n \setminus \{0\})$ is a strong deformation retract of $A \coloneqq S \times \mathbb{S}^{n-1}$. To do this, consider the inclussion $i \colon A \hookrightarrow X$ and the retraction $r \colon X \to A, r(s,v) \coloneqq \left(s, \frac{v}{\|v\|}\right)$. As the homotopy we can take

$$\begin{aligned} H \colon X \times [0,1] &\to A \\ (s,v,t) &\mapsto \left(s, (1-t)v + t\frac{v}{\|v\|}\right) \end{aligned}$$

Therefore by Corollary 11.0.3 for all $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_S)$ we have and $i \geq 0$

$$\mathrm{R}^i p_{X,*}\mathscr{F}|_X \cong \mathrm{R}^i p_{A,*}\mathscr{F}|_X$$

In particular if $n \geq 2$ we have

$$\mathrm{R}^i p_{S\times(\mathbb{R}^n\setminus\{0\}),*}\mathscr{F}|_{S\times(\mathbb{R}^n\setminus\{0\})} \cong \begin{cases} \mathscr{F} & i = 0 \\ \mathscr{F} & i = n-1 \\ 0 & i \notin \{0, n-1\} \end{cases}$$

The case $n = 1$ is different since $\mathbb{R} \setminus \{0\}$ has two connected components. We have a deformation retract $\mathbb{R} \setminus \{0\} \to \mathbb{S}^0 \cong \{\pm 1\}$. Therefore

$$\mathrm{R}^i p_{S\times(\mathbb{R}\setminus\{0\}),*}\mathscr{F}|_{S\times(\mathbb{R}\setminus 0)} \cong \begin{cases} \mathscr{F} \oplus \mathscr{F} & i = 0 \\ 0 & i \neq 0 \end{cases}$$

In particular taking $S = \{*\}$ since $\mathscr{F} \cong \underline{M}$ for some $M \in \mathsf{Mod}(\mathbf{k})$ we get

$$\mathrm{H}^i(\mathbb{R}^n \setminus \{0\}; \underline{M}) = \begin{cases} M & i = 0 \\ M & i = n-1 \\ 0 & i \notin \{0, n-1\} \end{cases}$$

$$\mathrm{H}^i(\mathbb{R} \setminus \{0\}; \underline{M}) = \begin{cases} M \oplus M & i = 0 \\ 0 & i \neq 0 \end{cases}$$

Chapter 13

de Rham Cohomology

Definition 13.0.1 (Partition of unity). Let $\mathscr{F}$ be a sheaf of $\mathbf{k}$-modules. A **partition of unity** of $\mathscr{F}$ subordinate to a locally finite open covering $\mathcal{U} = \{U_i\}_{i\in\mathcal{I}}$ is a collection of endomorphisms

$$\{f_i : \mathscr{F} \to \mathscr{F}\}_{i\in\mathcal{I}}$$

satisfying

1. For all $i \in \mathcal{I}$

$$\operatorname{supp} f_i \subset U_i$$

2. $$\sum_{i\in\mathcal{I}} f_i = \mathbb{1}_{\mathscr{F}}$$

   Note that this sum is well defined since for any point $x \in X$ there exists an open neighbourhood where all but finitely many of the maps f_i are the zero maps, i.e: the sum is locally finite.

Definition 13.0.2 (Fine sheaf). A sheaf of $\mathbf{k}$-modules $\mathscr{F}$ is **fine** if for every open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to this covering.

Proposition 13.0.1. Fine sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a fine sheaf. Using the Godement resolution we have that $\mathscr{F}$ admits a flasque resolution $\mathscr{F} \to \mathscr{I}^\bullet$.

Using Proposition 10.1.1 by Theorem 7.2.2 we have that

$$\mathrm{H}^k(X;\mathscr{F}) = \ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) \Big/ \operatorname{Im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$$

Let $\alpha \in \ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right)$. Since $\mathscr{I}^\bullet$ is locally exact in degree greater than 0 we have that α comes from $\mathscr{I}^{k-1}$, i.e: there exists an open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X such that $\alpha|_{U_i} = \mathrm{d}\beta_i$ with $\beta_i \in \mathscr{I}^{k-1}(U_i)$.

Consider a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to $\mathscr{U}$ and let

$$\beta := \sum_{i\in\mathcal{I}} f_i\beta_i$$

where the sum is locally finite. Note that $f_i\beta_i$ is a section of $\mathscr{I}^{k-1}(X)$ with values in $\mathscr{I}_p^{k-1}(X)$:

$$(f_i\beta_i)_p = \begin{cases} 0 & p \notin U_i \\ f_i\beta_i & p \in U_i \end{cases}$$

Since f_i has support in U_i this defines a section in $\mathscr{I}^{k-1}(U_i)$.

We have that $\mathrm{d}\beta = \alpha$ since $\alpha = \sum_{i\in\mathcal{I}} f_i\alpha|_{U_i}$. Therefore $\ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) = \mathrm{Im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$, which means

$$\forall k > 0 \quad \mathrm{H}^k(X,\mathscr{F}) = 0 \qquad \square$$

Definition 13.0.3. Let M be a smooth manifold of dimension m.
The **bundle of k-forms** $\Lambda^k(\mathrm{T}^*M)$ is the smooth vector bundle defined with:

- *Total space*: $\Lambda^k(\mathrm{T}^*M) = \coprod_{p\in M} \Lambda^k(\mathrm{T}_p^*M)$.
- *Projection map*: $\pi : \Lambda^k(\mathrm{T}^*M) \to M,\ \pi(p,\omega) = p$.
- *Fiber space*: For every $p \in M$ the fiber $E_p = \pi^{-1}(\{p\}) = \Lambda^k(\mathrm{T}_p^*M)$ is a real vector space of dimension $\binom{m}{k}$.

To define the local trivialization, consider a local chart $(U, \{x^i\}_{i=1}^m)$ of M. Then the differentials $\{\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}\}_{1\le i_1<\cdots<i_k\le m}$ form a basis for each fiber with base point in U and the map $\phi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^{\binom{m}{k}}$ is given by sending a form ω to its coefficients in this base.

Notation. Let $I = (i_1, \ldots, i_k)$ be a multi-index. We will write

$$\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$$

In this way the basis of the local trivialization is $\{\mathrm{d}x^I\}_I$.

Definition 13.0.4 (Smooth k-form)**.** Let M be a smooth manifold of dimension m.
The **sheaf of k-forms** $\Omega^k(M)$ over M is the sheaf of sections of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$.
More concretely, a k-form over an open set U is a map $\omega : U \to \Lambda^k(\mathrm{T}^*M)$ such that $\pi \circ \omega = \mathbb{1}_U$.

Example 13.0.1. Let M be a smooth manifold of dimension m.
Note that $\Lambda^0(\mathrm{T}^*M) = M \times \mathbb{R}$ is the trivial vector bundle. Therefore a 0-form is a map $\omega : M \to M \times \mathbb{R}$ such that $\pi \circ \omega = \mathbb{1}_M$, i.e: ω is a smooth function.
This means that $\Omega^0(M) \cong C^\infty(M)$.

Example 13.0.2. Let M be a smooth manifold of dimension m.
Suppose that $k > m$. Since the fibers of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$ have dimension $\binom{m}{k}$ this means that at least one index i in the multi-index I will be repeated. Therefore since $\mathrm{d}x^i \wedge \mathrm{d}x^i = 0$ we have that $\mathrm{d}x^I = 0$, i.e: $\Lambda^k(\mathrm{T}^*M) = 0$.
In particular $\Omega^k(M) = 0$ for all $k > m$.

Definition 13.0.5 (Exterior derivative)**.** Let ω be a k-form over a smooth manifold M.
If $(U, \{x^i\}_{i=1}^m)$ is a local chart then there exist functions $\alpha_I \in C^\infty(M)$ such that

$$\omega = \sum_I \alpha_I \mathrm{d}x^I$$

where $I = (i_1, \dots, i_k)$ is a multi-index and $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$.
We can define the **exterior derivative** of ω as the $k+1$-form

$$\mathrm{d}\omega = \sum_I \mathrm{d}\alpha_I \wedge \mathrm{d}x^I = \sum_I \left(\sum_{j=1}^m \frac{\partial \alpha_I}{\partial x^j} \mathrm{d}x^j \right) \wedge \mathrm{d}x^I$$

Lemma 13.0.1. Let M be a smooth manifold of dimension m.
Then for any k-form the exterior derivative verifies $\mathrm{d}^2\omega = 0$, i.e: the sequence

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

is a complex of sheaves of $\mathbb{R}$-modules.

Proof. It suffices to show that for any smooth function $f \in C^\infty(M) \cong \Omega^0(M)$ we have $\mathrm{d}^2 f = 0$. Indeed, if $\mathrm{d}^2 f = 0$ then for any basic form $f\mathrm{d}x^I$ we have

$$\mathrm{d}(f\mathrm{d}x^I) = \mathrm{d}(\mathrm{d}f \wedge \mathrm{d}x^I) = \mathrm{d}^2 f \wedge \mathrm{d}x^I - \mathrm{d}f \wedge \mathrm{d}^2 x^I = 0$$

Let $f \in C^\infty(M)$ be a smooth function. Its exterior derivative is

$$\mathrm{d}f = \sum_{i=1}^m \frac{\partial f}{\partial x^i} \mathrm{d}x^i$$

Differentiating again we get

$$\mathrm{d}^2 f = \sum_{j=1}^m \frac{\partial}{\partial x^j} \left(\sum_{i=1}^m \frac{\partial f}{\partial x^i} \right) \wedge \mathrm{d}x^j = \sum_{i,j=1}^m \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \wedge \mathrm{d}x^j$$

By Clairaut's theorem we have that $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ and by the antisymmetry of the wedge product $\mathrm{d}x^i \wedge \mathrm{d}x^j = -\mathrm{d}x^j \wedge \mathrm{d}x^j$. Therefore all the terms cancel leaving

$$\mathrm{d}^2 f = 0$$

□

Definition 13.0.6 (de Rham complex)**.** Let M be a smooth manifold of dimension m.
The **de Rham complex** of M is the complex of sheaves of $C^\infty(M)$-modules

$$\Omega^\bullet(M) := \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^m(M) \xrightarrow{\mathrm{d}} 0$$

where $\Omega^k(M)$ is the sheaf of k-forms over M and d is the exterior derivative.

Lemma 13.0.2 (Poincaré)**.** Let $U \subseteq \mathbb{R}^n$ be a contractible open set and $k \geq 1$. If $\omega \in \Omega^k(U)$ is a k-form such that $\mathrm{d}\omega = 0$ then there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(U)$ such that $\mathrm{d}\eta = \omega$.

Lemma 13.0.3. The de Rham complex is exact.

Proof. For $k \geq 1$ consider the relevant part of the complex

$$\Omega^{k-1} \xrightarrow{\mathrm{d}^{k-1}} \Omega^k \xrightarrow{\mathrm{d}^k} \Omega^{k+1}$$

We need to show that for any $p \in M$ the complex

$$\left(\Omega^{k-1}\right)_p \xrightarrow{\mathrm{d}_p^{k-1}} \left(\Omega^k\right)_p \xrightarrow{\mathrm{d}_p^k} \left(\Omega^{k+1}\right)_p$$

is exact, i.e: $\operatorname{Im} \mathrm{d}_p^{k-1} = \ker \mathrm{d}_p^k$.

For the $[\subseteq]$ inclusion, let $\mathrm{d}\eta \in \operatorname{Im} \mathrm{d}^{k-1} \subseteq \Omega^k$. By Lemma 13.0.1 we have that $\mathrm{d}(\mathrm{d}\eta) = 0$, i.e: $\mathrm{d}\eta \in \ker \mathrm{d}^k$.

For the $[\supseteq]$ inclusion, let $[\omega]_p \in \ker \mathrm{d}_p^k$, i.e: $[\omega]_p \in \left(\Omega^k\right)_p$ verifies $\mathrm{d}_p^k[\omega]_p = 0$. By definition there exists an open neighbourhood U of p and a k-form ω such that $\mathrm{d}^k\omega|_U = 0$.

Since M is a smooth manifold of dimension m we can choose a local chart (V, ψ) around p such that $V \subseteq U$ and $\psi(V)$ is an open ball in $\mathbb{R}^m$. Since open balls are contractible, by Poincaré's lemma there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(V)$ such that $\mathrm{d}^{k-1}\eta = \omega|_V$.

Therefore there exists $[\eta]_p \in \left(\Omega^{k-1}\right)_p$ such that $\mathrm{d}_p^{k-1}[\eta]_p = [\omega]_p$, i.e: $[\omega]_p \in \operatorname{Im} d_p^{k-1}$. □

Lemma 13.0.4. Let M be a smooth paracompact manifold. Then for all k the sheaf of k-forms $\Omega^k(M)$ is fine.

Proof. Let $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ be an arbitrary open cover of M.

Since M is paracompact there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}} \subset \mathcal{C}^\infty(M)$ subordinate to $\mathscr{U}$. This means that

$$\begin{aligned} \forall i \in \mathcal{I} \quad & \operatorname{supp} f_i \subseteq U_i \\ \forall p \in M \quad & \sum_{i\in\mathcal{I}} f_i(p) = 1 \end{aligned}$$

and this last sum is finite.

For all indices $i \in \mathcal{I}$ consider the map

$$\begin{aligned} \eta_i : \Omega^k(M) &\to \Omega^k(M) \\ \omega &\mapsto f_i \cdot \omega \end{aligned}$$

Note that since $\operatorname{supp} f_i \subseteq U_i$ we have $\operatorname{supp} \eta_i \subseteq U_i$. Furthermore for any k-form ω

$$\sum_{i\in\mathcal{I}} \eta_i(\omega) = \sum_{i\in\mathcal{I}} f_i \cdot \omega = \left(\sum_{i\in\mathcal{I}} f_i\right) \cdot \omega = 1 \cdot \omega = \omega$$

Therefore $\sum_{i\in\mathcal{I}} \eta_i = \mathbb{1}$, making $\{\eta_i\}_{i\in\mathcal{I}}$ a partition of unity subordinate to $\mathscr{U}$. □

Theorem 13.0.1. Let M be a smooth paracompact manifold. Then

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

Proof. Consider the resolution

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \Omega^\bullet(M)$$

where $\Omega^\bullet$ is the de Rham complex of M.

By Lemma 13.0.3 we know that $\Omega^\bullet(M)$ is exact, therefore we only need to show that this resolution is exact at $\Omega^0(M)$, i.e: that for all $p \in M$

$$0 \longrightarrow \underline{\mathbb{R}}_p \xrightarrow{\iota_p} \Omega^0_p \xrightarrow{\mathrm{d}_p} \Omega^1_p \longrightarrow 0$$

is a short exact sequence, i.e: $\operatorname{Im} \iota_p = \ker \mathrm{d}_p$.

- $\subseteq$ Let $[c]_p \in \underline{\mathbb{R}}_p$. By definition this is a locally constant function $c : U \to \mathbb{R}$ on some open neighbourhood U of p.
  Since c is locally constant we have that $\mathrm{d}c|_U = 0$, i.e $\mathrm{d}_p[c]_p = 0$.
- $\supseteq$ Let $[f]_p \in \Omega^0_p$ such that $\mathrm{d}_p[f]_p = 0$. This means that there exists an open neighbourhood U of p and a function $f \in \Omega^0(V) \cong C^\infty(M)$ such that $\mathrm{d}f|_U = 0$. Therefore f is locally constant in each of the connected components of U, i.e: $[f]_p \in \underline{\mathbb{R}}_p$.

Therefore since by Lemma 13.0.4 the resolution is made of fine sheaves and fine sheaves are $\Gamma(X, -)$ acyclic by definition of sheaf cohomology

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) = \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R}) \qquad \square$$

Corollary 13.0.1. Let M be a smooth paracompact manifold of dimension m. Then for all $k > m$

$$\mathrm{H}^k(M, \underline{\mathbb{R}}) = 0$$

Example 13.0.3 (The cohomology of S^1)**.** Consider the circle S^1 as a smooth manifold of dimension 1.
We are going to compute its sheaf cohomology with coefficients in $\underline{\mathbb{R}}$, $\mathrm{H}^\bullet(S^1, \underline{\mathbb{R}})$, using the de Rham complex

$$\Omega^0(S^1) \xrightarrow{\mathrm{d}^0} \Omega^1(S^1) \xrightarrow{\mathrm{d}^1} 0$$

On the one hand

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \ker \mathrm{d}^0 = \{f \in \Omega^0(S^1) \cong C^\infty(S^1) \mid \mathrm{d}f = 0\}$$

Since S^1 is connected, the only functions with 0 derivative are the constant functions. Therefore

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \mathbb{R}$$

On the other

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = \ker \mathrm{d}^1 \Big/ \operatorname{Im} \mathrm{d}^0 = \{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} \Big/ \{\mathrm{d}f \mid f \in \Omega^0(S^1) \cong C^\infty(S^1)\}$$

Note that since $\Omega^2(M) = 0$ all 1-forms $\omega \in \Omega^1(S^1)$ verify $\mathrm{d}\omega = 0$. Therefore

$$\{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} = \Omega^1(S^1)$$

Consider now the *integration* map

$$\begin{aligned} I : \Omega^1(S^1) &\to \mathbb{R} \\ \omega &\mapsto \int_{S^1} \omega \end{aligned}$$

Let $\mathrm{d}\theta$ be the differential of the angle. Since $I(\mathrm{d}\theta) = 2\pi$ we have that I is surjective. Furthermore by Stoke's theorem we have that $I(\omega) = 0$ if and only if $w = \mathrm{d}f$ for some $f \in C^\infty(S^1)$.

Therefore by the first isomorphism theorem we have that

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = {}^{\Omega^1(S^1)}\big/_{\ker I} \cong \operatorname{Im} I = \mathbb{R}$$

In summary, we have that

$$\mathrm{H}^k(S^1, \underline{\mathbb{R}}) = \begin{cases} \mathbb{R} & k \in \{0, 1\} \\ 0 & k > 1 \end{cases}$$

Chapter 14

Cohomology theories

In this chapter, we recap the properties of sheaf cohomology and compare them with the axiomatic approach to cohomology theories.

Definition 14.0.1. A **topological pair** is a tuple (X, A) where X is a topological space and A is a subspace of X. A morphism of topological pairs $(X, A) \to (Y, B)$ is a continuous map $f : X \to Y$ such that $f(A) \subseteq B$.
We denote by Top_2 the category of topological pairs.

We start by giving the definition of a cohomology theory following [ES45].

Definition 14.0.2. A **cohomology theory** (on Top) consists of

1. A sequence of contravariant functors $\mathrm{H}^n : \mathsf{Top}_2 \to \mathsf{Ab}$ for all $n \in \mathbb{Z}$.
   We denote by $\mathrm{H}^n(X) := \mathrm{H}^n(X, \varnothing)$. If $f : (X, A) \to (Y, B)$ is a continuous map we denote $f^* : \mathrm{H}^n(Y, B) \to \mathrm{H}^n(X, A)$ the induced map (where n is to be understood from the context).
2. A sequence of natural transformations $\delta^n : \mathrm{H}^n(A) \to \mathrm{H}^{n+1}(X, A)$.

Satisfying the following axioms:

1. **Homotopy**: If $f_0, f_1 : (X, A) \to (Y, B)$ are homotopic then its induced maps on H^n are equal
$$f_0^* = f_1^* : \mathrm{H}^n(Y, B) \to \mathrm{H}^n(X, A)$$
2. **Excision**: If $(X, A) \in \mathsf{Top}_2$ and $U \subseteq A$ is such that $\mathrm{ad}(U) \subset \mathrm{int}(A)$ then the inclussion map $i : (X \setminus U, A \setminus U) \to (X, A)$ induces an isomorphism in H^n:
$$i^* : \mathrm{H}^n(X, A) \xrightarrow{\sim} \mathrm{H}^n(X \setminus U, A \setminus U)$$
3. **Exactness**: Each pair $(X, A) \in \mathsf{Top}_2$ induces a long exact sequence via the inclussions $i : A \to X$ and $j : X \to (X, A)$:
$$\cdots \longrightarrow \mathrm{H}^{n-1}(X) \xrightarrow{i^*} \mathrm{H}^{n-1}(A) \xrightarrow{\delta^{n-1}} \mathrm{H}^n(X, A) \xrightarrow{j^*} \mathrm{H}^n(X) \xrightarrow{i^*} \mathrm{H}^n(A) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(X, A) \xrightarrow{j^*} \mathrm{H}^{n+1}(X) \longrightarrow \cdots$$

We also require the following axiom due to Milnor [Mil]

4. **Additivity**: If $X = \coprod_{i\in I} X_i$ is the disjoint union of topological spaces then

$$\mathrm{H}^n(X) \cong \prod_{i\in I} \mathrm{H}^n(X_i)$$

We say that a cohomology theory is ordinary if it further verifies

5. **Dimension**: If $\{*\}$ is the one-point topological space then for all $n \neq 0$

$$\mathrm{H}^n(\{*\}) = 0$$

In this case, $\mathrm{H}^0(\{*\})$ is called the coefficient group of the theory.

For sufficiently nice spaces, one can show that there exists only one ordinary cohomology theory with a given coefficient group.

Definition 14.0.3 (CW-complex).

Theorem 14.0.1 (Uniqueness of cohomology theories — [Hat02, Theorem 4.59]).

14.1 Relative sheaf cohomology

Definition 14.1.1 (Relative Sheaf Cohomology). Let X be a topological space and $A \subseteq X$ a closed subset. Let $j\colon X \setminus A \hookrightarrow X$ be the open immersion of the complement. For any sheaf of $\mathbf{k}$-modules $\mathscr{F}$ over X, the **relative cohomology** of the pair (X, A) with coefficients in $\mathscr{F}$ is:

$$\mathrm{H}^i(X, A; \mathscr{F}) := \mathrm{H}^i(X; j_! j^{-1}\mathscr{F})$$

If $A = \varnothing$ we recover the usual sheaf cohomology

$$\mathrm{H}^i(X, \varnothing; \mathscr{F}) = \mathrm{H}^i(X; \mathscr{F})$$

Remark 14.1.1. Relative sheaf cohomology is the right derived functor

$$\Gamma(X; j_! j^{-1}\mathscr{F}) := \{s \in \Gamma(X, \mathscr{F}) \mid s|_A = 0\}$$

Theorem 14.1.1 (Exactness). Let $A \subseteq X$ be a closed subset and $i\colon A \hookrightarrow X$ the closed immersion. For any sheaf of $\mathbf{k}$-modules $\mathscr{F}$, there exists a long exact sequence in cohomology:

$$\cdots \longrightarrow \mathrm{H}^i(X, A; \mathscr{F}) \longrightarrow \mathrm{H}^i(X; \mathscr{F}) \longrightarrow \mathrm{H}^i(A; \mathscr{F}|_A) \longrightarrow \mathrm{H}^{i+1}(X, A; \mathscr{F}) \longrightarrow \cdots$$

Proof. By Theorem 9.3.2 we have a short exact sequence of sheaves:

$$0 \to j_! j^{-1}\mathscr{F} \to \mathscr{F} \to i_* i^{-1}\mathscr{F} \to 0$$

Applying the long exact sequence of derived functors to $\Gamma(X, -)$ and using Lemma 10.0.3 for the term $\mathrm{H}^i(X; i_* i^{-1}\mathscr{F}) \cong \mathrm{H}^i(A; \mathscr{F}|_A)$, we obtain the result. □

Theorem 14.1.2 (Excision). Let $A \subseteq X$ be a closed subset. Let $Z \subseteq A$ be a subset such that $\mathrm{ad}(Z) \subseteq \mathrm{int}(A)$. Then for any sheaf $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$:

$$\mathrm{H}^i(X, A; \mathscr{F}) \cong \mathrm{H}^i(X \setminus Z, A \setminus Z; \mathscr{F})$$

Proof. Let $U \coloneqq X \setminus A$ and $j \colon U \hookrightarrow X$ the open immersion. By definition we have

$$\mathrm{H}^i(X, A; \mathscr{F}) = \mathrm{H}^i(X; \mathscr{G})$$

where $\mathscr{G} \coloneqq j_! j^{-1} \mathscr{F}$.

For simplicity, assume for now that Z is closed. Let $V = X \setminus Z$ and $W = \mathrm{int}(A)$. Since Z is closed V is open. Furthermore since $Z \subseteq \mathrm{int}(A) = W$ we have an open cover $X = V \cup W$.

By Mayer-Vietoris we have a long exact sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(V \cap W; \mathscr{G}|_{V\cap W}) \longrightarrow \mathrm{H}^i(X; \mathscr{G}) \longrightarrow \mathrm{H}^i(V; \mathscr{G}|_V) \oplus \mathrm{H}^i(W; \mathscr{G}|_W) \longrightarrow \mathrm{H}^i(V \cap W; \mathscr{G}|_{V\cap W}) \longrightarrow \mathrm{H}^{i+1}(X; \mathscr{G}) \longrightarrow \cdots$$

Note that the stalks of $\mathscr{G}$ are

$$\mathscr{G}_p = \begin{cases} \mathscr{F}_p & p \in U \\ 0 & p \notin U \end{cases}$$

Therefore since $W \subseteq A$ and $A \cap U = \varnothing$ we have $\mathscr{G}_p = 0$ for all $p \in W$. In particular $\mathscr{G}|_W = 0$ and thus $\mathscr{G}|_{V\cap W} = 0$. Plugging this into the above long exact sequence we get isomorphisms

$$\mathrm{H}^i(X; \mathscr{G}) \xrightarrow{\sim} \mathrm{H}^i(V; \mathscr{G}|_V)$$

To finish, we need to check that $\mathrm{H}^i(V; \mathscr{G}) = \mathrm{H}^i(X \setminus Z, A \setminus Z, \mathscr{F})$. Note that

$$V \setminus (A \setminus Z) = (X \setminus Z) \setminus (A \setminus Z) = X \setminus A = U$$

Consider the open immersion $j' \colon U \hookrightarrow V$. Since $\mathscr{G}$ was formed by extending $\mathscr{F}|_U$ by zero to X, restricting it to V (which fully contains U) yields the same extension by zero relative to V:

$$\mathscr{G}|_V = j'_! (\mathscr{F}|_U) = j'_! j'^{-1} (\mathscr{F}|_V)$$

Therefore

$$\mathrm{H}^i(V; \mathscr{G}|_V) = \mathrm{H}^i\left(V; j'_! j'^{-1} \mathscr{F}|_V\right) = \mathrm{H}^i(X \setminus Z, A \setminus Z; \mathscr{F}) \qquad \square$$

Theorem 14.1.3 (Homotopy). Let $f_0, f_1 \colon (X, A) \rightrightarrows (Y, B)$ be two homotopic maps of pairs. Let $\underline{M}_Y \in \mathsf{Mod}(\underline{\mathbf{k}}_Y)$ be a constant sheaf of $\mathbf{k}$-modules over Y. Then for all $i \geq 0$, the induced maps on relative cohomology coincide:

$$f_0^* = f_1^* \colon \mathrm{H}^i(Y, B; \underline{M}_Y) \to \mathrm{H}^i(X, A; \underline{M}_X)$$

Proof. Let $h \colon (X \times I, A \times I) \to (Y \times I, B \times I)$ be the homotopy between f_0 and f_1. We have

$$f_0^* = j_0^* \circ H \qquad f_1^* = j_1^* \circ H$$

Therefore we need to prove

$$j_0^* = j_1^* \colon \mathrm{H}^i\left(X \times I, A \times I; \underline{M}_{X\times I}\right) \to \mathrm{H}^i\left(X, A; \underline{M}_X\right)$$

Note that $p_{(X,A)} \circ j_0 = \mathbb{1}_{(X,A)} = p_{(X,A)} \circ j_1$. Furthermore

$$i_0 \circ p_{(X,A)} \colon (X \times I, A \times I) \to (X \times I, A \times I)$$

is homotopic to the identity.

Therefore by Theorem 11.0.1 we get isomorphisms

$$p_X^* \colon \mathrm{H}^i(X; \underline{M}_X) \xrightarrow{\sim} \mathrm{H}^i\left(X \times I; \underline{M}_{X\times I}\right)$$
$$p_A^* \colon \mathrm{H}^i(A; \underline{M}_A) \xrightarrow{\sim} \mathrm{H}^i\left(A \times I; \underline{M}_{A\times I}\right)$$

By the exactness axiom we have long exact sequences

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & \mathrm{H}^{i-1}(A; \underline{M}) & \longrightarrow & \mathrm{H}^i(X, A; \underline{M}) & \longrightarrow & \mathrm{H}^i(X; \underline{M}) & \longrightarrow & \mathrm{H}^i(A; \underline{M}) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle p_A^*}{\scriptstyle \sim} & & & & \downarrow{\scriptstyle p_X^*}{\scriptstyle \sim} & & \downarrow{\scriptstyle p_A^*}{\scriptstyle \sim} & & \\
\cdots & \longrightarrow & \mathrm{H}^{i-1}(A \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(X \times I, A \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(X \times I; \underline{M}) & \longrightarrow & \mathrm{H}^i(A \times I; \underline{M}) & \longrightarrow & \cdots
\end{array}$$

Therefore by the five lemma

$$p_{(X,A)}^* \colon \mathrm{H}^i(X, A; \underline{M}) \xrightarrow{\sim} \mathrm{H}^i(X \times I, A \times I; \underline{M})$$

By functoriallity of cohomology we have

$$j_0^* \circ p_{(X,A)}^* = \mathbb{1} = j_1^* \circ p_{(X,A)}^*$$

Therefore since $p_{(X,A)}^*$ is an isomorphism

$$j_0^* = j_1^* \qquad \square$$

Theorem 14.1.4 (Additivity). Let $X = \coprod_{\alpha\in\Lambda} X_\alpha$ be a disjoint union of topological spaces. Then for any sheaf of $\mathbf{k}$-modules $\mathscr{F}$ on X:

$$\mathrm{H}^i(X; \mathscr{F}) \cong \prod_{\alpha\in\Lambda} \mathrm{H}^i(X_\alpha; \mathscr{F}|_{X_\alpha})$$

Proof. By definition of the disjoint union topology, each X_α is an open subset of X.

Let $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$. Since $X_\alpha \cap X_\beta = \varnothing$ if $\alpha \neq \beta$ we have

$$\Gamma(X, \mathscr{F}) \cong \prod_{\alpha\in\Lambda} \Gamma(X_\alpha, \mathscr{F})$$

Furthermore if $\mathscr{F} \to \mathscr{I}^\bullet$ is an def:resolution]injective resolution since j_α^{-1} is right adjoint by Proposition 7.1.3 $I^\bullet|_{X_\alpha}$ is an def:resolution]injective resolution of $\mathscr{F}|_{X_\alpha}$.

Therefore

$$\begin{aligned}
\mathrm{H}^i(X;\mathscr{F}) &\cong \mathrm{H}^i\,\Gamma(X,\mathscr{I}^\bullet) \\
&\cong \mathrm{H}^i\left(\prod_{\alpha\in\Lambda}\Gamma(X,\mathscr{I}^\bullet|_{X_\alpha})\right) \\
&\cong \prod_{\alpha\in\Lambda}\mathrm{H}^i\,\Gamma(X_\alpha,\mathscr{I}^\bullet|_{X_\alpha}) \\
&\cong \prod_{\alpha\in\Lambda}\mathrm{H}^i(X_\alpha,\mathscr{F}|_{X_\alpha})
\end{aligned}$$

□

Theorem 14.1.5 (Dimension)**.** Let $\{*\}$ be the one point topological space. For all sheaves of $\mathbf{k}$-modules $\mathscr{F} \in \mathsf{Mod}(\underline{\mathbf{k}}_X)$ and $n \neq 0$

$$\mathrm{H}^n(\{*\};\mathscr{F}) \cong 0$$

Proof. We have

$$\mathrm{H}^n(\{*\};\mathscr{F}) = \mathrm{R}^n\,\Gamma(\{*\},\mathscr{F})$$

Note that

$$\Gamma(\{*\},\mathscr{F}) = \mathscr{F}(\{*\}) = \mathscr{F}_*$$

Therefore since the stalk functor is exact for all $n \neq 0$

$$\mathrm{H}^n(\{*\};\mathscr{F}) = \mathrm{R}^n\,\mathscr{F}_* = 0$$

□

Corollary 14.1.1. Let X be a CW-complex. For any constant sheaf of abelian groups $\underline{G} \in \mathsf{Mod}(\underline{\mathbb{Z}}_X)$ we have an isomorphism

$$\mathrm{H}^i(X;\underline{G}) \cong \mathrm{H}^i_{\mathrm{sing}}(X;G)$$

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